

A Nation of Emigrants: Transatlantic Emigration and Italian National Identity *

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Abstract

We study how intergroup contact among immigrants shapes national identity formation within migrant communities and in their places of origin. We assemble a new dataset on Italian migrants to the U.S. between 1892 and 1924 by linking population census records to Ellis Island passenger lists, which report migrants' municipalities of origin. In the U.S., exposure to more diverse Italian communities fosters shared national identity, as measured by national naming choices and intergroup marriage, while reducing assimilation into American society. In Italy, municipalities whose migrants were exposed to more diverse communities abroad exhibit lower cultural particularism. Our findings uncover a nation-building effect of intergroup contact within immigrant communities, extending from immigrant communities abroad to migrants' places of origin.

Keywords: National Identity, Migration, Age of Mass Migration

JEL Classification: J15, N32, N33, Z13

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1 Introduction

Nation-states transformed the scale at which political and economic life is organized. A prominent view in economics and political science holds that national identities arise from intentional state interventions deployed by political elites, such as mass schooling, bureaucratic centralization, military conscription, and propaganda (Mylonas and Tudor, 2021; Rohner and Zhuravskaya, 2024). This paper asks whether decentralized social interactions can create national identity in the absence of coordinated state intervention.

We examine this question in the context of the Italian migration to the United States (1892–1924). During this period, Italy was a recently unified, internally fragmented country, famously described by Austrian Chancellor Klemens von Metternich as little more than a “geographical expression.” The 4.5 million Italian migrants arriving in the United States constituted a heterogeneous group embedded in local identities who interacted within shared urban environments (Choate, 2008; Gabaccia, 2013). This context provides an ideal setting to examine whether decentralized social interactions among heterogeneous individuals can foster the emergence of a shared national identity, both within immigrant communities and, given the high rates of return migration, in their places of origin.

We construct a novel dataset linking U.S. census records of Italian immigrants to ship manifests recorded at the Ellis Island immigration station. This linkage allows us to trace the municipality of origin for nearly half a million migrants, yielding an unprecedentedly detailed view of the within-country composition of a historical immigrant community.

Our first finding is that Italian immigrants exposed to more diverse Italian communities in the U.S. were more likely to abandon local identities, measured through the local distinctiveness of their children’s names and marriages with immigrants from other parts of Italy, at the expense of slower cultural assimilation into American society. Our second finding is that identity formation within migrant communities had lasting effects on their places of origin. Italian municipalities whose emigrants were exposed to more diverse communities in the U.S. exhibit lower name particularism and greater commemoration of national events, as reflected in street-naming patterns and in monument construction. Evidence from emigrants’ letters, linguistic contamination of local dialects, returnees’ political activity, and voluntary enlistment in WWI suggests that return migration was a key mechanism for the diffusion of national identity into Italy. These findings provide novel evidence that decentralized social interactions among migrants can foster the consolidation of national identity and reshape both immigrant communities and their societies of origin.

Despite wide historical accounts, there is little quantitative evidence on the role of Italian emigrant communities in the formation of national identity. Existing data sources are not well-suited to addressing this question, as they do not provide information on migrants’ municipalities of origin and

¹The original epigraph reads “*Gli Italiani all'estero alla Madrepatria*” and is engraved on the two ceremonial braziers of the *Vittoriano* monument in Rome, whose perpetual flames symbolize national unity.

their locations in the U.S. To overcome this limitation, we develop an algorithm that links U.S. census records to ship passenger lists recorded at the Ellis Island immigration station, which report migrants' municipalities of origin. Our approach augments standard record-linkage methods (Abramitzky et al., 2021) with deep-learning text embeddings to flexibly match names that were frequently Anglicized after arrival. In our preferred specification, we link approximately 470,000 migrants, equivalent to a 20% matching rate.

We use this novel dataset to test two hypotheses. First, we examine whether exposure to Italians from different parts of the country at the neighborhood level shaped the formation of national identity in the United States. Second, we study whether inter-group contact among migrants influenced the formation of national identity in their municipalities of origin.

We conduct the first part of the analysis at the individual level. To measure the national identity of Italians in the U.S., we exploit the linguistic and cultural divides within Italy documented by De Mauro (1963) to construct a novel index of name particularism, following recent work (e.g., see Fouka, 2020). Names that are geographically concentrated reflect stronger local identities, whereas names that are more evenly distributed across the country capture national identity. We complement this measure with information on marriage patterns with individuals from other municipalities, provinces, and regions, as in Bazzi et al. (2019). We measure assimilation into American society using the name-based indicator developed by Abramitzky et al. (2020).

Estimating the effect of neighborhood diversity on cultural identity raises two distinct identification challenges. First, local diversity may reflect the selective sorting of immigrants across locations. Second, local diversity may be correlated with omitted factors that independently influence identity formation.

We address these concerns through two approaches. First, we construct an instrumental variable that leverages variation in ship composition, following the logic of Battiston (2018), Dippel and Heblich (2021), and Green (2025). Under the hypothesis that interactions during the transatlantic voyage shaped subsequent settlement patterns, variation in ship composition generates plausibly exogenous differences in immigrants' exposure to diversity upon arrival. Individuals traveling on more diverse ships are more likely to settle near their shipmates, and therefore to be exposed to more diverse Italian neighborhoods. We show that ship diversity predicts neighborhood diversity and is uncorrelated with observable individual and local characteristics, suggesting that the instrument plausibly isolates exogenous variation in exposure to within-Italian diversity.

Second, we implement a "movers" design in the spirit of Chetty and Hendren (2018). We estimate a difference-in-differences specification comparing individuals who move to more diverse neighborhoods with non-movers before and after the move. This design allows us to assess the parallel trends assumption by showing that movers and non-movers exhibit statistically indistinguishable trends in identity before relocation.

We find that exposure to more diverse Italian communities decreases cultural particularism and reduces Americanization. Moreover, Italians exposed to more diverse communities are more likely to marry individuals born in other municipalities and provinces, and less likely to marry natives.

Quantitatively, ten years after arrival, immigrants exposed to more diverse neighborhoods exhibit a 10% increase in national identity and show no evidence of assimilation into American society. In contrast, immigrants exposed to less diverse neighborhoods do not experience an increase in national identity and give their children 20% more American-sounding names.²

In the second part of the paper, we examine whether the national identity that formed within immigrant communities spilled over to migrants' places of origin. We use three indicators to proxy for the cultural relevance of national identity at the local level: naming patterns, street naming choices, and the construction of national monuments.

Using given names as our primary proxy, we show that Italian municipalities whose migrants were more exposed to diverse Italian communities in the United States experienced a persistent decline in name-based particularism. To address the absence of detailed historical microdata on names, we draw on individual-level casualty records for soldiers who died during World War 1 and on a newly constructed dataset of World War 2 casualties. In a difference-in-differences setting, we find that municipalities whose migrants were exposed to higher diversity in the United States shifted from locally concentrated names toward nationally shared names, consistent with the transmission of national identity from migrant communities back to their places of origin.

We corroborate these patterns using street-naming data. Municipalities with greater historical exposure to diversity among their U.S. emigrants are significantly more likely to commemorate national figures and events in street names. Conversely, we find no systematic relationship between emigrants' diversity exposure and placebo names related to religion, art, and literature. This contrast suggests that the results are not driven by a general propensity for commemoration or naming activity.

Third, we examine public monuments. We find that exposure to diversity through U.S. migration predicts a higher presence of monuments commemorating national events, but only in the decades following mass emigration. Exposure to diversity, by contrast, is not associated with national monuments in pre-migration decades, nor with monuments that do not commemorate national events.

How did emigrants concretely contribute to the formation of national identity in Italy? We find that emigration exerted a substantial cultural impact on Italy, plausibly through return migration.

First, we examine the content of letters that emigrants sent to their families in Italy. These reveal that emigrants learned Italian while abroad, interacted with Italians from other parts of the country, and developed nationalistic attitudes. Moreover, biographical data indicate that more than half of the emigrant population returned to their place of origin. Second, using newly digitized data from a detailed ethnolinguistic atlas of Sicilian dialects, we document a positive correlation between exposure to diverse emigrant communities in the United States and linguistic contamination across local dialects. Third, return migrants were heavily involved in public life. Compared with Italian-born individuals, first- and second-generation returnees were between 1.5 and 6 times more likely to be elected to local public office. Fourth, we document a positive correlation between municipality-level

²We formalize these results in a model of immigrant identity choice along the lines of Akerlof and Kranton (2000) and Shayo (2009). Under mild and plausible assumptions, this simple framework, displayed in Appendix Section D, yields predictions that closely align with our empirical findings.

death rates of U.S. emigrants who volunteered in the Italian Army during World War 1 and exposure to diversity in the United States. While suggestive, these patterns provide consistent evidence that emigrants returned, actively contributed to public life, and displayed attachment to the Italian state.

Benedict Anderson (1983) views nations as “imagined communities” formed through shared representations that do not rely on interpersonal contact. This paper documents a complementary perspective. Social interactions within migrant communities create national identity within fragmented groups. Our findings suggest that today’s diasporic communities—often originating from highly fractionalized societies (Alesina et al., 2003)—can play an important and underappreciated role in fostering nation-building.

Contributions to the Literature. First, a growing body of work examines the economics of nation-building (Rohner and Zhuravskaya, 2024).³ A unifying approach in this literature is to view national identity emerging in response to top-down interventions, such as mass education (Cantoni et al., 2017; Bandiera et al., 2019; Alesina et al., 2021; Blanc and Kubo, 2025), foreign state intervention (Dell and Querubin, 2018; Dehdari and Gehring, 2022), media propaganda (DellaVigna et al., 2014; Esposito et al., 2023), welfare spending (Caprettini and Voth, 2023), forced relocation (Bazzi et al., 2019; Carlitz et al., 2025), and military service and conflict (Ronconi and Ramos-Toro, 2025). Our contribution is to study a setting in which individuals from heterogeneous communities are insulated from coordinated policies. In this sense, our findings are related to Cagé et al. (2023), who link social interactions with charismatic leaders to preferences for authoritarianism. We provide novel evidence that sustained inter-group contact can foster the formation of national identity.

Second, we contribute to the literature on immigrant assimilation (e.g., Abramitzky et al., 2020; Fouka et al., 2022; Gagliarducci and Tabellini, 2025) and their influence on receiving communities (Bazzi and Fiszbein, 2025). A common focus of this work is on minority assimilation into a majority (native) population. We take a different perspective by studying how exposure to migrants from the same origin country, but from different regions, shapes group identity formation.⁴ Guided by a simple model of identity choice in the spirit of Akerlof and Kranton (2000), we show that greater within-origin diversity among migrants strengthens national identity and slows cultural assimilation into the host society.

Third, our findings relate to the literature on inter-group contact and identity. Most studies in this literature focus on majority–minority interactions (e.g. Bazzi et al., 2019; Bursztyn et al., 2024), in the workplace or across educational groups (Billings et al., 2021; Corno et al., 2022), and across ethnic, gender, or political cleavages (Dahl et al., 2021; Schindler and Westcott, 2021; Bagues and Roth, 2023;

³The economics of nation-building is a relatively recent field, but it builds on an established literature at the intersection of sociology, political science, and history. Modernist theses argue that nation-states emerged in response to industrialization (Gellner, 1983; Anderson, 1983), whereas Smith (1986) maintains that modern nations reflect pre-modern identities. Classic case studies of top-down nation-building processes within this tradition include Weber (1976) on education in France and Hobsbawm and Ranger (1983) on the use of invented tradition as a legitimizing device for national identity.

⁴In this sense, this work is related to the multidisciplinary field of diaspora studies (among others, see Gilroy, 1993; Edwards, 2003). Hall (2015) emphasizes that cultural identity in diasporic communities is produced in response to displacement and internal heterogeneity. Our findings echo these qualitative narratives, indicating that Italians in the United States developed an identity that is partially distinct from the local identity they acquired in Italy.

Okunogbe, 2024). We inform this literature by showing that *within*-minority interactions (among migrants) can shape group identity formation and reallocate identity away from the majority–minority assimilation path.

Fourth, we focus on the effects of out-migration on origin countries (Batista et al., 2025) and, specifically, contribute to the recent literature on so-called “social remittances.” Previous work documents how migrants diffuse social norms (Levitt, 1998; Levitt and Lamba-Nieves, 2011; Beine et al., 2013; Melki et al., 2024; Godlonton and Theoharides, 2025) and political preferences (Spilimbergo, 2009; Barsbai et al., 2017; Karadja and Prawitz, 2019; Manacorda et al., 2024) to origin communities, and contribute to increased collectivism among stayers (Knudsen, 2024). We document a novel effect of emigration: its contribution to the formation of national identity in migrants’ countries of origin.

Outline of the Paper. Section 2 provides a concise overview of the historical background. Section 3 introduces our novel dataset and the additional data used in the analysis. Section 4 focuses on the United States and documents that Italian immigrants developed a distinctive group identity, which we refer to as “national identity.” Section 5 examines how historical out-migration contributed to the rise of national identity in migrants’ communities of origin. Section 6 documents the concrete cultural and political effects of the emigration on Italian communities. Section 7 concludes.

2 Historical Background

2.1 The Italian Mass Migration to the United States, 1892–1924

Between 1876 and the early 1920s, approximately 14 million Italians emigrated abroad, constituting the largest episode of voluntary emigration in recorded history (Choate, 2008). By 1913, Italy was the world’s second-largest source of emigrants, and its emigration rate was comparable to countries such as Ireland and Norway in earlier decades (Foerster, 1919). The United States absorbed roughly 40 percent of this flow (Livi-Bacci, 1961). Migration was facilitated by liberal migration regimes on both sides of the Atlantic and remained unrestricted until the U.S. introduced a literacy test in 1917 and quota restrictions in 1921–1924, which abruptly curtailed inflows (Abramitzky and Boustán, 2017).

Italian migrants to the United States were predominantly young men from rural areas of Southern Italy (Foerster, 1919). Although migrants were negatively selected at the national level, they were positively selected within origin communities (Spitzer and Zimran, 2018). Persistent local ties within sending communities shaped migration decisions and settlement patterns in the United States. Historical accounts and recent empirical evidence emphasize the importance of origin-based networks in directing migrants toward destinations where relatives and those from the same municipality had already settled (Vecoli, 1983; Spitzer and Zimran, 2023). Chain migration produced substantial geographic concentration by place of origin. Stonecutters from Massa, for example, settled in Barre, Vermont; migrants from Molfetta concentrated in Hoboken, New Jersey; and emigrants from Floridia, Sicily, located in Hartford, Connecticut (Choate, 2008).⁵ Local clustering persisted even within cities and neighborhoods. In East Harlem, Neapolitans concentrated between 106th and 108th Streets, while

⁵Additional historical detail and rich anecdotal evidence are provided in Appendix Section C.

migrants from Basilicata settled primarily between 108th and 115th Streets (Orsi, 1985). In Chicago, Italians similarly settled across distinct neighborhoods according to their places of origin (Vecoli, 1983). These settlement patterns reproduced in the United States the strong localism that characterized Italian society at home.

Migration was also frequently temporary: many migrants were unmarried or left their families in Italy and viewed migration as a period of work abroad to support households at home and accumulate savings before returning (Cinel, 2002). Return flows were correspondingly large. U.S. statistics for 1908–1923 record return migration equivalent to about 60 percent of arrivals among Southern Italians, substantially above the rates observed for several other major European immigrant groups (Wyman, 1993; Cinel, 2002).

Emigrants typically returned to their home communities, often after spending several years abroad (Cinel, 2002). They maintained close connections with families and communities of origin through letters, remittances, visits, and information carried across the Atlantic by friends (Cinel, 2002; Wyman, 1993). In many Italian towns, news from America became part of local everyday life. Cinel notes that “American letters became almost common property,” with their contents widely discussed within the community (Cinel, 2002).

2.2 Cultural Fragmentation in Italy After Unification

Italian national identity emerged within political elites and only gradually diffused across the wider population in response to political unification in 1861 (Riall, 2002; Banti, 2014). This delay is captured by a famous 1867 epigram by Piedmontese statesman Massimo d’Azeglio, who remarked “Italy has been made; Italians remain to be made.” Regional divisions rooted in centuries of political fragmentation persisted along linguistic, cultural, and economic dimensions well into the twentieth century (Dickie, 1999).

At the time of unification, only a small minority of the population, estimated between 2.5 and 6.5 percent, spoke standard Italian (De Mauro, 1963). Italian functioned as a written language accessible to literate elites. Literacy rates were low, particularly in the South (Ciccarelli and Weisdorf, 2019).

Educational expansion gradually promoted linguistic homogenization but proceeded unevenly. Despite the introduction of compulsory elementary schooling (formally in 1861, but effectively in 1877), attendance was not universal, and dialects remained widespread, including as a language of instruction at schools. Linguistic fragmentation, therefore, remained pervasive on the eve of World War 1, especially in rural areas from which most migrants originated. A contemporary public inquiry documented that, as late as 1906, only 20% of the population could use Italian for everyday communication, and this proportion was likely lower in Southern rural areas (Corradini, 1910).

Geography and economic divergence reinforced these divisions. Mountainous terrain and limited transport infrastructure constrained cross-regional interactions, while the economic divide between the Center–North and the South widened after unification. The ratio of Southern to Northern GDP per capita declined between unification and the interwar period (Felice, 2019). This divergence reflected distinct development trajectories and was accompanied by social and political tensions, including the

post-unification civil war known as *brigantaggio* (Molfese, 1964; Dickie, 1999). Linguistic, economic, and cultural fragmentation thus remained central features of Italian society well into the Age of Mass Migration.

2.3 Cultural Convergence in Immigrant Communities in the United States

Italy's internal fragmentation was initially reproduced within Italian immigrant communities in the United States. Migrants tended to settle near individuals from the same municipality, province, or region, and social organizations were often structured around these local ties (Gabaccia, 2013). Over time, however, interactions within shared urban environments weakened some of these boundaries and contributed to the emergence of a broader Italian-American identity. Historians emphasize the role of shared experiences in the host country, including labor market interactions, associational life, religious institutions, and exposure to common media, in fostering this convergence (Vecoli, 1985; Luconi, 2003).

Language provides a particularly visible example. Most Italian immigrants did not speak standard Italian upon arrival and instead communicated in regional dialects. Contact among migrants from different parts of Italy generated linguistic convergence. Immigrants developed a creole that mixed Neapolitan with English, functioning as a *lingua franca* across regional lines (Carnevale, 2010). The Italian-language press circulated widely within immigrant communities and helped standardize language use across regional groups (Deschamps, 1996).

Mutual aid societies also illustrate cultural convergence. Early associations were frequently organized at the village or provincial level and restricted membership to migrants from specific localities. Covello (1967), for example, counted sixty-four associations with origin-based admission criteria in East Harlem alone. Beginning in the early twentieth century, however, such organizations increasingly federated across regional lines, eventually culminating in larger associations organized around a shared Italian identity (Vecoli, 1985).

Religious practices underwent an analogous process of convergence. Although Italian migrants were overwhelmingly Catholic, devotional traditions varied considerably across places of origin. In St. Louis, for instance, Sicilian immigrants boycotted worship of St. Ambrose, a cult that was widespread in Milan (Luconi, 2003). Over time, shared parishes and religious celebrations promoted interaction across regional groups and contributed to the emergence of common practices. Homogeneous religious practices emerged both endogenously, as a consequence of between-group interactions, as in the case of religious festivals in Manhattan, and in response to homogenizing efforts by the Catholic Church, particularly the Scalabrinians (Orsi, 1985; Choate, 2008).

3 Data

3.1 Linking Ellis Island Ship Passenger Lists to Census Records

3.1.1 Motivation and Overview

Our empirical analysis requires observing both the origin and destination of Italian immigrants. This information is necessary to measure the composition of Italian immigrant communities at a fine level

of spatial granularity and, in turn, migrants’ exposure to individuals from other parts of Italy. U.S. population census records report the enumeration district where individuals resided, but do not contain information on migrants’ places of origin beyond the country level.⁶

To circumvent this limitation, previous studies have linked U.S. census records to origin-country censuses for the United Kingdom (Coluccia and Dossi, 2025) and Norway (Abramitzky et al., 2012).⁷ In this paper, we link microdata from ship rosters recorded at the Ellis Island immigration station to the U.S. population census. These rosters report individuals’ names, surnames, years of birth, and years of immigration for approximately 4.2 million Italian migrants. Of these, about 2.5 million records are georeferenced to the coordinates of the municipality of origin in Italy and constitute our analysis sample. Coluccia and Spadavecchia (2025) show that migrants with a valid origin municipality are representative of the population of Italian immigrants.

Relative to prior approaches, this strategy offers two methodological advantages. First, all immigrants except for temporary movers should appear in both datasets.⁸ Second, and more importantly, the data include an additional blocking variable (year of immigration), which reduces the risk of false positives and multiple matches.

3.1.2 Description of the Linking Algorithm

Linking Ellis Island ship manifests to U.S. census records poses two challenges. First, census enumerators and Ellis Island officials typically did not speak Italian, leading to frequent transcription errors. Second, many Italians Anglicized their names after arrival. While spelling errors can be addressed using fuzzy string-matching techniques, name translations cause original and translated names to be orthographically distant.

To address this issue, we exploit the flexibility of neural networks to construct embeddings that preserve semantic proximity using bidirectional transformers (SBERT, Reimers and Gurevych, 2019).⁹ We fine-tune a multilingual encoder on a subset of names translated using a large language model (GPT-4o-mini) and use it to identify name translations. We complement this SBERT-based classifier with a traditional Jaro-Winkler classifier to handle spelling mistakes. In our application, we consider two names a match if either their Jaro-Winkler proximity exceeds .95, a threshold commonly used in the literature, or their SBERT similarity exceeds .65, an optimally chosen threshold. Surnames are not subject to translation. Hence, we consider two surnames a match if their Jaro-Winkler similarity exceeds .95.

Using this classification rule, we link the records in the Ellis Island dataset to the U.S. census. Consider an entry i in the Ellis Island dataset, and let n_i and s_i denote i ’s first and last names. Let b_i

⁶Appendix Section B provides additional details on the data sources, the implementation of the linking algorithm, and the validation exercises used to assess its reliability.

⁷Anelli et al. (2025) adopt a different approach and link census records to later social security records, which also contain the origin of the beneficiary. The main disadvantage of this approach relative to ours is that SSNs were issued in the 1970s, implying that they covered only migrants who did not return. This sample is likely non-representative, especially for Italians, who displayed return rates exceeding 60%.

⁸Approximately 95% of Italians who entered the United States between 1895 and 1924 passed through the Ellis Island station (Unrau, 1984).

⁹Figure F.1 provides a visual representation of the linking algorithm.

and y_i denote i 's year of birth and year of immigration, respectively. Define $\mathcal{M}_i = \{j \mid b_j \in [b_i - 2, b_i + 2] \wedge y_j \in [y_i - 5, y_i + 5]\}$ as the set of *candidate* matches among U.S. census records. Following Abramitzky et al. (2021), we allow for measurement error in reported birth and immigration years. Let $\overset{n}{\sim}$ denote the name-equivalence classifier defined above, such that $n_i \overset{n}{\sim} n_j$ if and only if the SBERT similarity between n_i and n_j exceeds 0.65 or their Jaro–Winkler similarity exceeds 0.95. Analogously, $\overset{s}{\sim}$ is the surname-equivalence classifier, such that $s_i \overset{s}{\sim} s_j$ if and only if their Jaro–Winkler similarity exceeds 0.95. The set of *effective matches* is then $\widetilde{\mathcal{M}}_i = \{j \in \mathcal{M}_i \mid n_i \overset{n}{\sim} n_j \wedge s_i \overset{s}{\sim} s_j\}$. The set $\widetilde{\mathcal{M}}_i$ may be empty, in which case i is unmatched; it may contain a single element (a “single” match); or it may contain multiple elements (a “multiple” match). Multiple matches are weighted by the inverse of the cardinality of $\widetilde{\mathcal{M}}_i$.

3.1.3 Evaluation of the Linking Approach and Validation

Figure I presents visual diagnostics of the performance of the linking algorithm.¹⁰ Figure Ia shows the number of immigrants by year of arrival in the full sample (gray bars) and in the linked sample (red dots). The two series exhibit a strong positive correlation, indicating that the algorithm does not disproportionately match migrants from particular arrival cohorts. Out of 2,526,024 immigrants in the sample, the algorithm successfully links 473,922 individuals to at least one census record.

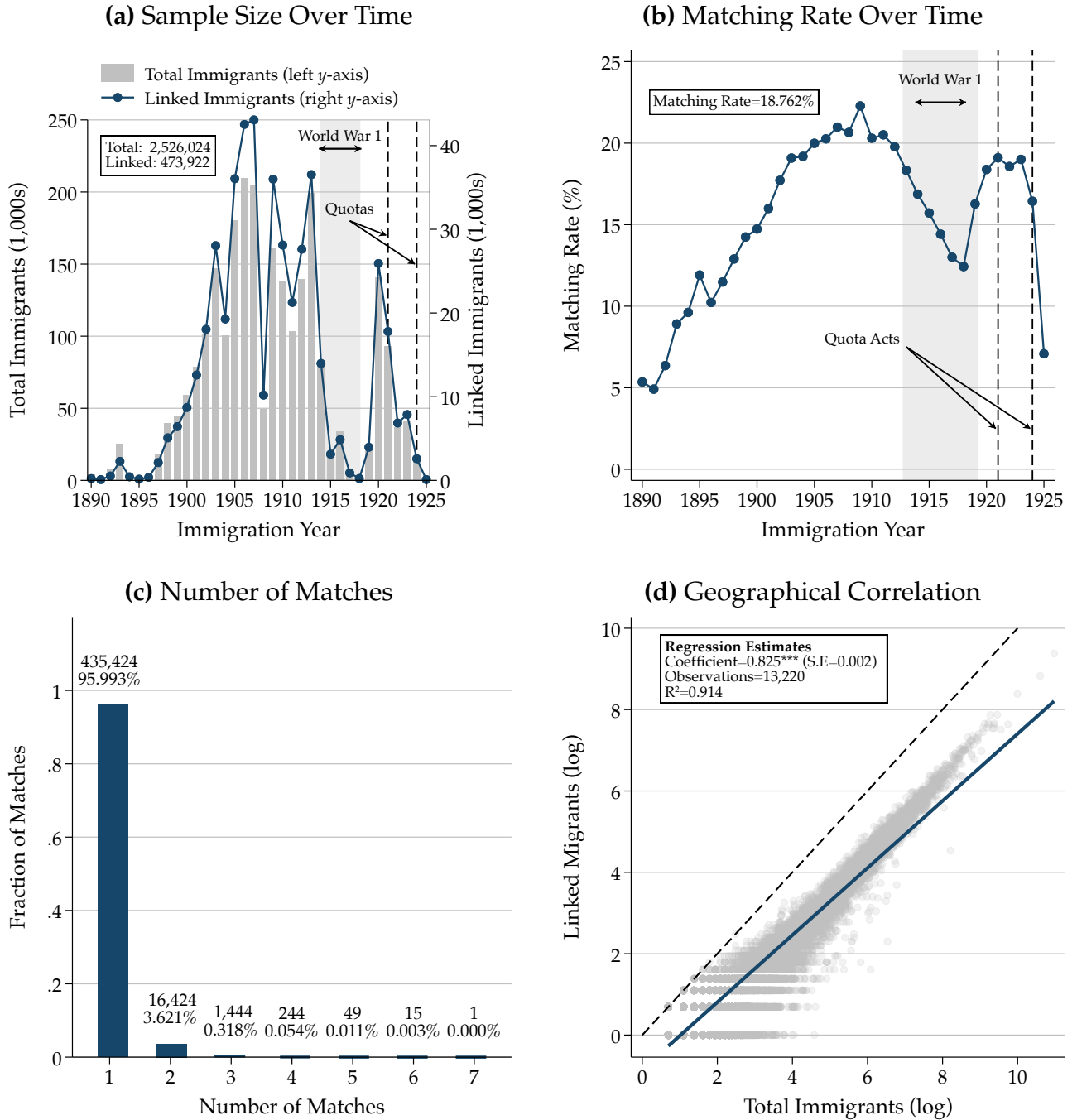
The average matching rate, shown in Figure Ib, is 19%. Matching rates are lower before 1900 because the quality of recorded names and surnames in the Ellis Island data improves between 1895 and 1900. Thereafter, the matching rate remains stable, except during the war years. Because immigrant inflows contracted during WW1, however, lower match rates during this period do not result in a large loss in sample size. The average matching rate is comparable to that reported in earlier record linkage studies, such as Abramitzky et al. (2021). However, when focusing on non-English individuals, Abramitzky et al. (2021) report lower rates (e.g., 10% for the Germans). More recent approaches, such as Althoff et al. (2025) and Green (2025), achieve higher matching rates. These figures are not directly comparable to ours, as both papers focus on samples composed predominantly of English-speaking individuals. Turning to non-English populations, Postel (2023) constructs a longitudinal linked sample of Chinese immigrants using U.S. census records and attains a 9% matching rate. Our algorithm thus performs on the higher end for relevant populations.

A high incidence of multiple matches may indicate spurious links. Figure Ic shows that 96% of the matches produced by the algorithm are single, meaning Ellis Island records linked to one census record. The share of records associated with more than two matches is less than 1%.

Figure Id explores the representativeness of the linked sample. We plot the log of the number of emigrants by municipality in the full and linked samples. The correlation between the two series is positive and statistically significant, indicating that the distribution of municipalities of origin in the linked sample mirrors that of the full sample. The estimated slope ($\hat{\beta} = 0.825$) suggests that migrants from smaller municipalities are slightly more likely to be linked, but the magnitude of this difference is modest.

¹⁰Appendix Section B.4 provides complementary evidence to evaluate the internal reliability of the linking algorithm.

Figure I. Descriptive Statistics on Linked Migrant Sample



Notes. This figure reports descriptive statistics for the linked sample of U.S. census records of Italian migrants matched to Ellis Island ship records. Panel [1a](#) shows the number of migrants in the Ellis Island data (gray bars) and the number successfully linked to U.S. census records (red bars). Panel [1b](#) reports the share of U.S. census records that match at least one Ellis Island record. Panel [1c](#) displays the distribution of the number of Ellis Island matches per U.S. census record, together with the corresponding shares relative to the total sample. Panel [1d](#) plots, at the municipality level, the number of migrants recorded in the Ellis Island dataset against the number appearing in the linked dataset, and includes a linear fit, the 45-degree line, and the estimated regression coefficient. Referenced on pages: [9](#), [A.22](#), [A.23](#).

Table [E.1](#) reports selected characteristics of migrants with a match (column 1) and without a match (column 2). Columns 3 and 4 present differences between the two groups. Panel A compares linked and non-linked migrants in the Ellis Island data, while Panel B reports the corresponding comparison using U.S. census records. In the Ellis Island data, linked migrants are younger than non-linked

migrants. They are also slightly less likely to originate from Northern Italy and more likely to come from the Islands, primarily Sicily, although these differences are small and not statistically significant. In the U.S. census, linked migrants are more likely to be male. This pattern is expected, as matching rates for women are lower due to surname changes after marriage. Conditional on labor force participation, the industry and occupation distributions of linked migrants closely resemble those of non-linked migrants. The geographic distribution of linked and non-linked migrants is also similar. Overall, the linked sample appears broadly representative of Italian immigrants in the U.S. census.

Validating the linked data against external sources is infeasible due to data limitations. In Figure F.4, we therefore implement two alternative procedures to assess the quality of the links.¹¹ We first construct an alternative sample, displayed in red, by providing gpt-4o-mini with the same information available to the linking algorithm and asking it to evaluate the reliability of each match. In 91% of cases, the model classifies the matches as certainly or very likely correct. All results are robust to restricting the analysis to this subset of matches. Second, we restrict the accepted matches to the highest-similarity match, and again find quantitatively unchanged results.

3.2 Measuring Local, National, and American Identity of Immigrants in the United States

National identity in contemporary settings is typically measured using survey data (Rohner and Zhuravskaya, 2024). Such data are not available in historical contexts. To address this limitation, prior work has proxied national identity using first names, focusing on subsets of names with clear ideological or ethnic connotations (Bazzi et al., 2019; Fouka, 2020; Assouad, 2025; Weigand et al., 2025)

In this paper, we propose a novel, disaggregated measure of the local-national dimension of identity based on first names. Our approach builds on historical research documenting the extreme geographic fragmentation of naming practices in Italy (Delille, 1985; Viazzo, 1989). First names were tightly linked to local kinship norms, patron-saint cults, and dialectal traditions, resulting in pronounced spatial concentration. We interpret geographically concentrated names as reflecting stronger local particularism, and geographically diffuse names as reflecting a broader national orientation.

The first measure of particularism is the Local Name Index (LNI), defined as

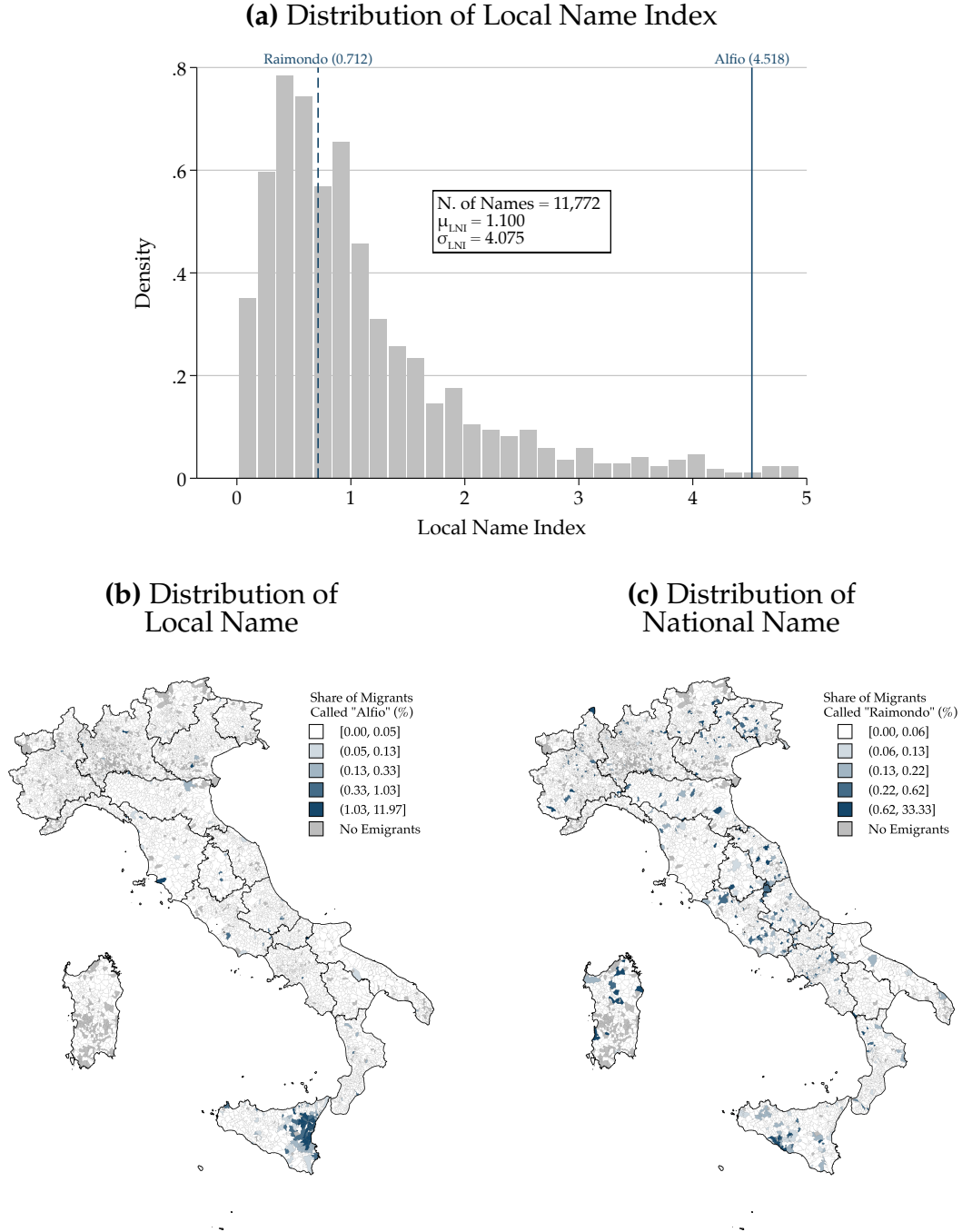
$$LNI_n \equiv 100 \times \sum_m (s_{mn} - p_m)^2, \quad (1)$$

where $s_{mn} = C_{mn}/N_n$, C_{mn} denotes the total number of individuals named n in municipality m , and $N_n = \sum_m C_{mn}$. Thus, s_{mn} is the share of individuals named n residing in municipality m . Further, let $p_m = B_m/B$, where $B_m = \sum_n C_{mn}$ and $B = \sum_m B_m$, so that p_m represents the population share of municipality m . Intuitively, p_m is the share that name n would exhibit in municipality m if it were evenly distributed across the territory. The LNI equals zero if name n is distributed in proportion to population across all municipalities (i.e., $s_{mn} = p_m$ for all m), and it approaches 100 as the mass of name n becomes increasingly concentrated.

We construct the LNI using municipalities of origin reported in the Ellis Island records. Figure IIa shows the resulting distribution. To illustrate the index, we highlight two similarly frequent names,

¹¹ Appendix Section B.5 describes these validation procedures in detail.

Figure II. Distribution of the Local Name Index and Sample Names



Notes. Panel [IIa](#) shows the distribution of the local name index defined in [\(1\)](#), along with the number of recorded names and the distribution's mean and standard deviation. The panel also reports the LNI values for two names—"Raimondo" and "Alfio"—chosen because they appear with similar frequency. Panel [IIb](#) (respectively, [IIc](#)) maps the spatial distribution of individuals named "Alfio" (respectively, "Raimondo") across municipalities, expressed as the share of individuals with that name relative to the 1881 population, in percentage terms. The maps superimpose regional borders; municipality and district boundaries are shown using 1991 borders. Referenced on page: [11](#).

"Alfio" and "Raimondo." Figure [IIb](#) plots the spatial distribution of "Alfio," while Figure [IIc](#) reports that of "Raimondo." The former ($LNI = 4.518$) is highly concentrated in western Sicily, plausibly reflecting the local cult of Saint Alfio, who is the saint patron of six towns in the area. By contrast, "Raimondo" ($LNI = 0.712$) is more evenly distributed throughout both Northern and Southern Italy.

The baseline particularism index captures the concentration of names, but it does not account for spatial proximity between municipalities.¹² To address this limitation, we construct a distance-adjusted index (DLNI):

$$DLNI_n \equiv \sum_m \sum_{m'} s_{mn} d_{m,m'} s_{m'n} , \quad (2)$$

where $d_{m,m'}$ denotes the geodesic distance between municipalities m and m' . The index $DLNI_n$ is equal to the expected distance between two randomly drawn individuals sharing name n . Accordingly, higher values of $DLNI_n$ indicate more geographically diffuse, and hence more “national,” names, while lower values indicate stronger local particularism.

We validate our approach using contemporary survey data and find that local naming practices correlate with lower trust in national institutions, such as the police, the judicial system, and public administration, but not with trust in unrelated institutions (the Catholic Church and labor unions) or with generalized trust.¹³ These patterns corroborate our hypothesis that names capture variation in national identity rather than reflecting more general cultural traits.

To measure assimilation into U.S. society, we follow Abramitzky et al. (2020) and assign to each name its associated Foreign Name Index (FNI). The FNI is normalized between zero and one and takes higher values for names that are more prevalent among immigrants than natives.¹⁴

Following Bazzi et al. (2019), we complement the name-based indices with measures of the probability of marrying Italians from other municipalities, provinces, and regions.

3.3 Measuring Local and National Identity in Italy

Unlike in the United States, individual-level records from Italian censuses did not survive. To the best of our knowledge, no existing sources allow us to observe the universe of names. Here, we outline one strategy to address this data limitation and present two complementary measures.

3.3.1 World War 1 and World War 2 Casualty Data as Individual-Level Repositories of Names

We collect names from two sources: the repositories of World War 1 (WW1) deaths and World War 2 (WW2) casualties. Italy suffered approximately 650,000 military deaths in WW1 and 500,000 casualties in WW2.

WWI casualty data report first and last name, father’s name, year of birth, municipality of origin, military rank, unit, year, place, and cause of death (Acemoglu et al., 2022).¹⁵ We georeference the municipality of origin and disambiguate first and last names. For 570,355 individuals, we observe complete information on name, year of birth, and municipality of birth.

¹²Consider two names, n_1 and n_2 , each observed in exactly two municipalities. If n_1 is present in two distant municipalities while n_2 is present in two nearby municipalities, it may be desirable for a measure of particularism to classify n_2 as more local than n_1 .

¹³Appendix Section A.2 provides additional details on the validation procedure. We use survey data to measure trust in national institutions and construct a contemporary local name index using vital records compiled by the Italian Statistical Office. Figure F.6 displays the correlation between LNI and DLNI and a synthetic measure of trust in national institutions using principal component analysis; Table E.2 reports the correlation between each variable and the name-based indices.

¹⁴Formally, $FNI_n \equiv \Pr(\text{Name} = n \mid \text{Immigrant}) / (\Pr(\text{Name} = n \mid \text{Immigrant}) + \Pr(\text{Name} = n \mid \text{Native}))$.

¹⁵The dataset was compiled by the Fascist regime for propaganda purposes and is available at cadutigrandeguerra.it.

WW2 casualty data have been recently digitized by the Italian Ministry of Defense and, to the best of our knowledge, have not been used before.¹⁶ We construct a dataset covering the universe of recorded casualties. The records report first name, surname, date and municipality of birth, and date and region of death or disappearance. As with the WWI data, we georeference the municipality of origin and clean first names. The final dataset contains information on name, municipality of origin, and year of birth for 408,940 individuals.

We use these data to construct a yearly series of municipality-level name-based indices. The time span of the resulting series corresponds to the birth cohorts eligible for military service during the two World Wars (1870–1925). The resulting sample consists almost exclusively of men, since women did not serve in combat roles in the regular army. Since conscription was mandatory and near-universal during both WW1 and WW2, these data plausibly constitute a representative sample of the male combat-age population, under the assumption that wartime mortality rates were not systematically correlated with naming patterns across municipalities.

In addition, we link the WW1 casualty data to the Ellis Island database to measure the number of U.S. emigrants who fought and died as soldiers in the Italian army.¹⁷ While conscription was universal, the enlistment of migrants in the United States was essentially voluntary (Choate, 2008). We thus view the willingness to serve (and die) in the Italian army as an additional variable capturing national sentiment, as in Caprettini and Voth (2023).

3.3.2 *Street Names*

We use street names as a complementary indicator of national and local identities (Azaryahu, 1996; Oto-Peralías, 2018). In Italy, municipal councils have complete discretion in naming new streets, subject to legal constraints, and the process often involves local communities. Street names thus capture the preferences of local legislators and community members.

We collect the universe of municipal streets in Italy in 2025 from OpenStreetMap, yielding a dataset with street names, georeferenced locations, and lengths for 849,124 streets.¹⁸ We use a large language model (GPT-5) to classify street names that reflect national and local identities. Examples of national-identity street names include figures associated with the independence movement, such as Giuseppe Garibaldi and Giuseppe Mazzini, prominent historical figures, such as Cristoforo Colombo and Giacomo Matteotti, and references to patriotic events, such as the Battle of Isonzo. By contrast, local identity street names honor notable individuals from the municipality or the surrounding area.

In addition, we identify street names belonging to three placebo categories: arts, literature, and religion. The same name may belong to more than one category. For example, Giosuè Carducci, a central figure in Italian literature at the end of the nineteenth century and winner of the Nobel Prize in 1906, was regarded as the “poet laureate” of the newly unified Italian state. As such, his name is

¹⁶Individual searches within the dataset are available at difesa.it.

¹⁷See Appendix Section A.7 for additional details on the linking approach and validation statistics.

¹⁸We exclude provincial, regional, and national roads, as well as highways, for two reasons. First, their names are not determined by municipal councils. Second, their names are typically administrative codes, as in the case of highways, or reflect only the places where they are located or the locations they connect.

classified as both national and literary. Moreover, the classification of a given street name generally depends on its location. A street named San Zeno is typically associated with religion. In Verona, however, it is also classified as local, since San Zeno is the city’s patron saint.¹⁹

3.3.3 Monuments

We use monuments as a third indicator of national sentiment (Johnson, 1995). We construct a dataset covering the universe of nationally recognized monuments compiled by the Italian Ministry of Culture.²⁰ The data report the year of construction and the location of each monument. We restrict the sample to monuments designated between 1861 and 2000, since coverage prior to unification depends on the legislation of pre-unitary states. We georeference each monument and assign it to a municipality.

Each monument is accompanied by a textual description. We apply a large language model (GPT 5) to these descriptions to distinguish monuments that signify national identity from all other monuments.²¹ National-identity monuments typically commemorate soldiers from the municipality in which the monument is located who died in combat, often during World War 1 and World War 2. The dataset includes information on 32,325 monuments designated between 1861 and 2000, of which 36% (11,774) are classified as reflecting national identity sentiment.

3.4 Construction of the Samples

3.4.1 Immigrants in the United States

First, we compile a panel of Italian migrants observed from 1900, or the first census in which they appear, through 1930. We link migrants over time using the longitudinal links produced by the Census Linking and Tree projects (Price et al., 2021; Abramitzky et al., 2025; Buckles et al., 2025). In this sample, name-based indices are computed by averaging them across all children born in the ten years preceding each census year. Migrants’ locations correspond to their place of residence in each census year.

Second, we construct a panel in which migrants are observed repeatedly at each child’s birth. In this case, a migrant’s location is taken from the census record closest to the child’s birth year.

In both samples, we measure the diversity of the Italian community at the enumeration district level. Enumeration districts were approximately ten times smaller than counties and, therefore, better suited to capture local immigrant communities (Eriksson and Ward, 2019). A limitation of enumeration districts is that their boundaries changed across censuses. To control for consistent sub-county characteristics, we construct a hexagonal spatial grid and assign individuals to grid cells using data from Berkes et al. (2023). Because the hexagons are time-invariant, hexagon-by-time fixed effects control for geographic variation at a finer level than counties and remain consistent over time. Panel A in Table I provides descriptive statistics at the neighborhood (Panel A.1) and individual (Panel A.2) levels.

¹⁹The full prompt used for classification is reported in Appendix Section A.3.

²⁰The dataset is available at dati.cultura.gov.it.

²¹As with streets, the prompt used for classification is reported in Appendix Section A.4.

Table I. Descriptive Statistics of Selected Variables

	Mean	Std. Dev.	Median	Min.	Max.	Observations
	(1)	(2)	(3)	(4)	(5)	(6)
Panel A. United States						
<i>Panel A.1 Neighborhood-Level Data</i>						
Italian Diversity	1.911	1.222	1.906	0.000	5.955	39,477
Population	1,679.481	843.645	1,553.000	13.000	32,766.000	39,477
Immigrants	24.232	13.691	22.607	0.000	84.400	39,477
Italian Immigrants	6.644	9.156	3.122	0.000	76.915	39,477
<i>Panel A.2 Individual-Level Data</i>						
Local Name Index	4.219	3.905	3.241	0.014	99.827	335,569
Distance-Adjusted Local Name Index	9.593	0.495	9.743	0.000	10.002	335,569
Foreign Name Index	0.759	0.153	0.786	0.000	1.000	330,247
Married to Italian from Other Municipality	0.180	0.384	0.000	0.000	1.000	317,885
log(Occupational Income)	2.458	1.356	3.045	0.000	4.394	335,569
Employed in Manufacturing	0.244	0.430	0.000	0.000	1.000	335,569
Skilled Manufacturing Job	0.154	0.361	0.000	0.000	1.000	335,569
Unskilled Manufacturing Job	0.165	0.371	0.000	0.000	1.000	335,569
Panel B. Italy						
Population	5,052.860	28,469.863	2,231.000	0.000	2,839,638.000	104,299
Number of US Emigrants	129.247	433.492	22.000	1.000	19,225.000	5,846
Share US Emigrants in Diverse Areas	0.175	0.164	0.168	0.000	1.000	5,846
Local Name Index	0.859	1.668	0.353	0.017	67.939	68,568
Distance-Adjusted Local Name Index	9.911	0.194	9.965	3.175	10.002	68,568
Number of Streets	107.014	281.237	53.000	0.000	17,873.000	8,176
Number of Streets by Designation:						
National Identity	19.327	27.333	10.000	0.000	680.000	8,176
Local Identity	35.738	118.161	14.000	0.000	5,905.000	8,176
Arts	4.396	14.274	1.000	0.000	774.000	8,176
Literature	5.807	13.449	1.000	0.000	548.000	8,176
Religious Figure	10.791	27.654	5.000	0.000	1,527.000	8,176
Number of Monuments	0.301	6.591	0.000	0.000	1,288.000	113,400
Number of National Identity Monuments	0.123	5.707	0.000	0.000	1,282.000	113,400

Notes. This table reports descriptive statistics for selected variables for the United States (Panel A) and Italy (Panel B). Panel A.1 uses census enumeration districts observed at decennial frequency from 1900 to 1930. Panel A.2 is the baseline individual-level dataset of Italian immigrants, observed in the first census in which they appear and, when possible, linked across subsequent censuses. Panel B consists of municipality-level data for Italy: population, name-based indices, and monuments are observed decennially, while all other variables are measured once per municipality. Referenced on pages: [15](#), [16](#).

3.4.2 Panel Dataset of Italian Municipalities

The analysis for Italy is conducted at the municipality level. We construct three datasets that differ in their temporal granularity.

First, name-based indicators are available at the cohort level. We therefore assemble a yearly panel tracking municipalities between 1870 and 1930. We collapse the data to coarser time intervals, using five- or ten-year windows depending on the specification, to reduce short-term noise and municipality-year observations with no recorded individuals. Second, the analysis of street names relies on a cross-sectional dataset in which each municipality is observed once. Third, for monuments, we observe both their location and year of construction. We aggregate these data at the municipality level by decade and follow municipalities from 1870, corresponding to the 1861–1870 decade, through 2000.

In all samples, we assign observations to municipalities at 1991 boundaries, as historical GIS data on municipal borders are unavailable. Panel B in Table I lists descriptive statistics for the main outcome and explanatory variables employed in the analysis on national sentiment in Italy.

4 Local, National, and American Identities of Italian Immigrants

4.1 Empirical Strategy

4.1.1 Measuring the Diversity of Italian Immigrant Communities

We measure the diversity of the Italian community by municipality of origin at the enumeration district level. Let $d(t)$ be a district at time t . Denote with $n_{d(t)}^m$ the number of immigrants from municipality m located in district $d(t)$, and let $p_{d(t)}^m = n_{d(t)}^m / \sum_m n_{d(t)}^m$ be the share of migrants from that municipality. The diversity of the Italian community is the Shannon entropy:

$$H_{d(t)} = - \sum_m p_{d(t)}^m \log(p_{d(t)}^m). \quad (3)$$

Entropy is weakly positive and increasing in the diversity of the Italian community. Shannon entropy reflects the idea that relatively infrequent encounters are more informative than frequent ones. In our context, this property is appealing because even limited exposure to immigrants from otherwise uncommon parts of Italy may matter more than repeated interactions with migrants from origins already well represented in the local community. Results, however, are robust to using the fractionalization index, a more standard metric that does not embed this feature.

Figure F.7 displays a measure of the geographic segregation of Italian immigrants. The blue line shows the distribution of the average distance between each immigrant’s municipality of origin and the municipality of origin of other Italian immigrants residing in the same enumeration district. We compare this distribution to counterfactual benchmarks generated under random sorting.²² The observed distribution exhibits excess mass at shorter distances, indicating that migrants from closer municipalities were more likely to settle together. However, segregation remains modest: the average observed distance between neighbors’ municipalities is only 10% lower than under random sorting.

Figure F.8 displays the variation in the average sub-national segregation of Italian communities over time and confirms that, in line with anecdotal evidence documented in Section 2.3, Italian communities became more diverse as new immigrants entered the United States. Figure F.9 displays the spatial distribution of $H_{d(t)}$ for districts in the New York boroughs of Manhattan and Brooklyn between 1900 and 1930, and highlights the substantial variation in the sub-national segregation of Italians across neighborhoods.²³

²²To benchmark observed segregation against random sorting, we construct counterfactual neighbor-distance distributions via permutation. For each individual i and district d , we define the pool of potential neighbors as all Italian immigrants observed in the same census year, excluding members of i ’s household. Let n_{id} denote the number of such neighbors in i ’s observed district. In each simulation draw, we sample n_{id} individuals at random from the pool and compute the mean distance between i ’s municipality of origin and those of the sampled individuals. We run 100 simulations per individual, yielding a counterfactual distribution under random assignment that preserves group sizes and household structure.

²³Unfortunately, enumeration-district GIS files are not available for the entire US territory. Figure F.9 displays boundary files

4.1.2 Baseline Specification

The baseline analysis employs the individual-level panel dataset described in Section 3.4.1. We estimate variations of the following specification:

$$y_{ihdt} = \alpha_{m(i)} + \alpha_h + \alpha_t + \beta \cdot H_{d(t)} + X_d' \Gamma_1 + X_i' \Gamma_2 + \varepsilon_{ihdt}, \quad (4)$$

where i , h , d , and t denote an individual, the hexagon of residence, the enumeration district, and the census year, respectively. Municipality-of-origin fixed effects ($\alpha_{m(i)}$) control for time-invariant origin characteristics such as population, emigration intensity, and geographic location. Hexagon fixed effects (α_h) absorb time-invariant local characteristics, including location and proximity to major ports of entry. Time-fixed effects (α_t) capture aggregate shocks, such as business-cycle fluctuations, that may affect immigration flows.

Because enumeration district boundaries change over time, we cannot include district fixed effects. We therefore estimate variants of (4) that include hexagon-by-time fixed effects ($\alpha_{h \times t}$). These absorb time-varying unobserved heterogeneity at sub-county spatial units and purge confounding variation driven by local factors such as economic conditions that evolve within counties. We also include municipality-of-origin-by-time fixed effects ($\alpha_{m(i) \times t}$), which absorb time-varying heterogeneity in migrants' places of origin, such as population growth, which generates larger emigrant flows, as well as local economic conditions in Italy. In addition, they account for the expansion of origin-specific migrant networks over time.

In various specifications, we include district-level controls (X_d) capturing demographic characteristics (population, the share of women, the share of African Americans, the share of immigrants, and the share of Italians) and economic characteristics (employment shares across occupations). In addition, some specifications include individual-level controls (X_i), namely occupation, gender, birth year, year of immigration, and the name index of the outcome variable computed on i 's own name.²⁴

The coefficient of interest (β) captures how individual identity outcomes y respond to changes in the diversity of the local Italian immigrant community. We consider three classes of outcomes: the local name index and its distance-adjusted variant; the foreign name index; and indicators capturing the probability that individual i married an Italian migrant originating from a different municipality, province, or region.

4.1.3 Identification Challenge and Instrumental Variable Design

Identification Challenges. Identification of the causal effect of diversity may fail for two reasons. First, the origin mix within a district may reflect selective sorting of immigrants across U.S. locations. If these groups differ in unobserved traits related to identity, then $\hat{\beta}$ may partly capture such selection.

Second, neighborhood diversity, $H_{d(t)}$, may respond endogenously to time-varying local conditions. Destination shocks can simultaneously affect outcomes and the composition of the Italian community by shifting migration flows or retention. Relatedly, residential mobility and marriage-market

from Shertzer et al. (2016), which, however, are only available for a few large cities.

²⁴The own-name local, distance-adjusted local, and foreign name indices control for observable differences in identity inherited by migrants' families.

choices may feed back into measured diversity, inducing simultaneity.

Instrumental Variable Design. To address these concerns, we construct an instrument based on exposure to Italian diversity induced by ship-level composition. Building on evidence that social ties formed aboard ships can persist after disembarkation and shape subsequent location choices (Battiston, 2018; Green, 2025), we hypothesize that interactions during the transatlantic voyage shaped migrants' post-arrival settlement patterns. This channel is particularly plausible in our setting because transatlantic voyages could last more than 50 days, creating sustained opportunities for interaction and the formation of social ties among passengers before arrival (Hatton, 2024). Under this hypothesis, variation in ship composition provides a source of variation in migrants' subsequent exposure to neighborhood diversity that plausibly does not reflect pre-migration networks.

Consider individual i from municipality $m(i) = m$ who traveled on ship $s(i) = s$.²⁵ Let x_{os} denote the number of immigrants from municipality o who traveled on ship s , and let p_{os} be the share of immigrants from municipality o on s , so that $\tilde{p}_{os} = p_{os} / (1 - p_{ms})$ is the associated normalized share.²⁶ The instrument for individual i is then given by

$$\hat{H}_i = - \sum_{o \neq m} \tilde{p}_{os} \log \tilde{p}_{os}. \quad (5)$$

We compute $\hat{H}_i$ only on ships with at least 100 immigrants (95% of the total) to avoid numerical instability. We estimate regression (4) using the instrument in (5) via two-stage least squares (2SLS).

For each individual, the instrument excludes shipmates born in the same municipality. This leave-out methodology ensures that $\hat{H}_i$ does not reflect pre-existing origin-specific networks. Results remain quantitatively unchanged when excluding immigrants from the same province, indicating that residual social networks not captured by excluding own-municipality shipmates are unlikely to drive the results.

Instrument Relevance. We implement two analyses to verify that ship diversity predicts diversity exposure at the neighborhood level.

First, we regress neighborhood diversity $H_{d(t)}$ on the ship-diversity instrument $\hat{H}_i$, conditional on the controls we include in the baseline regression specification. Figure IIIa shows a positive, strongly statistically significant association between the two variables.

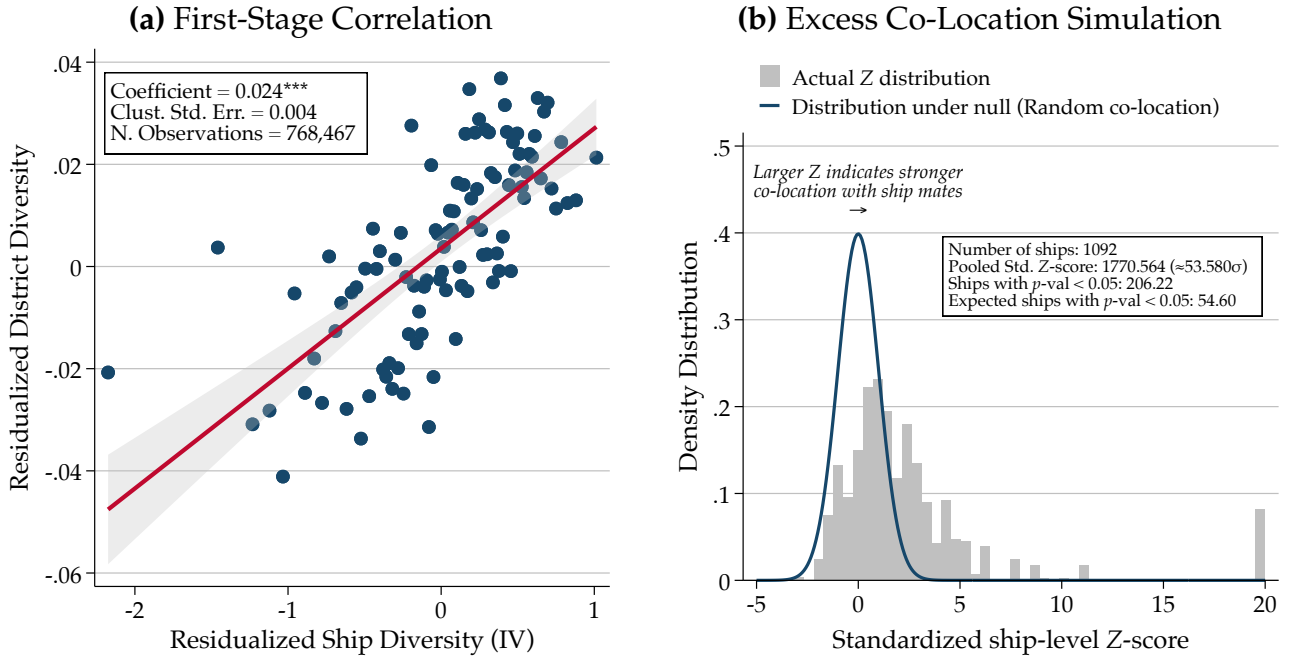
Second, drawing on the simulation approach of Basse et al. (2024), we test whether passengers traveling on the same ship are disproportionately likely to co-locate after arrival.²⁷ For each ship, we

²⁵For each immigrant, we know the year when they arrived and the name of the ship they traveled on. A "ship" is, therefore, defined as a unique ship-year pair. We omit the year variable and simply call it a "ship" to avoid notational burden. While it is possible that the same ship operated multiple voyages within the same year, this would introduce noise that likely attenuates the relevance of the instrument rather than compromise its validity.

²⁶Municipality shares on ship s are computed as $p_{os} = (x_{os} + \lambda) / (N_s + \lambda K)$, where x_{os} is the number of passengers from municipality o , $N_s = \sum_o x_{os}$ is the total number of passengers on the ship, and K is the number of distinct municipalities represented. Absent smoothing, these shares are the empirical frequencies from a multinomial distribution over municipalities induced by the ship's passenger list. We apply Laplace smoothing ($\lambda = 1$), which amounts to adding one observation per municipality to regularize the entropy measure in finite samples by shrinking the observed distribution, thereby reducing the influence of noisy, low-frequency ships.

²⁷Appendix Section A.8 discusses in greater detail the simulation-based approach.

Figure III. Instrument Relevance Tests



Notes. This figure reports evidence on the relevance of the ship-based instrument defined in (5). Panel IIIa shows the first-stage relationship between the instrument and observed diversity, as defined in (3). The unit of observation is an individual observed in the first census in which they appear and, when possible, linked across subsequent censuses between 1900 and 1930. To construct the binned scatterplot, both variables are residualized with respect to the baseline controls included in (6)—municipality, hexagon, and year fixed effects. The red line displays a linear fit; the corresponding regression coefficient and standard error are reported in the panel. Panel IIIb reports the ship-level distribution of the z-statistic measuring excess co-location with shipmates relative to the benchmark case of random location with shipmates. Under the null hypothesis of no co-location, the z-score distribution would be approximately normal, shown by the blue line. Values to the right indicate excess co-location with shipmates. The panel also reports the number of ships with a p -value below 0.05, together with the number that would be expected under the null of no co-location. Referenced on pages: 19, A.12.

compare the observed degree of co-location, measured by the probability that passengers from the same ship settle in the same district, with a simulated benchmark obtained by randomly assigning the same passengers to districts. Figure IIIb reports the distribution of the ship-level excess co-location statistics together with its asymptotic distribution under the null of no co-location (in blue). We find that passengers of the same ship are more likely to settle in the same district: the share of ships with a p -value below 5% is about four times larger than under random co-location.

Evidence in Support of the Identification Assumption. Identification relies on the assumption that, conditional on fixed effects and controls, ship composition influences identity outcomes only through its effect on post-arrival exposure to other Italian origins.

We implement two exercises to assess the plausibility of this assumption. First, we examine instrument balance by regressing predetermined area characteristics on the instrument, conditioning on the same fixed effects and controls as in the baseline specification. Figure F.10a reports the estimates. Because enumeration districts cannot be tracked consistently over time, we construct these predetermined characteristics at the county (in blue) and the hexagon (in red) levels. Across a wide set of variables, including the diversity of the Italian community, the estimates are small and statistically insignificant, indicating no correlation between the instrument and pre-migration characteristics.

Second, we regress the instrument on one-decade leads of individual characteristics that should not be affected by ship diversity. These include demographic variables, which are fixed by construction, and measures of employment status by industry and occupation. Figure F.10b reports the estimates. We find no statistically significant associations, suggesting that the instrument does not predict outcomes that could have been affected by immigrant networks or other confounders of diversity exposure.

While we find evidence against this possibility, ship composition could directly influence identity formation. In that case, the instrument would not be valid, but the reduced-form estimates could still be interpreted as evidence that intergroup contact affects identity formation. Relative to the structural interpretation, what would change is the *place* where such interactions occur: from the neighborhood to the ship. The main finding—that intergroup contact among Italian emigrants shaped national identity formation—would nonetheless remain true.

Movers' Design. As a complementary approach, we implement a movers' design along the lines of Chetty and Hendren (2018). This strategy takes advantage of individuals who move between neighborhoods and exploits differences in the diversity of Italian communities they are exposed to.²⁸ Because we compare movers and stayers over time, this framework allows us to evaluate the empirical plausibility of the parallel trends assumption through event-study graphs. We find that exposure to more diverse communities decreases name particularism (see Table E.3) and document no evidence of statistically significant pre-move differences between treated and control individuals (Figure F.11).

4.2 Local and American Identity and Neighborhood Diversity

4.2.1 Descriptive Patterns of Assimilation After Arriving in the U.S.

We begin our analysis with a descriptive exercise and display the evolution of the local name index (Figure IVa) and the foreign name index (Figure IVb) for children of immigrants by the number of years since their parents moved to the United States.²⁹ We split the baseline sample into two groups: red dots represent immigrants who settle in neighborhoods in the top quintile of the Italian community diversity distribution, whereas blue dots represent immigrants who settle in neighborhoods in the bottom 20%.

The trajectories of the LNI and FNI sharply diverge over time across neighborhood types. Migrants who settle in low-diversity areas exhibit no change in name particularism and experience name-based assimilation. Ten years after arrival in the United States, their children bear names that

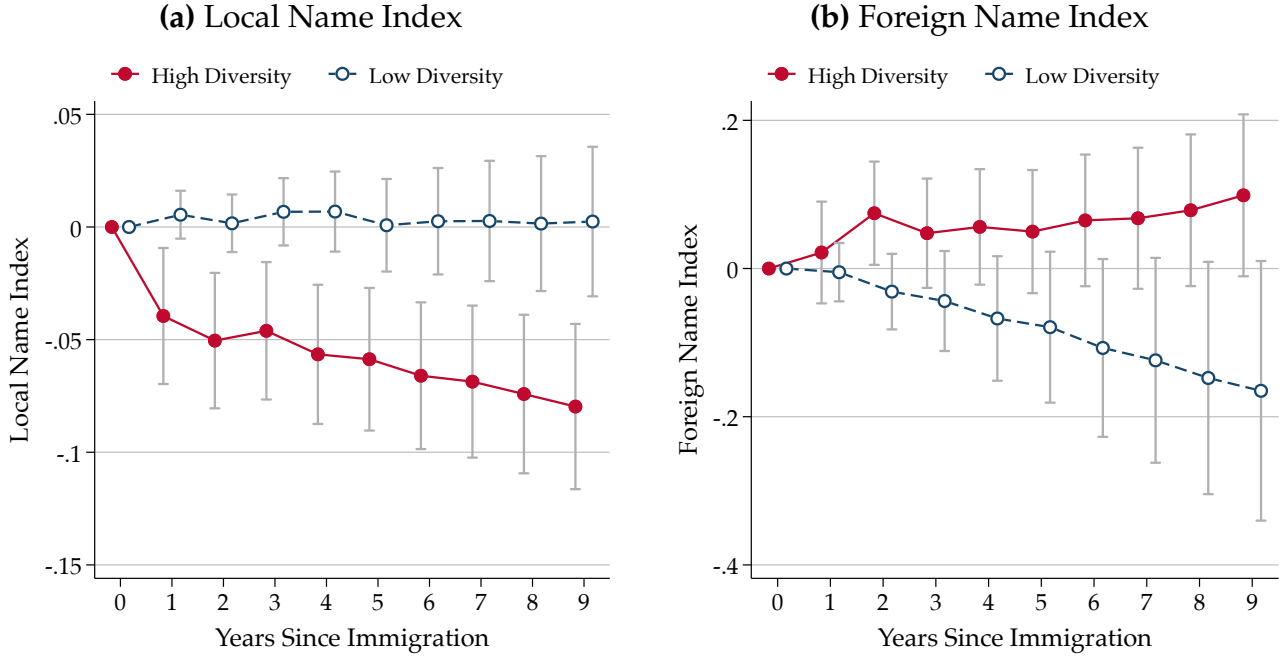
²⁸ Appendix Section A.9 describes the empirical specification and provides a detailed discussion of the results.

²⁹ Formally, we use the dataset that follows immigrants repeatedly at the time their children are born, and estimate the following regression:

$$y_{iht} = \alpha_i + \alpha_h + \alpha_t + \sum_{k=1}^9 \beta_k \times I\{t - \tau_i = k\} + \varepsilon_{iht}$$

where y_{iht} is one of LNI or FNI for a child born in year t to parent i . The regression includes individual, hexagon, and cohort fixed effects. Term τ_i is the immigration year of i , so the sequence $\{\beta_k\}_{k=1}^9$ captures the average LNI and FNI for children born k years after the parent moves to the US. Because we omit the $k = 0$ term, the estimates are relative to the year when migrants move to the US. Standard errors are clustered at the individual level. This specification is estimated separately for individuals who settle in the top and bottom quintiles of the diversity distribution (3).

Figure IV. Evolution of the Local and Foreign Name Indices



Notes. This figure reports the evolution of name-based indices by time since immigration, separately for immigrants settling in high- versus low-diversity Italian communities. The unit of observation is an individual Italian immigrant observed in the first census in which they appear. Each individual is observed every time they have a child. Time since immigration is defined as the difference between the year of birth of the child and the year of immigration of their parent, and individuals are split into the top 20% and bottom 20% diversity groups based on the municipality-level diversity measure (3). Panel IVa (resp. IVb) plots the event-time coefficients from a regression between the local name index (resp. the foreign name index) and years-since-migration time indicators, relative to the year of arrival. All regressions include individual, hexagon, and cohort fixed effects; Standard errors are clustered at the individual level. Bands report 90% confidence intervals. Referenced on page: 21.

are no longer differentially local, but are 20% more similar to English-sounding names. This pattern of gradual assimilation is consistent with evidence by Abramitzky et al. (2020).

Migrants who settle in high-diversity areas, by contrast, experience a decline in name-based particularism. Ten years after migration, their children bear names that are almost 10% less local. Differences in name particularism emerge as early as two years after arrival and compound over time. In contrast, even ten years after arrival, the children of migrants in high-diversity areas do not display less foreign-sounding names than at arrival, and, if anything, their names become more foreign-sounding. This pattern suggests slower, or even absent, cultural assimilation into U.S. society among migrants exposed to more diverse Italian communities.

4.2.2 Name-Based Assimilation: OLS and 2SLS Results

To provide more formal evidence for these descriptive patterns, Table II reports estimates of specification (4). Panels A, B, and C present results using the local name index (1), the distance-adjusted local name index (2), and the foreign name index as dependent variables, respectively. Columns (1–6) report OLS estimates. Columns (7–8) present the reduced-form effects of the instrument in (5) on each outcome, while columns (9–10) report the corresponding 2SLS estimates.

Depending on the specification, the OLS estimates indicate that a unit increase in the entropy of

Table II. Exposure to Immigrant Heterogeneity and National Identity Formation

	Ordinary Least Squares						Reduced Form		Two-Stage Least Squares	
	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)	(9)	(10)
Panel A: Local Name Index										
Italian Diversity	-0.170*** (0.014)	-0.164*** (0.015)	-0.154*** (0.016)	-0.163*** (0.014)	-0.121*** (0.017)	-0.126*** (0.020)			-2.210*** (0.686)	-4.939*** (1.330)
Ship Diversity							-0.049*** (0.009)	-0.047*** (0.010)		
Mean Dep. Var.	3.990	3.989	3.989	3.989	3.989	3.989	3.988	3.988	3.988	3.988
Panel B: Distance-Adjusted Local Name Index										
Italian Diversity	0.020*** (0.002)	0.020*** (0.002)	0.019*** (0.002)	0.019*** (0.002)	0.016*** (0.003)	0.015*** (0.004)			0.251** (0.114)	0.560** (0.232)
Ship Diversity							0.006*** (0.002)	0.005*** (0.002)		
Mean Dep. Var.	9.593	9.593	9.593	9.593	9.593	9.593	9.593	9.593	9.593	9.593
Panel C: Foreign Name Index										
Italian Diversity	0.017*** (0.001)	0.016*** (0.001)	0.016*** (0.001)	0.015*** (0.001)	0.015*** (0.001)	0.014*** (0.001)			0.105*** (0.024)	0.215*** (0.055)
Ship Diversity							0.002*** (0.000)	0.002*** (0.000)		
Mean Dep. Var.	0.758	0.758	0.758	0.758	0.758	0.758	0.758	0.758	0.758	0.758
Municipality FE	Yes	Yes	–	Yes	Yes	Yes	Yes	Yes	Yes	Yes
County FE	Yes	–	–	–	–	–	–	–	–	–
Year FE	Yes	Yes	–	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Hexagon FE	No	Yes	–	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Municipality-Year FE	No	No	Yes	No	No	No	No	No	No	No
Hexagon-Year FE	No	No	Yes	No	No	No	No	No	No	No
Individual Controls	No	No	No	Yes	No	No	No	No	No	No
Demographic Controls	No	No	No	No	Yes	Yes	No	Yes	No	Yes
Economic Controls	No	No	No	No	No	Yes	No	Yes	No	Yes
# Individuals	223,721	223,531	221,913	221,675	223,531	223,531	208,845	208,845	208,845	208,845
# Observations	334,668	334,303	330,615	331,519	334,303	334,303	311,848	311,848	311,848	311,848
K-P F-stat	–	–	–	–	–	–	–	–	24.327	30.750

Notes. This table reports the effect of exposure to diversity within the Italian community on name-based indices from specification (4). Panel A uses the local name index defined in (1) as the dependent variable; Panel B uses the distance-adjusted local name index defined in (2); Panel C uses the foreign name index of Abramitzky et al. (2020). The unit of observation is an individual observed in the first census in which they appear and, when possible, linked across subsequent censuses between 1900 and 1930. Columns (1–6) report ordinary least squares estimates; columns (7–8) report reduced-form instrumental-variable estimates using the ship-based instrument defined in (5); columns (9–10) report the corresponding two-stage least squares estimates. All specifications include municipality and census-year fixed effects. Column (1) includes county fixed effects; columns (2) and (4–10) replace county with more granular hexagon fixed effects. Column (3) replaces municipality fixed effects with municipality-year fixed effects and hexagon fixed effects with hexagon-year fixed effects. Individual controls (included in column 4) are occupation, gender, birth year, immigration year, and the own-name index. Demographic controls (included in columns 5, 8, and 10) are measured at the district level and include population, the share of women, African Americans, immigrants, and Italian immigrants. Economic controls (included in columns 6, 8, and 10) are measured at the district level and include employment shares by occupation. Standard errors, clustered at the district level, are reported in parentheses. Columns (9–10) also report the Kleibergen–Paap F -statistics. Referenced on page: 22.

***: $p < 0.01$, **: $p < 0.05$, *: $p < 0.10$

the Italian community is associated with a 0.12–0.17 decrease in the local name index. To put these magnitudes into perspective, a one-standard-deviation increase in diversity corresponds to a 4–5%-standard-deviation decline in the local name index. Panel B reports results using the distance-adjusted LNI. Because higher values of the DLNI indicate names that are more geographically widespread within Italy, increases in this index capture stronger national identity. A unit increase in diversity is

associated with a 0.02 increase in the DLNI, implying that a one-standard-deviation increase in diversity yields a 10%-standard-deviation increase in the distance-adjusted local name index. Panel C reports the complementary effect of diversity on the foreign name index, which, following Abramitzky et al. (2020), we interpret as a proxy for cultural assimilation in U.S. society. A unit increase in diversity is associated with a 0.02 increase in the FNI, which corresponds to an 11%-standard-deviation increase in the foreign name index for a one-standard-deviation increase in diversity.

These estimates are stable across a range of specifications. In column (2), we replace county fixed effects with hexagon fixed effects to control for more granular time-invariant local characteristics. In column (3), we include municipality-of-origin-by-time fixed effects, which absorb time-varying unobserved heterogeneity at the origin level. For instance, population growth may increase migrant outflows from a given municipality, potentially affecting the likelihood of settling near migrants from the same origin. In column (4), we add individual-level time-invariant controls from the U.S. census: occupation, gender, birth year, and year of immigration. Finally, columns (5) and (6) incorporate neighborhood-level time-varying controls to account for demographic factors—population size, the share of African Americans, immigrants, and Italian immigrants—and economic conditions, measured by employment shares across industries, that may influence assimilation.

These estimates are consistent with the predictions of a simple model of identity choice, along the lines of Akerlof and Kranton (2000) and Shayo (2009), which we outline in Appendix Section D. Under plausible assumptions, higher Italian diversity reduces the prevalence of local identities because the returns to national identity increase. Similarly, greater Italian diversity reduces the adoption of American identity because identity serves as a coordination device whose value depends on the level of diversity. Higher diversity thus reallocates identity *within* the immigrant group, rather than toward the host society.

As discussed in Section 4.1.3, OLS estimates may not recover the causal effect of exposure to Italian diversity due to selective sorting and simultaneity concerns. We therefore report reduced-form estimates using the ship-level instrument defined in (5) in columns (7–8), and the corresponding 2SLS estimates in columns (9–10). The instrumental-variable estimates corroborate the OLS results and are substantially larger in magnitude. Specifically, the 2SLS coefficients are approximately an order of magnitude larger than their OLS counterparts.

There are two reasons why the causal effect identified by the instrument is larger than the OLS correlation. First, migrants whose residential choices were influenced by shipmate networks were likely less socially embedded within their origin, making them particularly responsive to within-group diversity. Second, selective sorting and measurement error in neighborhood composition likely attenuate OLS estimates, as migrants with stronger pre-existing identity or assimilation preferences may self-select into particular locations. By contrast, the IV isolates variation in exposure that is orthogonal to such preferences and therefore recovers the effect for individuals whose social environment was plausibly exogenously assigned. Under this interpretation, the OLS and IV estimates are not directly comparable because the latter reflect a complying population that had lower initial attachment to local identity relative to the general population.

To test this “selection into compliance,” we regress the endogenous diversity measure on the instrument, interacted with proxies for emigrants’ social networks upon arrival. Under our hypothesis, the instrument should be particularly relevant for migrants who arrived in the US without substantial networks, and whose location choices could therefore be influenced by ship diversity. In Figure F.12, we consider three proxies for social networks at arrival: the number of prior migrants from the same municipality, those with the same surname, and their intersection.³⁰ The estimates support our hypothesis: the instrument is strongest for migrants with the weakest pre-existing networks, who constitute the relevant complier population. This pattern clarifies why the 2SLS estimates exceed the OLS estimates.

How does minority status impact identity formation within Italian communities? Evidence by Fouka and Tabellini (2022) indicates that the relative size of the community—and of other minorities—may influence assimilation and, more broadly, identity formation dynamics. In Table E.4, we display the results of heterogeneity analyses where we interact the baseline diversity measure (3) with the share of Italians (columns 2 and 6), the share of immigrants (columns 3 and 7), and the share of African-Americans (columns 4 and 8) in each enumeration district. The estimates indicate that diversity fosters national identity formation when Italians form smaller minority groups. In such settings, interaction between regional groups plausibly makes Italian identity more salient at the expense of Americanization. In line with Fouka and Tabellini (2022), the presence of larger communities of Black Americans fosters Americanization and does not attenuate the effect of Italian diversity on national identity formation.

4.2.3 Out-Group Marriage Patterns

Marriage patterns provide an additional proxy for group identity formation (Abramitzky et al., 2020). In our context, Gordon (1964) documents the central role of marriages between migrants and natives, as well as between migrants from different origin groups, as indicators of immigrant assimilation.

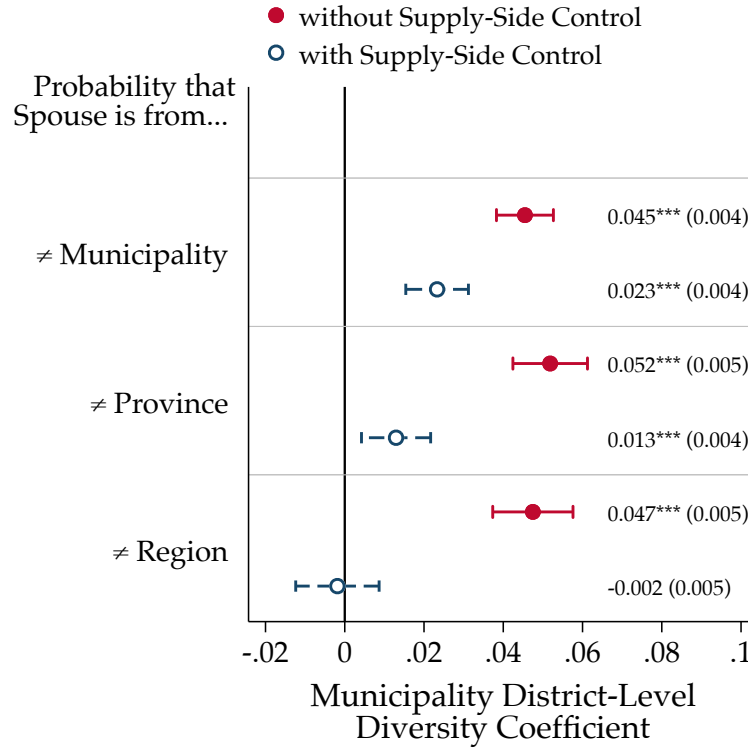
In more diverse immigrant communities, individuals are more likely to marry outside their municipality of origin because fewer potential spouses from the same origin are available. As a result, raw inter-municipality marriage rates may mechanically increase with diversity. To address this issue, we follow Bazzi et al. (2019) and construct supply-adjusted measures of inter-origin marriage. Specifically, we compare observed inter-origin marriages with the intergroup marriage probabilities that would arise under random matching between spouses irrespective of origin. We interpret the residual as reflecting identity-driven integration rather than mechanical supply effects.³¹

Figure V displays the results. Each dot reports estimates from equation (4), comparing specifications that do not correct for the mechanical association between diversity and intergroup marriage probabilities (red dots) with those that account for it (blue dots). Exposure to more diverse Italian

³⁰Importantly, all these measures are constructed “at baseline,” that is, they only include those who arrived before the given immigrant.

³¹For each married individual i , define the marriage market $\mathcal{M}(i)$ by census year and enumeration district, and let $\mathcal{S}_{-i}$ denote the set of opposite-sex individuals in $\mathcal{M}(i)$. The supply-only benchmark is constructed by repeatedly drawing a spouse $j \in \mathcal{S}_{-i}$ uniformly at random and computing $\mathbb{E}[I\{g(i) \neq g(j)\} \mid \mathcal{M}(i)]$, where $g(\cdot)$ denotes origin (municipality, province, or region). We net out this benchmark from observed inter-origin marriage outcomes in the estimating equation.

Figure V. Marriage Patterns and Exposure to Diversity in the United States



Notes. This figure shows regression estimates relating district-level Italian-community diversity to intermarriage outcomes. The unit of observation is an individual Italian immigrant observed between 1900 and 1930. All specifications include municipality, year, and hexagon fixed effects and control for demographic and economic district characteristics. Dots report the coefficient on Italian diversity as defined in (3). Dashed estimates additionally include supply-side controls capturing the predicted availability of same-origin spouses at the municipality, province, or region level, following Bazzi et al. (2019). Standard errors are clustered at the hexagon level and bands report 95% confidence intervals. Referenced on page: 25.

communities is associated with a higher probability of intergroup marriage. When controlling for supply-side effects, a unit increase in Italian community diversity raises the probability of marrying an immigrant from another municipality by 2.3%. Given that such marriages were rare, this corresponds to approximately 13% of the mean. Exposure to more diverse communities also has a positive, statistically significant effect on the probability of marrying an Italian from another province, but not from other regions.³²

4.3 Robustness Analysis

A first concern regarding our name-based indicators is that naming decisions for firstborn children may reflect family tradition rather than parents' identity choices. In Table E.5 and Table E.6, we therefore recompute all name-based measures excluding firstborn children. The results remain quantitatively unchanged.

The local name index is computed at the municipal level to maximize the granularity of informa-

³²Two reasons may explain why we do not find an effect of diversity on out-of-region marriages. First, regions may be too coarse to capture the identity. Given that there are only 20 regions in Italy and that most migrants originated from Southern regions, there is limited variation in marriages with individuals from other regions. Second, the supply-side correction is strongest at coarse aggregation levels, implying that once this benchmark is netted out, there is little residual variation in marriage patterns left.

tion contained in first names. However, municipalities may be too small to capture identity. In column (1) of Panels A and B in Table E.7 and Table E.8, we therefore recompute the LNI at the province-of-origin level. Provinces are larger administrative units—Italy has only 107 provinces, compared to more than 8,000 municipalities—so the province-level index reflects a coarser layer of identity. The results based on province-level and municipality-level LNI are quantitatively similar.

In the baseline definition of Italian community diversity in (3), we measure diversity at the enumeration district and municipality levels. Enumeration districts are small and, we argue, better suited than counties to proxy for the local communities immigrants interacted with in the U.S. Municipalities, in turn, provide an appropriate proxy for local social networks in the origin country. In Table E.7 and Table E.8, we replicate the analysis along two dimensions. First, in columns (3–4) and (5–6), we recompute the diversity measure at the province and region levels, respectively. Second, in columns (2), (4), and (6), we recompute diversity at the U.S. county level. Across these alternative definitions, we find a negative association between Italian diversity and name-based particularism. Estimates are less precise when diversity is measured at the county level, which is expected. Because counties are substantially larger than the relevant social units migrants interacted with, aggregating diversity at this level introduces attenuation bias due to measurement error.

A separate concern is that Italians in more diverse communities used less distinctive local names because they learned new names. It is worth noting, however, that this learning process could be a mechanism operating behind the main pattern we document, since their willingness to adopt these names could signify a shift from the local identity attached to their areas of origin. To explore the relevance of these mechanisms, in Table E.9, we estimate the baseline regression (4) and control for a term capturing the “expected” LNI and DLNI implied by the diversity of names in the municipalities of origin represented in each destination district. We find that higher diversity in the U.S. leads to less local names even after controlling for these terms, both in OLS (columns 2 and 6), in reduced-form (columns 3 and 7), and 2SLS (columns 4 and 8) specifications. This indicates that Italians living in more diverse neighborhoods chose disproportionately less locally distinctive names for their children, even relative to the greater diversity of names available in their neighborhoods.

On the independent-variable side, in Table E.10 and Table E.11, we replicate the baseline estimates using the fractionalization index and obtain qualitatively unchanged results.³³

Italian immigrants were concentrated in a small number of states—most importantly New York, Pennsylvania, New Jersey, and Massachusetts—raising concerns that our results may be driven by idiosyncratic characteristics of specific locations rather than by the diversity of Italian communities *per se*. In Figure F.13 and Figure F.14, we replicate the baseline analysis while sequentially excluding one state at a time from the estimation sample. The OLS and 2SLS estimates remain stable and statistically indistinguishable from the baseline coefficient regardless of the state we exclude.

Selective sorting across U.S. locations by migrants from specific areas of Italy may induce spa-

³³The fractionalization index is defined as one minus the Hirschman-Herfindahl index. Formally, let s_g denote the share of migrants originating from municipality g in a given district. The fractionalization index is given by $F = 1 - \sum_g s_g^2$. This measure captures the probability that two randomly drawn individuals belong to different origin groups and is widely used as a measure of ethnic or group diversity (e.g., see Alesina et al., 2003).

tial autocorrelation and potentially inflate statistical significance. To address this concern, we report confidence intervals in Figure F.15 and Figure F.16 obtained using a range of alternative variance estimators. In particular, we show that statistical significance is unchanged when adopting different clustering strategies across U.S. locations and when applying the Conley (1999) spatial correction with varying distance cutoffs.

5 Emigration and the Rise of National Identity in Italy

Historical evidence documents that transatlantic emigration had profound cultural implications for Italy. In the words of Cinel (2002, p. 91),

“Emigration and return migration were finally accomplishing what the Risorgimento had been unable to set in place.”

We now study this process empirically, examining how migration to the United States shaped national identity in emigrants’ communities of origin.

5.1 Name-Based National Identity

Our primary measure of national identity formation in Italy relies on given names, as described in Section 3.3.1. We leverage variation in the timing when different municipalities entered the mass migration to trace how name-based particularism evolved after migrants were exposed, while in the United States, to Italians from other parts of the country. To this end, we implement two empirical strategies.

5.1.1 Panel Regression

First, we estimate a panel regression of the following form:

$$y_{mt} = \alpha_m + \alpha_t + X'_{mt}\Gamma + \sum_{\tau=1885, 1890, \dots}^{1930} \beta_{\tau} \cdot \text{Diversity Exposure}_m \cdot I\{t \in [\tau, \tau + 5)\} + \varepsilon_{mt}, \quad (6)$$

where m and t respectively denote municipalities and years, with corresponding fixed effects given by α_m and α_t . The vector X_{mt} collects time-varying municipality-level controls. In some specifications, X_{mt} additionally includes the number of emigrants from m during the Mass-Migration period, interacted with year dummies. The coefficients of interest ($\{\beta_{\tau}\}$) trace the differential evolution of y_{mt} over time as a function of diversity exposure. The coefficient of the first five-year window (1880–1884) is omitted. We define $\text{Diversity Exposure}_m$ as the share of migrants from municipality m who settled in U.S. neighborhoods in the top 20% of the baseline entropy distribution.

Because Italian emigration to the United States was concentrated in the period 1895–1921, the coefficients for the pre-migration years in (6) provide a falsification test. If the treatment effect coefficients capture the effect of exposure to diversity through migration, we would expect no systematic differences between high- and low-exposure municipalities before large-scale emigration began.

5.1.2 Difference-in-Differences Regression

To implement a formal difference-in-differences design, we define a municipality's treatment time as the first year in which its annual U.S. emigration rate exceeds 1‰ of its 1880 population. This choice is motivated by Spitzer and Zimran (2023), who show that U.S.-bound emigration followed an S-shaped diffusion pattern. We take the 1‰ threshold as the inflection point when out-migration gains momentum. While the threshold is arbitrary, results are robust to alternative cutoffs between 1‰ and 5‰ (see Table E.12).

We estimate variations of:

$$y_{mt} = \alpha_m + \alpha_t + X'_{mt}\Gamma + \sum_{\substack{\tau=-10, -8, \dots \\ \tau \neq 0}}^{20} \beta_\tau \cdot \text{Diversity Exposure}_m \cdot I\{t - t_m^* \in [\tau, \tau + 2)\} + \varepsilon_{mt}, \quad (7)$$

where t_m^* denotes the first year in which the emigration rate exceeds 1‰. The control vector X_{mt} always includes an indicator $I\{t \geq t_m^*\}$ that equals one once municipality m enters the mass-migration period. The coefficients β_τ therefore trace event-time differences in outcomes for municipalities with higher *vis-à-vis* lower diversity exposure, before and after the mass emigration.

Identification in (7) relies on a parallel-trends assumption. Absent exposure to diverse Italian communities in the United States, municipalities with higher $\text{Diversity Exposure}_m$ should have exhibited the same evolution in y_{mt} around t_m^* as other municipalities. We assess the plausibility of this assumption through pre-treatment coefficients.

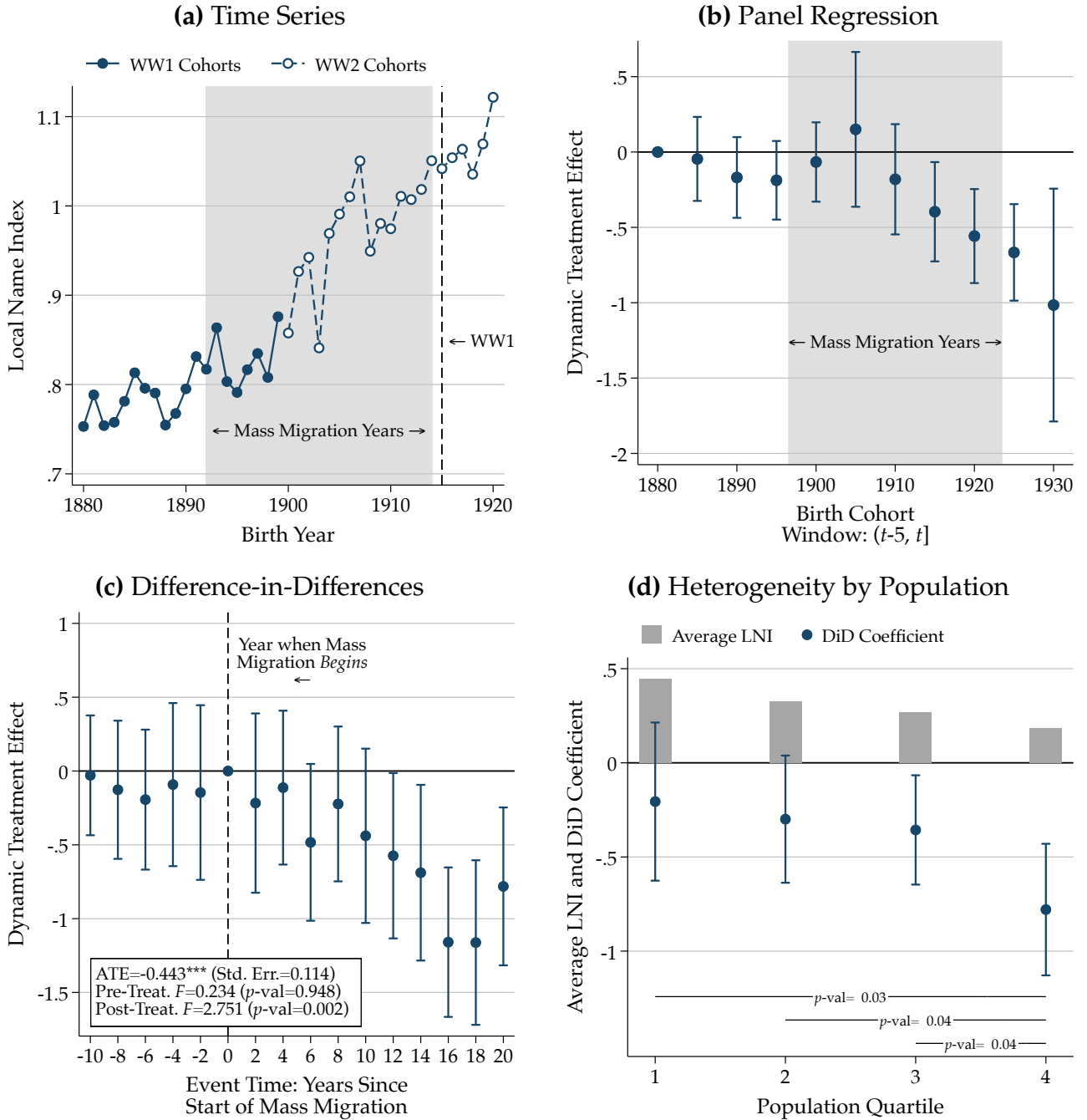
5.1.3 Results

In Italy, the mass migration period coincided with persistent localism and fragmentation (De Mauro, 1963). Figure VIa plots the average local name index of Italian men born between 1880 and 1920. Over this period, the LNI increased by more than 30%, indicating a shift toward more locally concentrated naming practices.

Figure VIb reports the estimates from equation (6). Italian municipalities whose migrants later settled in more diverse U.S. locations do not differ from other municipalities between 1885 and 1900. Because large-scale emigration began in the mid-1890s, this pattern indicates that treated and control municipalities were comparable before the onset of mass emigration. Beginning in the 1910s, however, municipalities whose migrants were exposed to relatively more diverse Italian communities in the United States exhibit significantly lower levels of local naming than control municipalities. The magnitude of this difference increases over time until the late 1920s.

In Table III, we report alternative specifications. Column (1) includes municipality and year fixed effects. Column (2) additionally controls for interactions between the total number of U.S. emigrants during the mass migration period and year fixed effects. This specification thus compares municipalities with similar emigration intensity and exploits variation in the destinations where migrants settled in the United States. The results confirm that the LNI declined in municipalities whose migrants were exposed to more diverse Italian communities in the U.S., while the pre-migration coefficients remain small in magnitude and statistically indistinguishable from zero. Column (3) documents a similar pattern when using the distance-adjusted local name index.

Figure VI. Name Particularism in Italy by Exposure to Immigrant Diversity



Notes. This figure reports dynamic estimates of the effect of exposure to diversity within the Italian migrant community on the local name index in Italian municipalities between 1880 and 1930. Panel VIa reports the average LNI across municipalities over time. In the rest of the figures, the unit of observation is a municipality. Panel VIb reports an event-study specification (6) that interacts diversity exposure with five-year windows, controlling for municipality fixed effects, time fixed effects, and interactions between the number of US emigrants and time indicators. Panel VIc reports a difference-in-differences event-study specification (7) where the treatment time is defined as the first year with at least one emigrant per 1,000 inhabitants. In both panels, time dummies are interacted with a treatment defined as the share of a municipality's US emigrants who settled in districts in the top quintile of the diversity measure (3). Panel VIId, the gray bars report the average LNI by municipality population quartiles; the blue dots report the coefficients of interaction terms between the baseline DiD coefficient (conflated in pre- and post-treatment periods) and population quartile dummies. Standard errors are clustered at the municipality level. Bands report 95% confidence intervals. Referenced on pages: 29, 32.

Figure VIc presents the dynamic treatment effects estimated from equation (7). Prior to the onset of mass emigration, differences in the average LNI between treated and control municipalities are

Table III. Name Particularism in Italy by Exposure to Immigrant Diversity

	Exposure Panel			Difference-in-Differences		
	(1) LNI	(2) LNI	(3) D-LNI	(4) LNI	(5) LNI	(6) D-LNI
Diversity Exposure $\times I(1880 < t \leq 1890)$	-0.002 (0.249)	0.086 (0.284)	0.014 (0.029)			
Diversity Exposure $\times I(1890 < t \leq 1900)$	-0.017 (0.249)	0.054 (0.283)	0.011 (0.029)			
Diversity Exposure $\times I(1900 < t \leq 1910)$	-0.268 (0.265)	-0.039 (0.302)	0.030 (0.031)			
Diversity Exposure $\times I(1910 < t \leq 1920)$	-0.505** (0.250)	-0.273 (0.284)	0.065** (0.029)			
Diversity Exposure $\times I(1920 < t \leq 1930)$	-0.798*** (0.258)	-0.596** (0.293)	0.087*** (0.030)			
Diversity Exposure $\times$ Post Mass Migration				-0.464*** (0.091)	-0.405*** (0.097)	0.045*** (0.010)
Post Mass Migration				0.010 (0.025)	0.055* (0.029)	-0.001 (0.003)
Municipality FE	Yes	Yes	Yes	Yes	Yes	Yes
Year FE	Yes	Yes	Yes	Yes	Yes	Yes
US Emigrants $\times$ Year FE	No	Yes	No	No	Yes	No
# Municipalities	7,823	5,715	7,823	7,823	5,715	7,823
# Observations	200,774	155,781	200,774	200,774	155,781	200,774
Mean Dep. Var.	0.879	0.886	9.909	0.879	0.886	9.909

Notes. This table reports the effect of exposure to diversity within the Italian migrant community on name-based indices in Italian municipalities. The unit of observation is a municipality observed at a yearly frequency between 1880 and 1930. The dependent variables are the local name index defined in (1) in columns (1–2) and (4–5) and the distance-adjusted local name index defined in (2) in columns (3) and (6). The baseline treatment is defined as the share of a municipality’s US emigrants who settled in districts in the top quintile of the diversity distribution (3). Columns (1–3) report event-study specifications of (6), where we interact diversity exposure with decade indicators. Columns (4–6) report difference-in-differences specifications of (7), where the post indicator equals one in years at or after the municipality-specific onset of mass migration (defined as the first year with at least one emigrant per 1,000 inhabitants); the table reports the coefficient of interaction terms between the diversity exposure measure and the post indicator. All regressions include municipality and year fixed effects. Columns (2) and (5) additionally include interactions between the number of US emigrants and year fixed effects. Standard errors are clustered at the municipality level and reported in parentheses. Referenced on pages: 29, 31.

***: $p < 0.01$, **: $p < 0.05$, *: $p < 0.10$

small and statistically insignificant. Following the start of the mass migration, naming practices in treated municipalities became progressively less local than in control municipalities. This gap widens over the subsequent two decades. On average, municipalities whose migrants settled in the top 20% most diverse areas of the United States experience a 52% decline in the local name index relative to the mean.

In Table III, we report results from alternative specifications of equation (7). Column (4) includes municipality and year fixed effects. Column (5) further controls for the total number of U.S. emigrants between 1892 and 1924, interacted with period dummies, thereby exploiting variation in emigrants’ settlement destinations among municipalities with comparable emigration intensity. Column (6) documents a similar pattern when we employ the distance-adjusted local name index as the outcome

variable.

We also report the coefficient of a secondary treatment equal to one after a municipality enters the mass migration period. Its estimate remains consistently small in magnitude and statistically insignificant, indicating that transatlantic emigration per se did not affect local identity. Increased national identity in treated municipalities was, therefore, plausibly driven by intergroup contact between emigrants from each municipality and Italians from other parts of the country.

In Figure [VIId](#), we examine heterogeneity in the effect of exposure to diversity by the size of the origin municipality. We estimate the baseline mean-shift specification in (7) and interact the DiD term with indicators for quartiles of municipality population.³⁴ The gray bars report the average local name index in the pre-treatment period for municipalities in each quartile. Smaller municipalities exhibit higher levels of name particularism, reflecting stronger local identities in smaller and more isolated communities. Exposure to more diverse Italian communities in the United States reduces the LNI across all population quartiles. However, the decline is quantitatively larger in larger municipalities, especially if compared to the mean, indicating that transatlantic emigration amplified pre-existing differences in particularism between large and small centers.

5.2 Street Naming Patterns

Street names provide an additional proxy for national identity formation. We classify street names into two identity-related categories, local and national, and three “placebo” categories, art, literature, and religion. The dataset consists of a cross-section covering the universe of municipal streets in Italy as of 2025.

We estimate variations of the following specification:

$$y_m = \alpha_{p(m)} + X'_m \Gamma + \beta_1 \cdot \text{Diverse}_m + \beta_2 \cdot \text{Homogeneous}_m + \varepsilon_m, \quad (8)$$

where $p(m)$ denotes the province of municipality m . The variable Diverse_m measures the number of U.S. emigrants from m who settled in neighborhoods in the top 20% of the entropy distribution, while Homogeneous_m captures the number of emigrants from m who settled in less diverse neighborhoods. Emigrants are expressed in thousands. The control vector X_m includes population levels before (1881) and after (1991) the mass-migration period, as well as geographic characteristics such as municipal area and altitude. Depending on the specification, we further include province or region fixed effects.

Table [IV](#) reports the results. In columns (1–3), the dependent variable is the share of streets bearing a national-identity designation, while columns (4–6) focus on streets associated with local identity. Columns (1) and (4) present the baseline specification; columns (2) and (5) add province fixed effects; and columns (3) and (6) further control for municipality characteristics. We find that the number of emigrants exposed to diverse Italian communities in the United States is positively associated with the share of streets commemorating national figures or events. By contrast, the number of emigrants exposed to relatively more homogeneous migrant communities is negatively associated with national-identity street naming.

³⁴In this specification, we additionally include population-quartile-by-year fixed effects to absorb time-varying shocks com-

Table IV. Street Names and Exposure to Immigrant Diversity

	National Identity Designation			Local Identity Designation			Placebo Designation		
	(1)	(2)	(3)	(4)	(5)	(6)	(7) Arts	(8) Literature	(9) Religious
Number of Emigrants by Neighborhood Diversity:									
Diverse (β_1)	0.326*** (0.079)	0.191** (0.084)	0.344*** (0.095)	-0.251** (0.115)	-0.177 (0.122)	-0.317* (0.192)	0.008 (0.027)	0.048 (0.034)	-0.024 (0.053)
Homogeneous (β_2)	-0.108*** (0.024)	-0.074*** (0.025)	-0.105*** (0.025)	0.108*** (0.036)	0.092** (0.039)	0.098** (0.049)	-0.001 (0.007)	-0.014 (0.009)	0.010 (0.014)
Region FE	Yes	–	Yes	Yes	–	Yes	–	–	–
Province FE	No	Yes	No	No	Yes	No	Yes	Yes	Yes
Controls FE	No	No	Yes	No	No	Yes	No	No	No
p -val: $\hat{\beta}_1 = \hat{\beta}_2$	0.000	0.002	0.000	0.004	0.006	0.001	0.543	0.079	0.266
# Municipalities	8,176	8,176	8,099	8,176	8,176	8,099	8,176	8,176	8,176
Mean Dep. Var.	0.192	0.192	0.194	0.298	0.298	0.299	0.027	0.041	0.105

Notes. This table reports cross-sectional estimates of specification (8) between emigration to diverse Italian communities in the United States and street-naming patterns in Italy. The unit of observation is a municipality. The dependent variables are the shares of streets whose eponyms reflect national identity (columns 1–3) and local identity (columns 4–6); and additional placebo categories (arts, literature, religious, in columns 7–9). The main explanatory variables are the number of emigrants to diverse neighborhoods and the number of emigrants to homogeneous neighborhoods. The outcomes are normalized by the total number of streets in the municipality. Columns (1–3) report national-identity eponyms; columns (4–6) refer to local-identity eponyms; columns (7–9) report placebo outcomes. Columns (1) and (4) include region fixed effects; columns (2) and (5) replace these with province fixed effects; columns (3) and (6) additionally include baseline municipality controls (population in 1881 and 1991, surface area, and elevation). Standard errors are clustered at the municipality level and reported in parentheses. The table also reports p -values from tests of equality between the coefficients on diverse and homogeneous emigration flows. Referenced on pages: 32, 33.

***: $p < 0.01$, **: $p < 0.05$, *: $p < 0.10$

Local-identity naming follows the opposite pattern. Exposure of emigrants to more diverse Italian communities is negatively correlated with the share of streets named after local figures or motives, whereas exposure to more homogeneous communities is associated with a higher prevalence of local-identity street names.

These patterns are quantitatively stable across specifications and underscore sharp differences in commemorative practices across municipalities that experienced similar emigration intensity but whose emigrants settled in more *vis-à-vis* less diverse communities. Table IV separately reports the p -value of a test of equality between the coefficients of emigration flows to diverse and homogeneous communities. Across all models, we find strong evidence that the two coefficients are statistically different ($p < 0.01$).

The estimates from (8) are correlational. However, placebo exercises corroborate an interpretation of these results that hinges on migrants' exposure to diversity. While we find a large and statistically significant association, with opposite signs, between historical exposure to diversity and national versus local identity in street names, we detect no systematic relationship with placebo categories. In columns (7–9), we employ the share of streets named after artistic, literary, and religious figures as the outcome variables. None of these outcomes is significantly associated with either the number of U.S.

mon to municipalities of similar size that may affect naming practices.

emigrants who settled in high or low-diversity communities. These patterns suggest that exposure to diversity is not merely correlated with a general propensity for street naming or commemorative practices, thus restricting the scope for alternative explanations based on unobserved municipal characteristics.

5.3 National Monuments

We use public monuments as a third proxy for national identity sentiment. We rely on a municipality-level dataset and aggregate the number of monuments to the decade level.

We estimate the following specification:

$$y_{mt} = \alpha_m + \alpha_t + X'_{mt}\Gamma + \sum_{\tau=1870,1880,\dots}^{1990} \beta_{\tau} \cdot \text{Diversity Exposure}_m \cdot I\{t = \tau\} + \varepsilon_{mt}, \quad (9)$$

where, as in the rest of the analysis, $\text{Diversity Exposure}_m$ denotes the share of U.S. emigrants from municipality m who settled in districts in the top quintile of the entropy distribution. In some specifications, we include region-by-decade fixed effects ($\alpha_{r(m) \times t}$) to absorb time-varying shocks at the regional level, such as differential exposure to wars and reconstruction. In this analysis, we restrict the sample to Southern municipalities for two reasons. First, national identity monuments largely commemorate casualties of the World Wars, and Northern regions were more exposed to combat, so monument construction there is more likely to reflect proximity to warfare rather than identity formation. Second, given the long time horizon of the analysis, comparisons between Northern and Southern regions would confound the estimates with multiple region-specific developments that unfolded over the twentieth century. As in (6), we interpret the coefficients for the 1870s, 1880s, and 1890s as evidence on pre-treatment differences between treated and control municipalities.

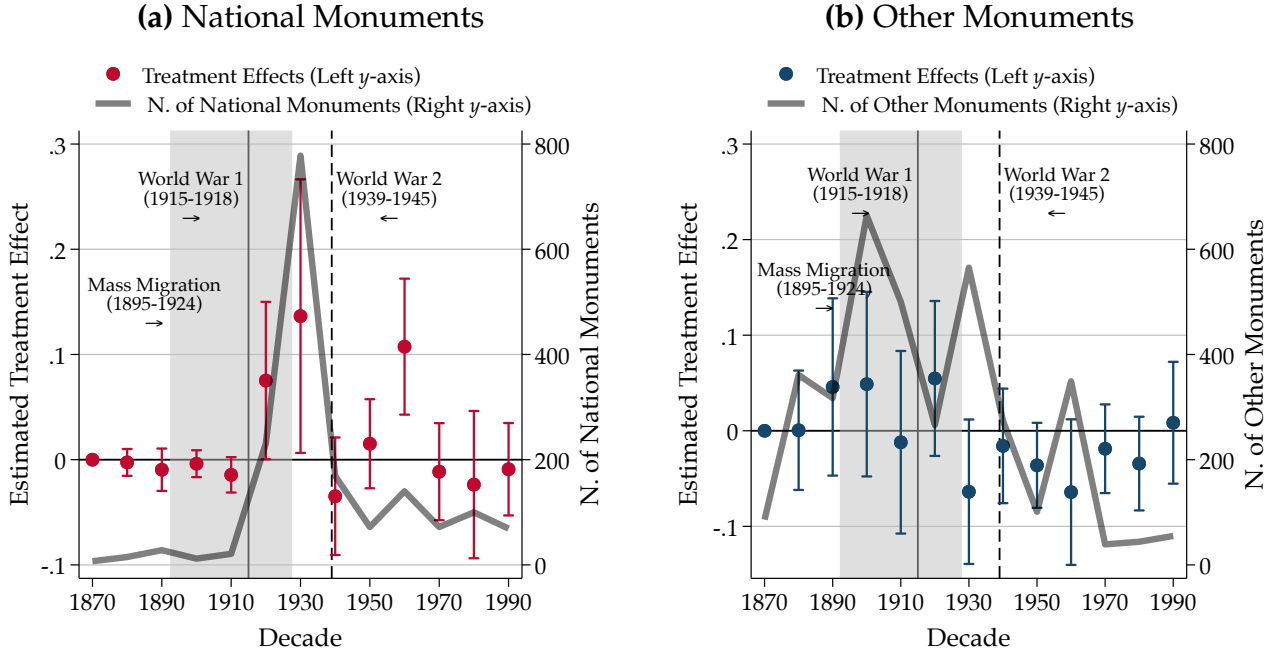
Figure VII reports the estimated effects. In both panels, the left y -axis plots the coefficients from equation (9), while the gray line on the right y -axis shows the number of monuments by year of construction.

Figure VIIa focuses on national monuments. Most national monuments were erected in the 1920s and 1930s and commemorate soldiers who died during WW1, with a second, smaller wave following WW2. Before the mass emigration, we find no statistically significant differences between municipalities whose U.S. emigrants later settled in more versus less diverse Italian communities.

Beginning in the interwar period, municipalities whose migrants settled in relatively more diverse Italian communities in the United States exhibit a higher likelihood of at least one national monument than control municipalities. We also estimate a statistically significant effect of exposure to diversity on the probability that at least one national monument is erected between 1950 and 1959, when monuments commemorated WW2 casualties. This persistence suggests that national identity formed through emigrants' historical exposure to diversity was activated by the experience of the Second World War.

Figure VIIb replicates the analysis for non-national monuments. We find no association between exposure to Italian diversity in the United States and the construction of non-national monuments prior to the mass migration period. Likewise, exposure to diversity has no effect on the construction

Figure VII. Monuments by Designation in Italy



Notes. This figure reports dynamic estimates obtained from (9) of the effect of exposure to diversity within the Italian migrant community on monument construction in Southern Italy. The unit of observation is a municipality-decade for municipalities observed between 1861 and 2000. In panel VIIa, the dependent variable is an indicator equal to one if the municipality has at least one national-identity monument in a given decade. In panel VIIb, the dependent variable is an indicator equal to one if the municipality has at least one non-national-identity monument in a given decade. In both panels, each dot reports, on the left y-axis, the coefficient of a set of interaction terms between decade indicators and the share of US emigrants from each municipality who settled in districts in the top quintile of the diversity measure defined in (3). In both panels, the specification includes municipality fixed effects, decade fixed effects, and interactions between the number of US emigrants and decade indicators. The gray line reports, on the right y-axis, the total number of monuments in each category. Standard errors are clustered at the municipality level. Bands report 95% confidence intervals. Referenced on page: 34.

of non-national monuments in the post-migration period. This pattern suggests that our estimates are unlikely to capture general shifts in commemorative activity unrelated to exposure to diversity.

Table E.13 reports the estimates in tabular form under a range of alternative specifications where we control for time-varying shocks at the region (columns 2 and 7) and province (columns 3 and 8) levels. In columns (5) and (9), we control for the number of U.S. emigrants over the migration period interacted with period dummies to test whether the results are driven by the overall intensity of the migration outflows rather than by its direction. In columns (6) and (10), we add municipality-level time-invariant controls—including the number of deaths during WW1 and WW2. The baseline results remain quantitatively robust.

5.4 Robustness Analysis

Most Italian migrants in the United States originated from Southern regions. A potential concern is that our results may be driven by specific areas. In Figure F.17, Figure F.18, and Figure F.19, we replicate the baseline analyses for names, streets, and monuments while sequentially excluding one region at a time from the estimation sample. In each panel, the first row reports the full-sample baseline estimate, and the following rows report the treatment effects obtained when excluding each region. Across all outcomes and specifications, the estimates remain statistically comparable to the baseline.

A related concern is that spatial clustering of emigrants across regions or municipalities may induce spatial autocorrelation, inflating statistical significance. In Figure F.20, Figure F.21, and Figure F.22, we replicate all analyses using alternative standard error estimators, including the spatial correction proposed by Conley (1999) with different distance cutoffs. Statistical significance remains unchanged across outcomes and specifications.

We assess the sensitivity of our naming and monument results to potential violations of the parallel trends assumption using the framework developed by Rambachan and Roth (2023). In our application, robust confidence intervals reported in Figure F.23 and Figure F.24 indicate that the post-treatment departure from parallel trends would have to be at least as large as the largest pre-treatment deviation to overturn the results. These patterns indicate that the results remain robust to sizable departures from the parallel trends assumption.

6 How Did Emigrants Influence National Identity Formation?

This section provides evidence of the cultural impact of transatlantic emigration on emigrants' communities of origin. According to politician Leopoldo Franchetti (1950, p. 230),

“As we were busy writing books, [...] southern peasants who emigrated to the Americas and returned proved to be the true agents of change.”

Historical accounts suggest that these cultural remittances were mediated by multiple channels, including return migration, correspondence, and economic connections with the communities of origin (Choate, 2008).

The available data do not allow us to disentangle the relative importance of these channels. Instead, we provide evidence that ideas, practices, and national attachment developed abroad could reach and influence emigrants' communities of origin.

6.1 Illustrative Evidence from Primary Sources: Emigrants' Letters

The correspondence of emigrants with their families and communities of origin provides vivid, although quantitatively modest, insight into their migration experience and the information they disseminated in Italy. We collect data from a dataset of letter correspondence contributed by emigrants and their descendants.³⁵ We employ a large language model (gpt-5.4-mini) to analyze the content of letters. Figure F.25a (resp. Figure F.25c) reports the number of all (resp. U.S.) emigrants in the dataset by region of origin.

Figure F.25b displays the content of the letters of all emigrants. In line with historical evidence documenting the linguistic fragmentation of Italy during this period, the proportion of migrants reporting learning the language of their host country (43%) is similar to the proportion reporting learning Italian (34%). Intergroup contact with Italians from other parts of the country was even more

³⁵Appendix Section A.5 provides details on the data source and the methodology. The dataset only contains information on 162 emigrants, which precludes us from using letters in a quantitative analysis.

prevalent than with natives (81% versus 74%). Emigrants' letters very often (68%) contain nationalistic content related to Italy, whereas a smaller share of individuals mention nationalism in favor of the host country (22%). While marriage does not feature prominently, mentions of marriages with Italians from other regions are more than twice as likely as with natives (15% versus 6%). In line with historical evidence, the majority of migrants express a desire to return (62%), and approximately 60% explicitly mention their wish to return to their place of origin. According to their biographies, 41% of emigrants returned.

Data on U.S. emigrants, albeit more scattered, convey a similar picture (see Figure F.25d). If anything, Italians in the U.S. are more likely to mention contact with Italians from other regions and less likely to report contact with natives. They are more likely to send letters containing nationalistic messages and to mention marriages with Italians from other regions. Consistent with evidence by Bandiera et al. (2013), the majority (71%) of Italians in the U.S. express the desire to return to their place of origin, and approximately 64% returned.

While descriptive, this analysis illustrates the relevance of inter-group contact within Italian communities and the emergence of nationalism among their members. Moreover, the letters echo historical evidence that Italians typically viewed migration as temporary, with most expressing a desire to return to their place of origin.

6.2 Linguistic Contamination of Sicilian Dialects

Language is a central component of culture (Lazear, 1999). Cultural fragmentation in Italy was evident in the linguistic fragmentation of spoken dialects, especially in the South (De Mauro, 1963). In this section, we ask whether transatlantic emigration influenced dialects, a deep-rooted component of local culture.

To test this hypothesis, we digitize a vast ethnolinguistic atlas, the *Vocabolario Siciliano*. The *Vocabolario* is the most comprehensive endeavor to reconstruct the linguistic geography of local dialects in Italy (Piccitto, 1977; Tropea, 1985, 1990). The *Vocabolario* contains information on the diffusion of words in local dialects in Sicily at the *municipality* level. We digitize the first three volumes, covering letters A to Q, and parse information on 65,074 words into a machine-readable database.³⁶

Using this dataset, we construct multiple measures of linguistic contamination at the local (municipal) level and estimate cross-sectional regressions between these measures and transatlantic emigration. We discretize the baseline diversity exposure measure (i.e., the number of emigrants who settle in districts in the top quintile of the diversity distribution) and the number of U.S. emigrants from each municipality into above- and below-median categories. Table V reports the baseline results.

In columns (1–3), we employ the log-number of Americanisms in the *Vocabolario* as a measure of the contamination between Sicilian dialects and English.³⁷ In all specifications, we find that districts with above-median U.S. emigration display a larger presence of Americanisms in their dialect.

³⁶Additional details on the data source and the extraction methodology are provided in Appendix Section A.6, and Figure F.26 shows a sample page and the locations of the enumerated municipalities. Only the first three volumes of the *Vocabolario* are digitally available.

³⁷Table E.14 lists the Americanisms we search for in the *Vocabolario*.

Table V. Dialect Contamination and U.S. Out-Migration

	(log) Number of Americanisms			(log) Number of Words Spoken in Other Municipalities			(log) Number of Recorded Words
	(1)	(2)	(3)	(4)	(5)	(6)	(7)
High US Emigration	0.081** (0.041)	0.082** (0.039)	0.078** (0.039)				
High Diversity Exposure				0.679*** (0.166)	0.622*** (0.170)	0.615*** (0.165)	
Diversity Exposure (Cont.)	0.204 (0.161)	0.231 (0.164)	0.266 (0.176)				-1.697 (2.970)
US Emigration (Cont.)				-0.469** (0.186)	-0.248 (0.159)	-0.214 (0.134)	0.295 (0.660)
Demographic Controls	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Geographic Controls	No	Yes	Yes	No	Yes	Yes	Yes
Province FE	No	No	Yes	No	No	Yes	Yes
# Municipalities	378	378	378	378	378	378	378
Mean Dep. Var.	0.153	0.153	0.153	4.606	4.606	4.606	4.781

Notes. This table reports cross-sectional associations between various measures of dialect contamination and emigration to diverse emigrant communities in the United States. The unit of observation is a municipality. The sample comprises the universe of Sicilian municipalities. The dependent variables are: the zero-inflated log number of Americanisms (columns 1–3), the zero-inflated log number of words that appear in each municipality and in other municipalities (columns 4–6), and the zero-inflated log number of entries in the *Vocabolario Siciliano* (column 7). The main explanatory variables are: in columns (1–3), an indicator for municipalities with an above-median number of US emigrants; in columns (4–5), an indicator for municipalities with an above-median number of US emigrants who settled in districts in the top quintile of the diversity distribution (3). In columns (1–3), the regressions control for the number of US emigrants who settled in districts in the top quintile of the diversity distribution (3). In columns (4–6), the regressions control for the number of US emigrants. Column (7) includes both controls. Each regression includes population in 1901 as a control; columns (2) and (5) further include geographic controls—elevation, an indicator for municipalities on the coast, latitude, and longitude; columns (3), (6), and (7) further include province fixed effects. Standard errors are clustered at the municipality level and reported in parentheses. Referenced on pages: 37, 38.

***: $p < 0.01$, **: $p < 0.05$, *: $p < 0.10$

Compared to below-median municipalities, above-median areas display 8% more words of English origin.

In columns (4–6), we use the zero-inflated log-number of words shared by more than one municipality as a crude measure of linguistic contamination among Sicilian dialects. Across all specifications, we find a positive correlation between diversity exposure and linguistic contamination. Quantitatively, the dialects spoken in municipalities with above-median diversity exposure have 85% more words that are shared with other municipalities. Table E.15 shows that the same pattern holds when placing more weight on more distant municipalities (column 2) and that this pattern does not depend on the distance between municipalities (columns 3–5). Similarly, the results are not driven by words shared by only a few dialects (see columns 6–8).

In column (7) of Table V, we use the number of words that appear in the *Vocabolario* as a placebo outcome and find no differential coverage in areas more exposed to diversity or with larger U.S. emigration outflows.

These patterns suggest that transatlantic emigration exerted substantial influence on spoken lan-

guage in Sicily. Larger outflows are associated with a greater presence of Americanisms, while exposure to diverse communities in the U.S. is associated with greater dialect contamination. Since dialects plausibly represent a deep-rooted component of local identity, as well as the predominant form of spoken language, this analysis provides suggestive evidence of cultural remittances from transatlantic emigration to emigrants' communities of origin.

While the data only allow for a cross-sectional analysis, we perform two exercises to gauge the credibility of these results. First, in Table E.16, we employ three “placebo” outcomes which confirm that (i) areas more exposed to diversity did not receive better coverage in the *Vocabolario*, and (ii) exposure to diversity is not associated with a higher number of words that are only present in the municipality's dialect. Second, in Figure F.27–Figure F.28, we conduct a randomization-inference exercise by randomly permuting the baseline treatments across municipalities. The actual coefficients lie in the upper tail of the placebo distribution, indicating that the results are unlikely to be explained by chance.

6.3 Emigrants' Political Activity

Historical evidence documents the active involvement of return migrants in local civic and political life, including village organizations, labor unions, and political parties (Wyman, 1993). To obtain a more direct measure of emigrants' engagement in their local communities and in Italian national institutions, we examine their representation in local elected offices. We collect information on the mayors, municipal board members, and municipal council members for the universe of Italian municipalities in 1986, the first year when such information is available.³⁸ For each of the 134,936 individual elected officials, we have information on their name, surname, birth city and year, their rank and party, and the municipality where they were elected.

First, we focus on elected officials born in the United States. We restrict the sample to individuals born before 1930 to avoid including later-generation returnees and children of emigrants who settled in the United States after WW2.³⁹ Table E.17a displays the results. We identify 179 foreign-born and 29 U.S.-born elected officials among 14,613 elected officials. While small, these numbers indicate a substantial overrepresentation of foreign-born and U.S.-born politicians relative to the broader population. Elected officials among foreign-born and U.S.-born individuals are respectively 0.085% and 0.156% of the population, while they are only 0.026% among Italian-born individuals (see columns 3 and 6). Thus, second-generation return migrants are three times more likely to be elected to local administrative offices, while U.S.-born returnees are more than 6 times more likely than native-born to hold elected office.

For first-generation emigrants, we focus on elected officials who we match to Ellis Island emigrant records.⁴⁰ We restrict the sample to individuals born before 1924, the last year covered in the ship

³⁸The data were collected by the Italian Minister of the Interior and are publicly available at this [link](#).

³⁹All elected officials born in the U.S. bear Italian-sounding last names, indicating that they are of Italian ancestry.

⁴⁰We match official records to Ellis Island entries on first and last name, birth year, and region of origin. We use the same similarity threshold of Section 3.1. We disambiguate multiple matches by selecting emigrants originating from the official's closest municipality.

records. Table E.17b displays the results. We identify 51 matched U.S. emigrants among 14,613 elected officials. Because the linking algorithm incurs attrition, this raw count should be interpreted as a conservative lower bound. Assuming an attrition rate of 85%, i.e., the same as the baseline linked sample described in Section 3.1, the implied share of U.S. emigrants among elected officials is 3.95%. Compared with Italian-born individuals, first-generation emigrants are thus 1.46 times more likely to hold elected office.

6.4 Emigrants' Participation in World War 1 in the Italian Army

As an additional measure of emigrants' participation in Italian society, we follow Caprettini and Voth (2023) and use World War 1 death rates among individuals who emigrated to the United States and served in the Italian army. Military participation among emigrants was quantitatively important: more than 300,000 emigrant reservists returned to Italy to serve during World War 1 (Choate, 2008). Enlistment constituted a high-stakes expression of national attachment, exposing migrants to the risk of death in the service of the Italian state. Although military service was formally universal, Italian consulates in the United States lacked enforcement capacity, implying that enlistment for emigrants was voluntary in practice (Choate, 2008).

We estimate variants of the following regression:

$$y_m = \alpha_{p(m)} + X'_m \Gamma + \beta \cdot US\ Emigrants_m + \gamma \cdot Diversity\ Exposure_m + \varepsilon_m, \quad (10)$$

where, as in the previous specifications, $Diversity\ Exposure_m$ denotes the share of U.S. emigrants from municipality m who settled in districts in the top quintile of the entropy distribution. In our preferred specification, we control for province-level fixed effects, as in (8), and for the number of U.S. emigrants.

Table E.18 displays the results. Exposure to more diverse environments in the United States is associated with higher death rates among returning migrants serving in the Italian army (columns 1–4). The magnitude is large and stable across specifications: moving from the 25th to the 75th percentile of diversity exposure is associated with a 1.5 percentage points increase in the death rate among migrants, which corresponds to 80% of the mean.

By contrast, diversity exposure is uncorrelated with overall municipal death rates (column 5), suggesting that the relationship is specific to migrants rather than reflecting broader differences in wartime mortality. Moreover, the intensity of U.S. emigration is uncorrelated with World War 1 death rates (columns 3–5), indicating that the estimates plausibly reflect the effects of exposure to diversity. This pattern is consistent with greater national attachment among migrants exposed to more diverse environments abroad.

7 Conclusions

This paper investigates whether national identity can emerge in response to intergroup contact among individuals in the absence of top-down state interventions. We study the Italian diaspora at the turn of the 1900s, when 4.5 million emigrants from diverse linguistic, political, and socioeconomic back-

grounds settled in a handful of neighborhoods across America.

We construct a novel, highly detailed dataset that allows us to observe the municipality of origin of half a million Italian migrants in the United States by linking the U.S. census records and the Ellis Island ship passenger lists. We document that American neighborhoods were areas where Italians from different parts of the country met and interacted. These contacts contributed to the emergence of a sense of shared national identity even in the absence of coordinated state interventions. Increased national identity among migrants hampered cultural assimilation into U.S. society.

We then show that the national identity that emerged in immigrant communities diffused to Italy. Municipalities more exposed to intergroup interaction through emigration experienced an increased sense of national identity. These patterns are difficult to reconcile with purely domestic explanations and instead point to a transnational process through which identity formation abroad reshaped cultural outcomes at home.

We document this cultural influence in emigrants' letters to their families, in cross-dialect contamination and in the presence of Americanisms in Sicilian dialects, in the political activity of return migrants, and in higher volunteering rates in WWI among return migrants.

Our findings suggest that today's diasporic communities can play an important and underappreciated role in fostering nation-building in migrants' countries of origin.

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ONLINE APPENDIX

A Nation of Emigrants: Transatlantic Emigration and the Rise of Italian National Identity, 1890–1940

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A Data Appendix and Robustness Checks

A.1 Measurement of Completed Fertility

The decennial U.S. Census provides cross-sectional snapshots of households and therefore does not directly observe completed fertility. To recover completed fertility histories, we combine household structure information within each census year with longitudinal individual links across censuses to reconstruct parent-child relationships over time.

We begin by identifying Italian migrants and second-generation individuals in each census year (1900, 1910, 1920, and 1930), defined as individuals either born in Italy or with at least one parent born in Italy. Using household identifiers and relationship-to-household-head variables, we classify individuals as parents or children within each household and retain only nuclear-family relationships. In each household, we identify up to two parents and link each child to the corresponding parent identifiers. This procedure yields a set of parent-child observations for each census year.

Because children may appear in different households or census years as families age, relying on a single census snapshot would underestimate fertility. We therefore exploit longitudinal links across censuses to aggregate children observed in different years (Price et al., 2021; Buckles et al., 2025; Abramitzky et al., 2025). Specifically, for each census year taken as a reference, we employ the longitudinal crosswalks that link individual identifiers across census waves to map parents observed in other census years to the reference year. We then pool all parent-child links across census waves and collapse them at the parent level, removing duplicate children identified by year of birth and first name.

The resulting dataset assigns to each parent the full set of children ever observed co-residing with them in any census between 1900 and 1930. This approach recovers fertility histories that are substantially closer to completed fertility than those obtained from a single census snapshot, while relying solely on observed household relationships and longitudinal links. Nonetheless, we do not observe children who were not recorded in any census.

A.2 Validation of the Name-Based Measure of National Identity

To validate our name-based measure of national identity, we use contemporary survey data (ITANES) from Bellucci et al. (2008). The ITANES survey was conducted at regular intervals between 2001 and 2023; we focus on the first three waves (2001, 2008, and 2013), which are the only ones with municipality identifiers. The data provide individual-level information on demographics, electoral behavior, and, most importantly for our analysis, trust, for a representative sample of 7,699 individuals. We aggregate the data at the municipality level and exclude municipalities with fewer than five respondents.

From the ITANES data, we construct measures of trust in national institutions: the army, parliament, the head of state, the police, the judicial system, the public administration, and the European Union. We also collect measures of trust in “placebo” dimensions: the Catholic Church, labor unions, and other citizens. All variables are coded on a scale from 1 (lowest trust) to 5 (highest trust). For

each municipality, we compute the average response.

We combine the ITANES data with vital statistics reporting the number of children born between 1999 and 2024 in each municipality by name.¹ Using these data, we construct contemporary counterparts to the historical LNI and DLNI defined in (1)–(2). We then compute the average municipality-level LNI as

$$LNI_m = \sum_n \frac{N_{nm}}{N_m} \cdot LNI_n, \quad (\text{A.1})$$

where LNI_n denotes the local name index of name n , N_{nm} is the number of individuals named n in municipality m , and N_m is the total number of individuals born in municipality m . The distance-adjusted LNI is defined analogously.

First, we construct a synthetic measure of trust in national institutions as the first principal component of all trust measures. Figure F.6a reports the correlation between this measure and the local name index. We find a negative and statistically significant relationship, indicating that trust in national institutions is lower in municipalities where the names of newborn children reflect more localized naming practices. Figure F.6b shows a similarly positive—i.e., lower name localization—and statistically significant correlation with the distance-adjusted local name index. In both cases, we control for log population, as larger cities are expected to exhibit lower name particularism, and for an indicator equal to one for Southern regions.

Column (1) of Table E.2 presents the tabular counterpart of the baseline exercise. Columns (2–8) report the correlation between the LNI (Panel A) and the distance-adjusted LNI (Panel B) and each of the variables entering the PCA-based measure of trust in national institutions. In all cases, we find a negative (resp. positive) and, in most cases, statistically significant correlation between the outcome and the local name index (resp. the distance-adjusted local name index). By contrast, columns (9–11) show no statistically significant relationship between measures of trust unrelated to national institutions and the name-based measures of cultural particularism. This pattern suggests that these measures are specifically associated with national identity, rather than with more general trust.

A.3 Street Data

We collect the universe of street names in Italy from the online repository OpenStreetMaps. Streets are grouped into municipal, provincial, regional, and state roads and highways. The national government is responsible for state roads and highways, while provinces and regions are responsible for provincial and regional roads, respectively. Since our analysis leverages local street naming decisions, we exclude non-municipal roads from the sample. We thus have information on 849,124 streets, which correspond to approximately 93.7% of the entire sample of roads.

The dataset contains the name and coordinates of each road. We geo-reference roads to the municipality where they are located in 1991, i.e., the GIS file we employ throughout the analysis on Italian municipalities.

¹We are grateful to the Italian Statistical Institute for assembling this dataset. Due to privacy constraints, we observe only names with at least two children born in a given municipality-cohort cell.

We employ a large language model (GPT-5) to infer the motive for the designation of each road. We employ the following prompt to categorize designation motives into local or national identity. Importantly, we give the LLM both the eponym and the municipality of each road and instruct it to identify local names using that information. Hence, names of local saints are identified as signals of local identity only where they are the subject of specific veneration.

Second, we ask the LLM to identify three “placebo motives” for designation, namely, artistic, literary, and religious figures. We employ this information as a falsification exercise in the analysis. Importantly, we do not constrain the model to assign road names to a single category. Hence, local saints, for example, are flagged as signifying both local identity and a religious motive. Similarly, we do not impose that each road must be classified in one category.

Overall, the algorithm identifies 33% of streets as signaling local identity, 18% as national identity, 4% as designated after (visual) artistic figures, 5% after literary figures, and 10% after religious figures. Approximately 40% of all roads are not classified under any such motive.

gpt-5 Prompt for Street Classification

You are an expert on Italian cultural and political history and street toponymy.

Input:

- Street name: {street}
- Municipality: {comune}

Task: Interpret what this name commemorates in the context of Italian public toponymy and evaluate how strongly it expresses different collective identities.

Use historical and cultural knowledge about Italy to understand what the name symbolizes to Italians — for example, “Via Garibaldi” signals national identity because Garibaldi is recognized as a hero of unification.

Definitions of dimensions:

1. *national_identity* — reference to the Italian nation, its unification, or explicitly national heroes, institutions, or symbols.
Examples: Garibaldi, Cavour, Mazzini, Vittorio Emanuele II, Italia, Unità d'Italia, XX Settembre, 25 Aprile.
2. *local_identity* — reference to regional or municipal belonging, local saints, local landmarks, or figures of local history.
Examples: San Gennaro (in Naples), Sant’Ambrogio (in Milan).
3. *religious_identity* — reference to religion, spirituality, or ecclesiastical institutions.
Examples: San Pietro, Santa Maria, Papa Giovanni XXIII, Madonna, Convento, Santuario.
4. *literature_identity* — reference to literary figures, authors, poets, or literary works.
Examples: Dante, Petrarca, Manzoni, Leopardi, Pirandello.

5. `art_identity` — reference to artists, painters, sculptors, or art movements.

Examples: Michelangelo, Leonardo, Raffaello, Caravaggio, Bernini.

Rules:

- Base scores on how the name would be understood by an informed Italian observer aware of national culture and history.
- Scores = 1 if the name clearly evokes the dimension; 0 if it does not.
- Multiple 1 for one street are possible (e.g., a saint who is also a local patron has `local_identity` = 1 and `religious_identity` = 1).
- If the meaning cannot be inferred confidently, set all scores = 0 and `named_entity` = null.
- If the commemorative meaning is unclear or generic, assign all zeros.

Return:

```
{  
  "street_name": "{street}",  
  "comune": "{comune}",  
  "named_entity": "<person, place, event, or concept commemorated>",  
  "national_identity": float,  
  "local_identity": float,  
  "religious_identity": float,  
  "literature_identity": float,  
  "art_identity": float  
}
```

A.4 Monuments Data

We collect buildings of significant cultural significance as identified by the Italian Ministry of Culture. This repository covers the universe of heritage sites across the Italian territory. Overall, the dataset contains 128,086 sites and includes, among other information, construction and designation years, addresses and coordinates, site names and typologies, a historical text documenting the monument's historical significance, and an artistic description of the site's aesthetics.

We exclude 24,369 sites with a missing construction year and 5,702 sites with missing coordinates. Moreover, we exclude sites erected before the Italian unification (1861) and after 2000. We exclude sites older than 1861 because regional coverage before unification is uneven, possibly reflecting the heterogeneous designation criteria among pre-unitary states. This leaves us with 32,369 sites. Among these, 44 are georeferenced in areas that are no longer part of Italy (Fiume, Dalmatia, and various islands that are now part of Greece). We discard these sites, yielding a sample of 32,325 monuments.

As with the street data, we assign monuments to municipalities at the 1991 borders to ensure

geographical consistency.

We employ the following prompt to infer whether each monument is associated with a national identity motive. In most cases, national identity monuments commemorate soldiers from the municipality who died during World Wars 1 and 2.

gpt-5 Prompt for Monument Classification

You are an expert in Italian history and culture.

You will be provided with a description of a monument that commemorates a historical event, figure, or cultural aspect.

Your task is to determine whether the monument commemorates Italian national identity.

Information:

- Monument Category: {category}
- Historical Information: {historical_information}
- Description: {description}

Return a JSON object with the following structure:

```
{  
  "national_identity": 1 if the monument commemorates Italian national identity, 0 otherwise  
}
```

A.5 Analysis of Emigrants' Letters

We collect data on emigrants' letters from the publicly available dataset idiariraccontano.org, a crowd-sourced repository of letters sent by emigrants to their families in Italy and collated by the Italian Ministry of Foreign Affairs. The dataset includes biographical information for each migrant (name and surname, municipality of origin, occupation, education level, year of departure, and country of emigration), as well as detailed biographies and the full texts of the letters. Because the materials were contributed by emigrants and their descendants, the sample may over-represent individuals who eventually returned to Italy.

We focus on individuals who migrated between 1861 and 1939. The sample covers 162 emigrants and 14 individuals who migrated to the United States. While the small sample size does not permit quantitative analysis, the information in the letters nonetheless constitutes a valuable primary source for illustrating broader characteristics and attitudes of Italian migrants. In total, we analyze 878 letters.

We use a state-of-the-art large language model (gpt-5.4-mini) to analyze the content of the letters. We use the following prompt.

gpt-5.4-mini Prompt for Analysis of Emigrants' Letters

You are a rigorous historical researcher. Respond in valid JSON adhering to the schema. Do not invent quotations: use literal passages from the provided text.

User: Letter in Italian. Classify the text into the following migration-destination-country categories:

1. `learning_italian`: learning/use of the Italian language (learning, studying, speaking, writing Italian).
2. `learning_destination_language`: learning/use of the destination-country language (e.g., English in the US, French in France).
3. `contact_other_italians`: contact/social interaction with Italians from other Italian regions/provinces.
4. `contact_with_destination_natives`: contact/social interaction with natives of the destination country (not Italians, not other foreign immigrants).
5. `nationalism`: homeland, nation, patriotism, symbols or nationalist discourse.
6. `destination_country_nationalism`: patriotism/nationalism referring to the destination country (not Italy).
7. `marriage_other_italians`: marriage with an Italian emigrant from another Italian region/province.
8. `marriage_with_destination_natives`: marriage/conjugal relationship with a native person from the destination country (not Italians, not other immigrants).
9. `return_to_italy`: mention of return/re-entry/repatriation to Italy.
10. `settle_permanently_in_destination`: explicit intention to remain permanently in the destination country (settle down, put down roots, not return to Italy).
11. `return_to_origin_place`: mention of return to the town/city/region/province of origin in Italy.
12. `return_to_non_origin_place`: explicit intention to return to Italy to a place different from the town/city/region/province of origin.

Rules:

- Use ONLY explicit textual evidence.
- If label=1, include a short, meaningful passage taken from the text.
- If label=0, evidence passage must be an empty string.

- Confidence must be between 0 and 1.
- `contact_with_destination_natives=1` only if the text clearly indicates natives of the destination country.
- `marriage_with_destination_natives=1` only for marriage/conjugal relationship with a native of the destination country.
- `destination_country_nationalism=1` only if patriotism/nationalism refers to the destination country, not Italy.
- `settle_permanently_in_destination=1` only with explicit and stable intention to remain permanently in the destination country.
- `return_to_non_origin_place=1` only if the text indicates intention to return to Italy and a destination in Italy different from the place of origin.
- If the text only says “return to Italy” without a specific destination, `return_to_non_origin_place=0`.

Full text: <entry_body_text>

In addition, we run the large language model on the biography of each individual to identify those who effectively return to Italy using the following prompt.

gpt-5.4-mini Prompt for Analysis of Emigrants' Biographies

You are a rigorous historical researcher. Respond in valid JSON adhering to the schema. Do not invent quotations: use literal passages from the provided text.

User: You are given an author biography (in Italian). Determine whether there is explicit evidence that the author returned/re-entered/was repatriated to Italy.

Rules:

- `label=1` only with explicit evidence of return to Italy.
- `label=0` if absent or implicit/uncertain.
- If `label=1`, include a short literal passage from the biography.
- If `label=0`, return an empty string.
- Confidence must be between 0 and 1.

Full text: <biography>

Figure F.25a and Figure F.25c report, respectively, the distribution of the regions of origin of all emigrants and U.S. emigrants. Emigrants from the center and northern regions are overrepresented

relative to the emigrant population. However, we have data on the two largest sending regions to the United States, Campania and Sicily.

Figure F.25b reports the coverage of selected topics in the letters of all emigrants. Blue solid bars refer to Italy-specific topics, whereas the black-contoured bars report the coverage of topics related to the destination country for comparison. Approximately 34% of all emigrants mention learning Italian while abroad. This compares to 43% mentioning learning the language of their host country, which could plausibly yield higher labor market returns. Nearly 81% emigrants mention intergroup contact with Italians from other regions, which is similar to the share of those reporting interacting with natives of the host country (74%). Italian nationalism is a recurrent theme in emigrants' letters, as almost 68% mention it. By contrast, less than 22% of letters contain passages related to nationalism of the host country. Marriage is a relatively rare topic in migrants' letters. However, marriages with Italians from other regions are more than twice as likely to appear compared to marriages with natives (15% against 6%). In line with historical evidence, the vast majority of migrants mention the willingness to return to Italy (62%), with 58% explicitly mentioning the desire to return to their place of origin, while 10% state their wish to return to Italy but to a place other than their origin. Based on the reported biographical information, 41% of emigrants returned to Italy.

Data for U.S. emigrants convey a similar picture (see Figure F.25d). Italians in the United States were even more likely than other migrants to report contact with Italians from other regions and less likely to mention contact with natives. This pattern is consistent with higher economic segregation in the United States compared to other destinations, such as Argentina (Pérez, 2021). Italians in the United States were also more likely to report the desire to return to their place of origin, and a larger proportion of them (64%) effectively returned.

A.6 The *Vocabolario Siciliano* Ethnolinguistic Atlas

We digitized the first three volumes (letters A to Q) of the *Vocabolario Siciliano*, henceforth VS (Piccitto, 1977; Tropea, 1985, 1990). The VS represents the most ambitious and comprehensive attempt to map the lexicon of all Sicilian dialects. It is the result of a large-scale collaboration between more than 1,000 enumerators coordinated by linguists Giorgio Piccitto and, later, Giovanni Tropea. The coordinating team prepared 14 questionnaires containing approximately 1,200 questions on a wide range of subjects.² Each Sicilian municipality was assigned to a team of enumerators who administered the questionnaire to locals. The questionnaires were administered between 1955 and 1975, while the volumes were published in waves between 1977 (letters A to Q) and 2002 (letters Sg to Z).

The vocabulary tracks the spatial distribution of more than 100,000 words at the municipality level. We digitize the first three volumes, which are available as scanned PDFs. We apply an open-source optical character recognition (OCR) software (*tesseract*) to convert the scanned images into machine-readable text. We then parse the entries of the digitized text through regular expressions. We identify 65,074 single entries. Each entry is associated with one or more sub-entries. Each sub-entry has a unique meaning. For each sub-entry, the VS lists the municipalities where the local dialect

²For example, questionnaire 2 concerned body parts, while questionnaire 4 investigated moral attitudes.

employs it. We parse all entries and find that 69,33% (45,118) of the sub-entries mention one or more municipalities. The remaining 30.67% refer to words that are common to all dialects. Figure F.26a displays one sample page of the first volume of the VS. Entry “bbossu,” for example, is an Americanism from the word “boss,” and it is adopted in thirteen municipalities. Figure F.26b shows the locations of all municipalities listed in the VS.

To validate the municipality extraction method, we also implement an algorithm that leverages the flexibility of an LLM (gpt 5.4 nano) to parse the municipality codes. The regular-expression (regex) method attaches at least one municipality to 45,118 sub-entries (69,33%), while the LLM method attaches at least one municipality to 42,137 sub-entries (64.75%). Only 77 sub-entries feature a municipality identified by the LLM but not by regex. Conversely, 3,058 sub-entries matched to a municipality by the regex method are not identified by the LLM approach. Among sub-entries with at least one municipality code, the LLM and regex methods identify the same municipalities in 86.85% of cases. A manual verification indicates that most matches identified by regex but not by the LLM are accurate. Multiple LLM-only matches are instead spurious matches of sub-entries in which no municipality is mentioned. Taken together, we view the regex results as reliable and comprehensive.

We construct several variables to measure the contamination of Sicilian dialects. First, we compile the count of Americanisms that appear in each municipality’s dialect. The full list of Americanisms is available in Table E.14. Second, for each municipality, we count the number of words that appear in that municipality and (i) in other municipalities, (ii) in 2 to 4 other municipalities, (iii) in 5 to 9 other municipalities, (iv) in 10 or more other municipalities, (v) in no other municipalities, and (vi) the total number of entries. We exclude from these counts all entries that are adopted in the general Sicilian language rather than in specific municipal dialects. Third, we construct a weighted version of measure (i) where words receive a weight that is proportional to the distance between the municipalities that share them, so that more distant contamination receives a higher weight. Fourth, we distinguish linguistic contamination by the distance between the municipalities and compile counts of (i) the number of words shared by municipalities within 50 Km, (ii) the number of words shared by municipalities between 50 and 100 Km, and (iii) the number of words shared by municipalities more distant than 100 Km.

A.7 Linking WW1 Death Records with the Ellis Island Database

We link World War 1 death records to the matched Ellis Island–Census database described in Section 3.1. Because we do not observe whether emigrants returned to Italy, we adopt a conservative approach that combines fuzzy string matching with information on individuals’ municipalities of origin and years of birth.

Consider an individual i in the World War 1 death records, with first name n_i , surname s_i , year of birth b_i , and with origin municipality coordinates (x_i, y_i) . Let the Ellis Island–Census database contain individuals j with standardized names (n_j, s_j) , birth year b_j , and origin municipality coordinates (x_j, y_j) . We begin by restricting attention to a set of candidate matches, retaining only individuals j in the Ellis Island–Census data whose year of birth falls within a two-year window of b_i , whose first

and last names share the same initial letters, and who immigrated before i 's recorded year of death. These restrictions are employed in the census-linking literature to allow for misreporting and reduce computational time (Abramitzky et al., 2021).

Within this candidate set, we implement fuzzy string matching to allow for transcription errors and minor spelling differences. For each pair (i, j) , we compute the Jaro–Winkler similarity between first names and surnames, denoted $JW(n_i, n_j)$ and $JW(s_i, s_j)$. We classify a pair as a potential match if both similarities exceed 0.95.

We further refine the set of matches using spatial proximity. For each pair (i, j) , we compute the great-circle distance d_{ij} between (x_i, y_i) and (x_j, y_j) . Starting from a radius of one kilometer around municipality m_i , we iteratively expand the search radius until at least one candidate satisfies the distance criterion. Let r^* denote the smallest radius such that $d_{ij} \leq r^*$ for at least one candidate. We retain all candidates with $d_{ij} \leq r^*$ and discard those located farther away.

This procedure yields, for each World War 1 record i , a set of linked Ellis Island–Census records that are close in birth year, highly similar in both first and last name, and geographically consistent with the reported municipality of origin. In the final sample, we discard all matches with $r^* > 50$ Km, although results are robust to alternative thresholds between 20 and 100 kilometers. Furthermore, we retain only single matches, as multiple matches may reflect spurious links due to common names and surnames.

The final sample comprises 16,847 U.S. emigrants who died while serving in the Italian army during World War 1, corresponding to roughly 3% of all Italian military deaths. Approximately 2.2% of individuals who migrated to the United States died during the war, compared to 1.9% in the overall population. This comparison, however, partly reflects compositional differences, as the emigrant population was disproportionately young and male.

A.8 Instrumental Variable: First-Stage Simulation Test

This appendix describes the logic and implementation of the ship-level co-location diagnostic used to assess the plausibility of the first-stage mechanism underlying the instrumental variable strategy.

The instrumental variable exploits variation in ship-level origin composition to generate quasi-exogenous differences in post-arrival exposure to other Italian origins. The underlying rationale is that interactions formed during the transatlantic voyage affected subsequent settlement decisions. Under this hypothesis, passengers traveling on the same ship should display excess residential co-location after arrival, relative to what would be expected based solely on aggregate destination sizes.

The diagnostic implemented here provides a test of this implication. It evaluates whether shipmates from different municipalities are disproportionately likely to settle in the same U.S. enumeration districts, after accounting for the distribution of district sizes and removing the mechanical contribution of the ship itself. This test applies the logic of Basse et al. (2024) to construct a finite-sample randomization test under a null of conditional exchangeability, in which post-arrival district assignments are independent across shipmates.

Consider a ship-year s with n_s passengers indexed by $j = 1, \dots, n_s$. Let $d(j)$ denote the enumer-

ation district in which passenger j is observed in the census, and let $m(j)$ denote the municipality of origin. For each district d , define n_{sd} as the number of passengers from ship s residing in district d , and n_{smd} as the number of passengers from ship s residing in district d and originating from municipality m .

We define the observed cross-origin co-location statistic as

$$T_s^X = \sum_d \left[\binom{n_{sd}}{2} - \sum_m \binom{n_{smd}}{2} \right], \quad (\text{A.2})$$

which counts the number of pairs of shipmates who settle in the same district but originate from different municipalities. By construction, this statistic excludes within-municipality pairs and therefore does not mechanically capture pre-existing origin-based networks. This follows the “leave-out” logic we employ in the instrument construction.

To construct a benchmark under the null of no ship-based settlement ties, we simulate counterfactual district assignments for the passengers of ship s . Let N_{dt} denote the total number of individuals observed in district d in census year t , and let n_{sdt} denote the contribution of ship s to district d . We define a leave-one-out district capacity vector

$$N_{dt}^{(-s)} = N_{dt} - n_{sdt}, \quad (\text{A.3})$$

which removes the mechanical contribution of ship s from the destination distribution. Under the null hypothesis, the n_s passengers of ship s are assigned to districts without replacement, with probabilities proportional to $N_{dt}^{(-s)}$. Municipality labels $m(j)$ are held fixed across simulations. For each simulated assignment, we recompute the statistic T_s^X , yielding a simulated distribution $\{T_{s,r}^{X,MC}\}_{r=1}^R$.

For each ship-year s , we compute the upper-tail Monte Carlo p -value

$$p_s^{MC} = \Pr \left(T_s^{X,MC} \geq T_s^X \right), \quad (\text{A.4})$$

as well as the associated standardized statistic based on the simulated mean and variance. A systematic excess of low p_s^{MC} values across ship-years indicates that shipmates from different municipalities co-locate more than would be expected based on district sizes alone.

Figure IIIb provides strong empirical support for the first-stage. The distribution of ship-level standardized co-location statistics exhibits excess right-tail mass relative to the null distribution implied by random assignment. In the pooled sample, the aggregate standardized statistic is large, reflecting pervasive excess co-location across ship-years, and the number of ships with p -values below conventional significance thresholds substantially exceeds what would be expected under the null. This strongly suggests that ship-level interactions shaped post-arrival settlement patterns, thereby validating the relevance of ship-based variation as a source of exogenous shifts in local exposure to Italian diversity.

A.9 Movers Design

We complement the analysis presented in Section 4 with a design along the lines of Chetty and Hendren (2018), which leverages individuals who move across neighborhoods (“movers”) to identify the

effect of exposure to diversity of the Italian community on national identity formation.

We employ the dataset described in Section 3.4.2, in which each individual is observed whenever they have a child. Consider individual i , who has a child at times t and t' . A mover is anyone who moves between two hexagons between t and t' .³ We denote the “diversity shock” i is exposed to upon moving as $D_i \equiv H_{d'(i)} - H_{d(i)}$, which is the difference in diversity between the new and the old neighborhood. We binarize D_i into a categorical variable, *Diversity Shock* _{i} , equal to one if D_i is in the top 20% of the distribution and zero otherwise, and denote t_i^* as the first year when i moves between two hexagons. Never-movers are included as the residual control group.

We estimate variations of the following specification:

$$\begin{aligned} LNI_{idht} = & \alpha_i + \alpha_{h \times t} + \sum_{\substack{k=-10 \\ k \neq -1}}^{+10} \beta_k \cdot I\{t - t_i^* = k\} \cdot \text{Diversity Shock}_i + \\ & + \sum_{\substack{k=-10 \\ k \neq -1}}^{+10} \gamma_k \cdot I\{t - t_i^* = k\} + \varepsilon_{idht}, \end{aligned} \quad (\text{A.5})$$

where α_i denote individual fixed effects and α_{ht} denote hexagon-by-time fixed effects. The treatment effect coefficients $\{\beta_k\}$ capture differences in the local name index between movers and non-movers before and after movers change their neighborhood.

Figure F.11 presents the results. Identification relies on a standard parallel trends assumption requiring that, conditional on fixed effects, treated and control individuals would not have experienced diverging LNI trajectories if the former had not moved to a different neighborhood. While this assumption is untestable, we find that pre-move coefficients $\{\beta_k\}_{k=-10}^{-1}$ are never statistically significantly different from zero, indicating that movers and non-movers were not systematically different before the former moved to a different neighborhood.

Over the decade following the move, we estimate that migrants who experienced a “diversity shock” gave their children names that were 10% less local compared to individuals who did not move or moved to neighborhoods with diversity comparable to their origin district. This effect builds over time and is largest 6 to 7 years after migrants move. Quantitatively, exposure to diverse neighborhoods upon moving decreases LNI by approximately 9% of a standard deviation.

In Table E.3, we report estimates from a mean-shift specification in which the event-time indicators are replaced by a categorical term $I\{t \geq t_i^*\}$ that equals one for all observations after the treatment period and zero otherwise. In the baseline specification, shown in column (1), we include individual, area, and event-time fixed effects. Individual fixed effects imply that identification relies on within-individual changes in LNI before and after treatment. In column (2), we add birth-year fixed effects to control for cohort-specific shocks that may affect trends in name choice and correlate with the local name index. In column (3), we replace area and cohort fixed effects with area-by-cohort fixed effects, absorbing neighborhood-level time-varying shocks such as changes in local labor market conditions or immigrant inflows. Column (4) replaces the binary diversity indicator with a continuous treatment

³We use hexagons rather than districts because we cannot consistently follow districts over time. The diversity of the local community, however, is measured at the district level, as in the rest of the analysis.

measure and confirms the qualitative results.⁴

In column (5), we add an interaction between the age at which migrants moved and the baseline treatment to test whether the effect of exposure to diversity is age-specific. We do not find evidence in this direction. The estimated difference-in-differences coefficient remains negative, although it is less precisely estimated, while the interaction term between the treatment and age at move is not statistically significant.

Last, in column (6), we replicate the analysis using the distance-adjusted LNI metric, which confirms the result obtained from the unweighted LNI indicator. Exposure to more diverse neighborhoods has a positive and statistically significant effect on the distance-adjusted local name index, indicating *less* name particularism, equal to approximately 15% of its standard deviation.

A.10 Definition of the Supply-Side Correction of Naming Choices

In Table E.9, we control for a “supply-side” expected local name index and distance-adjusted local name index based on the diversity of the destination enumeration district. The underlying idea is that part of the effect we estimate may reflect the fact that Italians who lived in diverse areas learned about new names that were not present in their areas of origin. It is worth noting that this learning process could be part of the mechanisms underlying our results, since their willingness to adopt these names would signify a shift from the local identity attached to their areas of origin. To understand the extent to which this process explains our results, we define two supply-side corrections, which quantify the expected LNI and DLNI if the names that immigrants gave to their children reflected the ones they encountered in their enumeration district of destination.

For a given district d in census year t , we observe a set of origin municipalities $\mathcal{M}_{dt} = \{m_1, \dots, m_k\}$. Let $s_{m dt} = N_{m dt} / \sum_{m' \in \mathcal{M}_{dt}} N_{m' dt}$ denote the share of Italians from origin m . Let L_i denote the LNI or DLNI associated with child i living in district d in year t . Let $\bar{L}_{m dt}$ denote the average L_i across children from origin m in the district-year. The expected LNI (or DLNI) for origin municipality m is defined as

$$E_{dt}^{-m} = \sum_{\substack{m' \in \mathcal{M}_{dt} \\ m' \neq m}} \frac{s_{m' dt}}{\sum_{q \neq m} s_{q dt}} \cdot \bar{L}_{m' dt} \quad (\text{A.6})$$

where $\bar{L}_{m' dt}$ is the average score of children from other municipalities of origin living in the same district. Then, the district-level expected LNI (or DLNI) for municipality is simply the average across origin municipalities:

$$E_{dt} = \sum_{m \in \mathcal{M}_{dt}} s_{m dt} \cdot E_{dt}^{-m}. \quad (\text{A.7})$$

The expected index (A.7) is equal to the LNI and the D-LNI that we would observe if immigrants gave their children names aligned with those they observed in the destination districts they lived in.

Table E.9 documents that while the learning component plays an important role in driving immigrants’ naming choices, diversity retains substantial explanatory power when (A.7) is included as a control, both when looking at the LNI (columns 1–4) as well as the DLNI (columns 5–8), and, similarly,

⁴Because non-movers do not experience a cross-sectional diversity shock, we apply a monotone increasing transformation to the continuous diversity measure (exponential) and impute zero exposure for non-movers.

both in the OLS specifications (columns 2 and 6) as well as in the instrumental variable regressions (columns 3–4 and 7–8).

A.11 Conley Standard Errors with High-Dimensional Fixed Effects

As a robustness check, we estimate standard errors that allow for spatial autocorrelation following Conley (1999). Although Conley standard errors are widely used in applied work, existing implementations, particularly in Stata, become computationally prohibitive in large samples and in the presence of high-dimensional fixed effects (HDFE, Colella et al., 2023). In our application ($N \approx 300,000$), this constraint is binding. We have developed a fast Python implementation of OLS and exactly identified 2SLS with multi-way HDFE and Conley-type spatial correlation, using blocked neighbor searches and just-in-time compilation to ensure memory safety and computational feasibility, to tackle this software limitation. The code is available upon request at davide.coluccia@bristol.ac.uk.

B Linked Sample of Ellis Island and Population Census Records

This section provides a detailed discussion of the primary data and the algorithm we developed to link the Ellis Island immigration directories and the federal census records. Then, we assess the linked sample’s reliability and representativeness and present key descriptive statistics.

B.1 The Ellis Island Immigration Data

The individual-level dataset on Italian immigrants in the United States is from Coluccia and Spadavecchia (2025). The dataset comprises 4,194,728 uniquely identified records of Italians who entered the United States through the Ellis Island immigration station between 1892 and 1924. The data contain information on the name and surname, age, immigration year, ship name, origin municipality, and literacy of each immigrant. We defer the interested reader to the original paper for a detailed methodological discussion of the data.

Origin municipalities are missing in 12% of cases. The remaining 88% are geo-coded using commercial software. Approximately 32% are geo-coded to coordinates outside Italy. These cases conflate reporting errors and migrants returning from temporary stays in Italy who had previously traveled and settled in the United States. Ultimately, our analysis sample thus comprises 2,526,024 individuals who are located at the coordinates of their municipality of origin in Italy.

B.2 Motivation and High-Level Description of the Approach

Linking the population census and the Ellis Island individual records shares several challenges faced by standard intergenerational census-linking algorithms (e.g., see Abramitzky et al., 2021). As is common in (historical) record linkage, frequent misspellings and transcription errors do not allow exact matching. Instead, automated algorithms implement “fuzzy” string matching methods that allow the linking of strings with minor discrepancies.

Our task, however, presents additional, more specific obstacles. First, the Ellis Island records were collected by American officers of the shipping companies. Hence, Italian names and surnames display frequent phonetic misspellings, translations, and abbreviations. Second, when completing their census enumeration, Italians commonly Americanized—i.e., translated—their names (e.g., see Biavaschi et al., 2017). Both circumstances are problematic for standard fuzzy string-matching algorithms because translations and phonetic errors frequently result in vastly different string records.

We design a linking algorithm to deal with these challenges. Our approach leverages recent advances in Natural Language Processing (NLP) and Large Language Models (LLM) to circumvent the algorithmic shortcomings of common string similarity metrics. More specifically, we capture the similarity between translated versions of the same name through a fine-tuned sentence bidirectional encoder representations from transformers (SBERT) model (Reimers and Gurevych, 2019). SBERT models produce string embeddings that encode the contextual representation of tokens. We complement the SBERT similarity metric with a standard Jaro-Winkler string similarity measure to identify spelling and phonetic mistakes.

We use the SBERT-Jaro-Winkler distances and the immigration year to pair each Ellis Island record with a set of plausible links in the U.S. Census. We label the resulting sample as “*Sample 1*.” We propose two methodologies to reduce the prevalence of “multiple matches,” i.e., Ellis Island records paired with more than one Census entry. First, in dataset “*Sample 2*,” we employ a state-of-the-art LLM to identify the most likely match(es) among those that appear in *Sample 1*. Second, in “*Sample 3*,” we take the match(es) that maximize an arithmetic average of the SBERT-Jaro-Winkler similarities within each match. Our analysis is based on *Sample 1*, but we show that all our results remain unchanged using the other samples. As is standard in the literature, we weight multiple matches by the inverse of the number of matches to avoid multiple counting.

B.3 Technical Description of the Algorithm Implementation

Figure F.1 provides a visual overview of the algorithm. We now describe each component in greater technical detail.

B.3.1 Construction of the Training Data

The starting point is the individual-level data contained in the Ellis Island ($N \approx 2,500,000$) and the Census ($N \approx 4,400,000$). We extract the universe of first names listed in these records and harness the flexibility of a state-of-the-art LLM (OpenAI’s gpt-4o-mini) to assign each name to an existing Italian name. This first step produces a set of approximately 40,000 names that appear in the original data and that the LLM pairs with their correct Italian counterpart. We split the sample into two equal-sized sub-samples. The following panel replicates the prompt that the LLM receives to classify name pairs.

gpt-4o-mini Prompt for Name Cleaning

You are an expert in reconstructing Italian **names** that were altered by U.S. immigration officials.

Task:

Return ONLY a valid JSON object with the most likely original Italian **first name** and a confidence score (-100 and 200).

Instructions:

- You must return an existing Italian first name.
- If you cannot find an Italian name, return the original and set the confidence to -100.
- If the name is already an Italian **real** first name, return as is and set the confidence to 200.

Rules for the confidence level:

- 200 = Name is a perfect match with no discrepancies.
- 90 = Extremely confident match (name clearly matches, considering translation/mis-spelling).
- 80 = Likely match but with minor discrepancies (plausible misspelling).
- 70 = Possible but uncertain match; significant discrepancies exist.

- 60 = Unlikely match; substantial doubts remain.
- 50 = Most likely match, but still very uncertain.
- 0 = Uncertain match; no clear match found.
- -100 = No Italian name found; return the original name.

Output Format:

```
{{
original: <original recorded name>,
inferred italian: <most likely Italian name>,
confidence: <confidence score>
}}
```

We restrict the attention to names with a confidence score of at least 80 to ensure that the training set comprises high-quality matches.

B.3.2 Training of the NLP Embeddings

Using the first subsample, we fine-tune a multilingual pre-trained SBERT model from the huggingface repository (paraphrase-multilingual-mpnet-base-v2). The training data contains both translations and minor spelling and phonetic mistakes. The embeddings produced by the fine-tuned SBERT model thus replicate the high contextual similarity between recorded names that map onto the same Italian name.

B.3.3 Filtering Classification Rule

Consider two first names, a and b . Name a appears in the Ellis Island data, and b is recorded in the Census. Let $\mathbf{e}_a$ and $\mathbf{e}_b$ denote the embeddings produced by the fine-tuned SBERT model. We say that b is a candidate match for a if the cosine similarity between the two embeddings is above a threshold $\tau_1 \in [0, 1]$:

$$\sigma_{ab}^{SBERT} \equiv \frac{\mathbf{e}_a^T \mathbf{e}_b}{\|\mathbf{e}_a\| \cdot \|\mathbf{e}_b\|}, \quad (\text{B.1})$$

or if the Jaro-Winkler similarity between a and b is above a certain threshold $\tau_2 \in [0, 1]$. The Jaro-Winkler similarity between strings a and b is based on the Jaro similarity,

$$\sigma_{ab}^J = \begin{cases} 0 & \text{if } m_{ab} = 0 \\ \frac{1}{3} \left(\frac{m_{ab}}{\|a\|} + \frac{m_{ab}}{\|b\|} + \frac{m_{ab} - t_{ab}}{m_{ab}} \right) & \text{otherwise,} \end{cases} \quad (\text{B.2})$$

where m_{ab} is the number of matching characters and t_{ab} is half the number of transpositions. The Jaro-Winkler similarity is then

$$\sigma_{ab}^{JW} = \sigma_{ab}^J + \ell_{ab} p (1 - \sigma_{ab}^J), \quad (\text{B.3})$$

where ℓ_{ab} is the length of the common prefix, typically capped at four characters, and p is a scaling parameter, conventionally set to 0.1. The number of transpositions is defined as the number of

matching characters that are not in the right order divided by two. The Jaro-Winkler similarity is a widely adopted edit metric in record linkage algorithms because it effectively captures minor typographical errors, transpositions, and abbreviations in short strings, and its prefix-weighted structure prioritizes matches that share the same initial characters, which are typically more stable. The score is normalized such that zero means no similarity, and one denotes exact matches.

We employ the SBERT and Jaro-Winkler similarities disjointly because they capture different instances of imperfect linking. A low cosine similarity between two SBERT embeddings reveals that the two names are likely translations of a single name. Hence, they should be considered as candidate matches. A low Jaro-Winkler similarity between two names indicates that the two names are imperfect transcriptions of the same name and, as such, should be considered candidate matches. Since these are different cases, we do not take an average “SBERT-Jaro-Winkler” similarity. Instead, we populate the set of possible matches as those b ’s that have *either* a “high” SBERT-cosine *or* a “high” Jaro-Winkler similarity with a

B.3.4 Estimation of the Classification Hyperparameters

To estimate the thresholds τ_1 and τ_2 , we employ the remaining half of the names matched to their proper Italian counterpart. Specifically, we split the 20,000 samples into a training set composed of 14,000 pairs and a test set of 6,000 names, over which we evaluate the performance of the composite distance classifier. We augment the train and test sets with a random sample of false matches drawn to reflect the frequency distribution of names in the population. Let $\sim$ denote an equivalence class whereby $a \sim b$ implies that names a and b constitute a match. Given the classification rule

$$a \sim b \Leftrightarrow \sigma_{ab}^{SBERT} \geq \tau_1 \vee \sigma_{ab}^{JW} \geq \tau_2, \quad (\text{B.4})$$

we search for the values $\hat{\tau}_1$ and $\hat{\tau}_2$ that maximize the $F1$ -score over the training set.⁵ The $F1$ -score is defined as the harmonic average between the precision and recall of the classifier. Precision is defined as the number of true positive results divided by the number of all samples predicted to be positive, including those not identified correctly. Recall is the number of true positive results divided by the number of all samples that should have been identified as positive. The $F1$ -score is a standard measure to target when fine-tuning classifiers, as it balances the tradeoff between accuracy and coverage.

In our application, the optimization routine selects $\hat{\tau}_1 = .65$ and $\hat{\tau}_2 = .95$. We provide additional details on the performance of the fine-tuned classifier in the next section.

B.3.5 Construction of Sample 1

The first sample comprises all matches obtained using the standard classification rule that satisfies the optimal thresholds. Formally, let B denote the universe of records in the Census. For each Ellis Island record a , the set of matches in *Sample 1* is denoted by

$$m_a^1 = \left\{ b \in B \text{ such that } |y_b - y_a| \leq 2 \wedge |t_b - t_a| \leq 5 \wedge a \sim b \wedge \tilde{\sigma}_{ab}^{JW} \geq \tau_3 \right\}, \quad (\text{B.5})$$

⁵Specifically, we run a grid search over the $[0, 1]^2$ space for τ_1 and τ_2 for evenly spaced values of the thresholds using a 0.05 pass.

where y_a and y_b denote the birth year recorded in the Ellis Island and the census registers, t_b and t_a denote the immigration year recorded in the census and in the Ellis Island registers, and the classification rule $\sim$ is computed using the couple optimal thresholds $(\hat{\tau}_1, \hat{\tau}_2) = (.65, .95)$ estimated in the previous step. The term $\tilde{\sigma}_{ab}^{JW}$ denotes the *surname* Jaro-Winkler similarity between records a and b . Since census enumerators do not translate surnames, it is not necessary to apply NLP tools on surnames because the Jaro-Winkler similarity alone ensures that we include misspellings in the set of potential matches. We select $\tau_3 = .95$ to ensure high-quality surname matches.

B.3.6 Construction of Sample 2

In *Sample 2*, we refine the set of matches obtained in the previous step using a state-of-the-art LLM (OpenAI’s gpt-4o-mini). We run the following prompt using the API.

gpt-4o-mini Prompt for Match Disambiguation

You’re an expert historian matching Italian immigrant records to U.S. census data.

Original immigrant details:

- Name: {original_name}

Census candidates:

- Candidates: {candidates}

Task

Carefully assess if any candidate matches the immigrant, considering:

- Italian-to-English translations (e.g., Giovanni $\leftrightarrow$ John, Giuseppe $\leftrightarrow$ Joseph).
- Phonetic errors, abbreviations, and misspellings from historical American records.

Prefer translations over phonetic matches.

Provide your best candidate selection clearly reflecting your certainty.

If there is one match that is clearly the best, only return that match.

If there is no clear match, return the most likely match(es) and set confidence to 6.

Rules for assigning confidence:

- 1 = Name is a perfect match with no discrepancies.
- 2 = Extremely confident match (name clearly matches, considering translation/misspelling).
- 3 = Likely match but with minor discrepancies (plausible misspelling).
- 4 = Possible but uncertain match; significant discrepancies exist.
- 5 = Unlikely match; substantial doubts remain.
- 6 = Most likely match, but still very uncertain.

Output Format:

```

{{
best matches: <list of candidate names>,
confidence: <confidence score>
}}
```

In the prompt, the set of candidate names that gpt-4o-mini is asked to disambiguate is given by, for the general record a , the set m_a^1 . Thus, if we let $\succsim$ denote the operator that maps potential matches into GPT-validated matches, i.e., matches with a confidence score of three or lower. The set of matches in *Sample 2* is then given by

$$m_a^2 = \left\{ b \in m_a^1 \text{ such that } b \succsim b' \ \forall \ b' \in m_a^1 \right\} \subseteq m_a^1, \quad (\text{B.6})$$

which simply implies that, in m_a^2 , all matches must be within m_a^1 , so, in this sense, *Sample 2* is a refinement of *Sample 1*, and they must be valid according to GPT.

B.3.7 Construction of Sample 3

The set *Sample 3* is an alternative refinement of *Sample 1*. Let $\hat{\sigma}_{ab}$ be the similarity metric that determines the name-level match. Thus, if $\sigma_{ab}^{SBERT} \geq \hat{\tau}_1$ and $\sigma_{ab}^{JW} < \hat{\tau}_2$, it shall be $\hat{\sigma}_{ab} = \sigma_{ab}^{SBERT}$, and the reverse if $\sigma_{ab}^{JW} \geq \hat{\tau}_2$ and $\sigma_{ab}^{SBERT} < \hat{\tau}_1$. In the residual case where both similarities are above the threshold, we pick $\hat{\sigma}_{ab} = \sigma_{ab}^{JW}$. Then, the set of matches included in *Sample 3* for the generic record a is

$$m_a^3 = \left\{ b \in m_a^1 \text{ such that } \alpha \cdot \hat{\sigma}_{ab} + (1 - \alpha) \cdot \tilde{\sigma}_{ab} \geq \alpha \cdot \hat{\sigma}_{ab'} + (1 - \alpha) \cdot \tilde{\sigma}_{ab'} \ \forall \ b' \in m_a^1 \right\} \subseteq m_a^1, \quad (\text{B.7})$$

where $\alpha \in (0, 1)$ is a weight that determines the importance assigned to the name relative to the surname. We select $\alpha = 0.5$ to assign equal weight to name and surname similarities. In this case, the refinement relative to *Sample 1* is given by taking the most similar match(es) in the set of plausible candidate matches m_a^1 , according to a weighted average of the name and surname similarities $\hat{\sigma}_{ab}$ and $\tilde{\sigma}_{ab}$.

B.4 Internal Validation and Representativeness of the Linked Sample

This section complements Section 3.1.

When constructing candidate matches for each Ellis Island record, we restrict the census population to individuals born within a ± 2 -year window around the reported birth year and who immigrated within a ± 5 -year window around the reported immigration year. Such buffers are standard in historical record linkage as they reflect reporting errors in birth—and plausibly immigration—years, partly due to low literacy and recall issues (e.g., Abramitzky et al., 2021).

Figure F.2 shows the difference in birth year (Figure F.2a) and immigration year (Figure F.2b) between linked census and Ellis Island records. Three patterns emerge. First, in both dimensions, the modal match occurs at zero gap, indicating that a relative majority of links involve identical reported years. Second, the distributions are symmetric around zero, ruling out systematic biases toward matching individuals reported as born or immigrated earlier or later in the census. Third, the linking

probability declines with the absolute gap between reported years. This pattern is mechanically expected in a well-behaved linkage procedure: if matches were driven by noise or by the blocking windows alone, the distribution of gaps would be approximately uniform within the admissible range, or even exhibit excess mass in the tails. The observed monotonic decline, therefore, indicates that the linkage algorithm assigns greater weight to records that agree more closely on blocking variables, and that spurious matches at the boundaries of the admissible windows are relatively rare.

An additional indicator of match quality is the similarity between names and surnames in the Ellis Island and census records. For each accepted link, the name similarity must exceed either 0.95 under the Jaro–Winkler metric or 0.65 under SBERT. Figure F.3a reports the distribution of Jaro–Winkler similarities, whereas Figure F.3b reports the distribution of SBERT similarities. There are very few matches with SBERT similarity below 0.65, while approximately 30% of the sample has Jaro–Winkler similarity below 0.95. This pattern is expected. The SBERT similarity is designed to capture semantic equivalence between name translations, which often appear orthographically distant and therefore score poorly under Jaro–Winkler. By contrast, high Jaro–Winkler matches generally also exhibit high SBERT similarity, indicating that the two distances are mutually consistent in identifying plausible matches. Importantly, a substantial share of the linked sample (approximately 40%) is constituted by nearly-perfect matches, i.e., matches with a name-similarity above 0.95. This pattern suggests that a large fraction of links is by near-exact agreement on names, rather than relying on the more permissive translation-aware metric.

Figure F.3c displays the distribution of surname similarities. By construction, accepted matches must have a Jaro–Winkler surname similarity above 0.95. We do not apply the SBERT correction to surnames because it was extremely rare for Italians to translate their surname (Carnevale, 2010). Interestingly, the majority of surnames ($\approx 60\%$) exhibit a near-perfect surname match, i.e., a surname similarity of one. This may reflect the fact that surnames were less likely than given names to be altered or Anglicized upon arrival, and census enumerators tended to record them more consistently across sources.

B.5 External Validation of the Linked Sample

In Figure F.4, we compare the baseline linked migrant sample (Sample 1 in Appendix Section B.3) with two refinements. In Sample 2, we use `gpt-4o-mini` to validate the matches produced by the baseline algorithm and retain only those deemed most plausible. In Sample 3, for each Ellis Island record with at least one accepted link, we retain only the match with the highest combined name and surname similarity. Both samples are refinements in the sense that they impose additional restrictions on Sample 1, and therefore are weakly smaller by construction.

In Figure F.4a, we plot the matching rate of Sample 2 (in red) and 3 (in blue) over time. The two series co-move and are qualitatively very similar to the pattern displayed in Figure Ia for Sample 1. The average matching rate of the baseline sample (18.762%) is also similar to Sample 2 (16.142%), indicating that GPT-4 deems 86% of the matches in Sample 1 as extremely plausible.

In Figure F.4b, we report the distribution of the number of matches in Sample 2 (red-contoured

bars) and 3 (gray lines). In both cases, as in Sample 1 (Figure 1b), single matches account for the vast majority of the accepted links in the sample.

Finally, in Figure F4c, we report the correlation between Sample 1 and Samples 2 (in red) and 3 (in blue). To construct this figure, we aggregate the matches at the municipality level; hence, each dot displays the municipality-level log-number of Ellis Island migrants from each municipality that are paired with at least one valid census record in Sample 1 (on the x -axis) and in Samples 2 and 3 (on the y -axis). By construction, both scatter plots lie below the 45-degree line because Samples 2 and 3 are refinements of Sample 1. In both cases, the correlation between the matching rates is large (0.87 for Samples 1–2 and 0.98 for Samples 1–3) and highly statistically significant, indicating that both refinements rescale the matching rate without modifying its spatial coverage.

Validating the linked dataset using external sources at the individual level is not feasible, as this is the first dataset to recover individual-level information on Italian immigrants' municipalities of origin. More aggregated bilateral data, such as municipality-to-county migration flows, are also unavailable. In Figure F5, we compare out-migration flows constructed from the Ellis Island records with official figures compiled by the Italian Commissioner General for Emigration and published in the *Annuario Statistico della Emigrazione Italiana dal 1876 al 1925*. As described in Coluccia and Spadavecchia (2025), these statistics report the annual number of emigrants to the United States by region, based on passports issued for transatlantic travel.

Figure F5a (resp. Figure F5c) reports the number of emigrants by year compiled using data from the *Annuario* (solid red line, on the left y -axis) and the Ellis Island full (resp. linked) dataset (dashed blue line, right y -axis). The Ellis Island dataset closely replicates the outflow constructed from Official Statistics data throughout the sample period. The number of emigrants in the Ellis Island dataset is lower than in the official statistics because the municipality of origin contains missing or invalid values, but, as explained by Coluccia and Spadavecchia (2025), this is unlikely to generate systematic selection into the dataset.

In Figure F5b, we report the correlation between the full Ellis Island dataset and data from the *Annuario*. To construct this figure, we collapsed the full Ellis Island dataset, i.e., irrespective of whether migrants are linked to the census, at the region-year level. Hence, each dot reports the number of U.S. emigrants in each year between 1892 and 1924 according to the official statistics (on the x -axis) and the Ellis Island dataset (on the y -axis). We find a positive and highly statistically significant correlation between the two datasets, as evidenced by the regression coefficient reported separately. In all but one case (Calabria, 1902), the difference between the Ellis Island and the official statistics is small relative to the number of emigrants

Figure F5d replicates the exercise using the linked Ellis Island dataset. Since, as we documented in Figure 1d, the linked sample preserved the spatial distribution of the full Ellis Island dataset, we expect to recover a positive correlation between the linked dataset and official statistics. The figure confirms this pattern: the regression coefficient remains highly statistically significant (though scaled by a factor approximately equal to the matching rate).

Together with the validation results documented in Section 3.1, these exercises indicate that (i) the

Ellis Island dataset replicates less granular data compiled from official sources, (ii) the linked Ellis Island dataset preserves the representativeness of the sample, and (iii) the reliability of the linking algorithm is confirmed by multiple internal validation refinements.

C Additional Historical Background

This appendix provides additional historical detail supporting the discussion in Section 2. It elaborates on three aspects only summarized in the main text: (i) the geographic and social organization of Italian migrant communities in the United States; (ii) the depth of linguistic and regional fragmentation within Italy at the time of mass migration; and (iii) specific mechanisms through which cultural convergence emerged within immigrant communities and fed back to Italy.

C.1 Geographic Concentration and Localism in Migrant Communities

Italian migration to the United States was shaped by strong local networks that generated highly localized settlement patterns. Migrants from specific towns or provinces frequently clustered together in destination cities, often following earlier migrants from the same locality (Vecoli, 1983). These patterns of chain migration produced sharp geographic concentration across and within U.S. cities.

Numerous examples illustrate this localization. Stonecutters from the marble-producing area of Massa settled in Barre, Vermont; migrants from Molfetta concentrated in Hoboken, New Jersey; and emigrants from Floridia, Sicily, in Hartford, Connecticut (Corradini, 1923). Residential concentration persisted even within neighborhoods. As Luigi Villari (1912, p. 216), the Italian consul in New York, observed,

"[...] we can find only Sicilians in a street, only people from Calabria in another street, and immigrants from Abruzzi in a third one. There are even streets where only individuals from a single town live: a colony from Sciacca here, a colony from San Giovanni in Fiore there, a colony from Cosenza somewhere else."

In East Harlem, New York City, even though most Italian migrants originated from the South, segregation by place of origin remained widespread. Neapolitans clustered between 106th and 108th Streets, while migrants from the nearby region of Basilicata settled primarily between 108th and 115th Streets (Orsi, 1985). In Chicago, segregation was similarly stark, as Italians settled across sixteen distinct neighborhoods according to their origins (Vecoli, 1983). Such patterns of segregation within Italian communities reflected the deep fragmentation that characterized Italy itself. Over time, however, these localized identities gradually gave way to broader forms of shared identity, as Italian communities coalesced into a more unified "Italian American" group.

These patterns mirrored the strong localism that characterized Italy itself and initially reproduced regional divisions within migrant communities. Interaction across groups nevertheless increased over time as migrants shared workplaces, institutions, and public spaces within American cities.

C.2 Cultural Fragmentation in Italy

C.2.1 Linguistic Fragmentation

At the time of unification, standard Italian was spoken by only a small minority of the population, estimated between 2.5 and 6.5 percent (De Mauro, 1963). Fewer than one percent of individuals outside

Tuscany and Rome spoke Italian as a first language. Italian functioned largely as a written language accessible to literate elites, and literacy rates were low nationwide (Ciccarelli and Weisdorf, 2019).⁶ Four decades later, a public inquiry reported that roughly 50 percent of the population had some knowledge of Italian, yet only 20 percent could use it for everyday communication (Corradini, 1910). The *legge Casati* (1861) introduced compulsory elementary schooling up to age eight (increased to nine with the *legge Coppino*, in 1877) in an effort to promote the “national language,” but implementation remained weak. In 1870, between 40 and 65 percent of school-age children did not attend school, and as late as 1906, half of children aged six to ten had received no formal schooling.

Despite state efforts at linguistic homogenization, local dialects remained widespread, even among educated elites. King Vittorio Emanuele II, for instance, conferred with his ministers in his native Piedmontese dialect. Biographers testify that, when, after unification, he confronted non-Piedmontese politicians, he felt uneasy and often unable to communicate effectively (Chabod, 1965). Camillo Benso, Count of Cavour, the architect of unification, preferred French to Italian and thought Sicilians spoke Arabic (Smith, 1985). Dialect use was common even in schools, where, in 1910, 50 percent of teachers reportedly conducted lessons in dialect rather than in Italian, despite formal requirements to the contrary (Corradini, 1910). On the eve of World War 1, linguistic fragmentation therefore persisted, particularly in rural areas, from which the majority of Italian migrants to the United States originated.

C.2.2 Geography

Geography reinforced fragmentation. The Apennine mountain chain limited interaction across regions, and travel across the peninsula remained slow prior to the expansion of the railway network (Gilmour, 2011). Economic divergence between the Center–North and the South further deepened regional cleavages. Quantitative evidence indicates that the ratio of Southern to Northern GDP per capita declined markedly between unification and the interwar period (Felice, 2019). These linguistic, economic, and geographic barriers shaped the identities migrants carried abroad.

Economic divergence reinforced cultural distance between the two areas, as illustrated by the civil conflict that followed unification. This conflict, fought largely by the Southern peasantry and commonly referred to as *brigantaggio*, at its peak required the mobilization of two-fifths of the regular army (Molfese, 1964; Dickie, 1999). Linguistic and educational disparities further deepened these divisions: school attendance remained low in the South, where the use of local dialects persisted well into the postwar period (De Mauro, 1963).

C.3 Cultural Convergence in Immigrant Communities

Historians of the Italian diaspora emphasize that Italy’s cultural fragmentation produced highly localized immigrant communities in the United States (Gabaccia, 2013). Over time, these parochial identities blended into an increasingly homogeneous “Italian American” identity, particularly in response

⁶Fewer than 1 percent of the population outside Tuscany and Rome spoke “Italian” as a first language. Italian functioned largely as a written language, accessible primarily to literate individuals. In 1861, the literacy rate averaged around 20 percent nationwide (and was closer to 10 percent in the South), placing a strict upper bound on the share of Italian speakers (Ciccarelli and Weisdorf, 2019).

to World War 1 (Vecoli, 1985; Luconi, 2003). Reflecting this shift, in 1934, the leading Italian-American newspaper, *Il Progresso Italo-Americano*, wrote:

"[...] there cannot be differences between a Lombard and a Roman, between a Venetian and a Sicilian, between a Piedmontese and a Neapolitan. The increasing difficulties in our lives require solidarity, not divisions."

To underscore the importance of migrant group identity and its influence on their country of origin, Choate (2008) refers to Italy as an "emigrant nation."

C.3.1 Associational Life

Fragmentation and subsequent homogenization are evident in the evolution of mutual aid societies, grassroots associations that provided financial and social support to immigrants. Initially, these associations were organized along narrow local and regional lines: only migrants from Abruzzo, for instance, could join the *Fratellanza Abruzzese* in Providence, while membership in the *Ateleta Beneficial Association* in Pittsburgh was restricted to migrants from the small village of Ateleta (Luconi, 2003). In East Harlem alone, Covello (1967) counted as many as sixty-four mutual aid societies with origin-based admission criteria. Beginning in the 1910s, however, these organizations increasingly federated across local origins (Gabaccia, 2013). In Milwaukee, for example, all associations merged into a single "Italian Societies of Milwaukee" confederation. The first national Italian association, the "Order of the Sons of Italy," was established in 1905, and, in the 1920s, its membership increased to 300,000 (Vecoli, 1985).

C.3.2 Linguistic Convergence

Language barriers among Italians from different parts of the country were substantial. In 1888, for example, the Irish archbishop of New York sought Italian clergy to minister to newly arrived immigrants, but the priests available to him, largely from Northern Italy, were unable to communicate with Southern Italians (Choate, 2008). As noted above, most Italian immigrants did not speak standard Italian upon arrival and instead developed a syncretic creole that blended regional dialect—predominantly Neapolitan—with English (Carnevale, 2010). This Italian-American creole became the primary medium of communication across regional lines, and its cultural significance was such that it survived even among returnees in Italy.

Linguistic homogenization was further reinforced by the Italian-American press, which predominantly published in standard Italian (Carnevale, 2010). Between 1900 and 1930, the Italian-language daily press ranked second only to German-American newspapers. As Deschamps (1996) argues, these newspapers promoted the use of Italian among migrants more than they did in Italy.⁷

⁷Newspapers were more effective in spreading Italian among migrants than among stayers primarily because migrants gained unprecedented access to basic education upon arriving in the U.S., thanks to public and denominational (Catholic) schools.

C.3.3 Religious Practices

Although virtually all Italian immigrants were Catholic, religious practices varied markedly across regions, and devotion to local saints remained strong. In St. Louis, for instance, Sicilians boycotted the Church of St. Ambrose, which had been erected by Lombards to honor a saint traditionally associated with Milan (Luconi, 2003).

Over time, however, the Catholic Church emerged as a force of homogenization. The Missionaries of St. Charles Borromeo, commonly known as the “Scalabrinians,” were founded to minister to Italian migrants abroad and explicitly sought to preserve and cultivate both Catholic practice and a shared Italian cultural identity (Choate, 2008).⁸

Convergence in religious practices also emerged endogenously within increasingly diverse immigrant communities. In his ethnography of East Harlem, Orsi (1985) documents that Italian immigrants initially venerated a wide array of local saints and patrons brought from their communities of origin. As boundaries between regional groups blurred, however, few survived. Among them, the Madonna of Mount Carmel was the most relevant one. Originally associated with immigrants from parts of Campania and Sicily, devotion to the Madonna gradually became a shared religious experience among community members, irrespective of their origins. This convergence did not result from clerical intervention but instead arose through repeated collective practice in the common urban space, especially during the annual *fiesta*.

C.3.4 Cuisine

A final dimension of cultural integration concerns cuisine (Chiaricati, 2020; Soffiati and Grandi, 2024). Although Italian cuisine is a defining component of Italian culture, recent work shows that many contemporary “Italian” foods emerged within immigrant communities in the United States. This process involved two steps: the diffusion of regional foods across immigrant groups and the invention of new culinary practices.

Pasta and pizza exemplify the first channel. Originally concentrated in Naples, where it was eaten by hand, pasta spread widely among Italian immigrants to the point that Fascist authorities considered it an American invention. Until the early 1900s, pizza remained a similarly marginal food confined to the slums of Naples and to Neapolitan emigrants. Its modern form emerged in the U.S., where immigrants introduced toppings and commercialized it through specialized shops. When American troops landed in Sicily in 1943, they were surprised by the absence of pizza shops.

The second channel involved invention. Tomato sauce entered Italian cookbooks only in the 1930s after becoming common among emigrants, while dishes such as meatballs and spaghetti Alfredo remain distinctive of Italian-American cuisine.

⁸As their founder, Giovanni Battista Scalabrini, put it, “Religion and fatherland [...] become intertwined. [...] May Italy, sincerely reconciled with the Apostolic See, emulate its ancient glories and add a new, undying glory, setting even its faraway children on the shining paths of true civilization and progress” (Scalabrini, 1887, p. 145–161). In doing so, Scalabrini explicitly advocated against the “melting pot” and immigrant assimilation: “Not many years ago, in the United States, there were enormous efforts to Americanize [...] Let us also work so that all Italians abroad possess the same motto, the same steadfastness, the same courage: for Religion and for the Fatherland” (Scalabrini, 1898, p. 39).

C.4 Return Migration and Feedback Effects to Italy

Did cultural convergence among Italian immigrants in the United States influence Italy? Qualitative historical evidence suggests that it did (Choate, 2008). Partly because of high return migration rates, values and practices acquired abroad plausibly diffused back to migrants' communities of origin. Return migrants founded mutual aid associations that transcended village boundaries and often evolved into political organizations. In the Apennine village of Montefegatesi, in 1902, returnees financed the construction of a monument to Giuseppe Garibaldi, a symbol of national unity (Wyman, 1993). Cultural feedback also operated through consumption practices: dishes developed within Italian-American communities later entered Italian culinary traditions, although their diasporic origins were frequently obscured (Chiaricati, 2020). More generally, qualitative studies indicate that Italy's identity as an "emigrant nation" developed partly through interactions among local identities in emigrant communities and the influence that return migrants exerted over their origin communities.

D A Model of Immigrant Group Identity Choice

We develop a parsimonious framework to rationalize how exposure to within-group diversity shapes identity formation. We derive two implications which we test in the empirical analysis.

A continuum of individuals resides in a local community. Each individual originates from a municipality m . The community is characterized by a measure of within-Italian diversity $D \in [0, 1]$, which increases with the heterogeneity of origin municipalities.

Agents allocate a unit of cultural investment across three identities, $j \in \{L, N, A\}$, where L , N , and A respectively denote local, Italian, and American identity. Let $x_j \geq 0$ denote investment in identity j , with $x_L + x_N + x_A = 1$.

Following Akerlof and Kranton (2000), identity yields utility if it is socially recognized. We let the effective return to identity j be $a_j(D) = v_j s_j(D) e^{-c_j}$, where $v_j > 0$ captures the intrinsic value of identity j , $s_j(D) \in [0, 1]$ denotes its social salience, and $c_j \geq 0$ captures the cost of adopting it. We assume $s_L(D) = 1 - D$, $s_N(D) = D$, and $s_A(D) = \bar{s}_A$. Thus, local identity becomes less useful as local origins fragment, while national identity becomes a more effective coordination device across immigrants from different municipalities. Exposure to natives is orthogonal to within-Italian diversity, so the salience of American identity is not affected by D . We assume $0 = c_L < c_N < c_A$, reflecting that local identity is inherited, national identity requires some adaptation across local cultural boundaries, and American identity entails greater linguistic and social effort.

Agents choose x_j to maximize $U = a_L(D) \log x_L + a_N(D) \log x_N + a_A(D) \log x_A$. This structure is analogous to coordination-based models in which the payoff to identity depends on group composition (e.g., Shayo, 2009). Let $S(D) \equiv v_L(1 - D) + v_N D e^{-c_N} + v_A \bar{s}_A e^{-c_A}$. The optimal investments are $x_L^* = v_L(1 - D)/S(D)$, $x_N^* = v_N D e^{-c_N}/S(D)$, and $x_A^* = v_A \bar{s}_A e^{-c_A}/S(D)$. Differentiating with respect to diversity gives $\partial x_L^*/\partial D = -v_L(v_N e^{-c_N} + v_A \bar{s}_A e^{-c_A})/S(D)^2 < 0$ and $\partial x_N^*/\partial D = v_N e^{-c_N}(v_L + v_A \bar{s}_A e^{-c_A})/S(D)^2 > 0$. Greater diversity, therefore, shifts cultural investment from local toward national Italian identity.

Proposition D.1. (*Local-to-National Identity*) *Higher within-Italian diversity reduces investment in local identities and increases investment in national Italian identity.*

Consider, next, investment in American identity. Americanization efforts in response to diversity are given by $\partial x_A^*/\partial D = -v_A \bar{s}_A e^{-c_A}(v_N e^{-c_N} - v_L)/S(D)^2$. We assume $v_N e^{-c_N} > v_L$. This condition has a simple interpretation. The total effective return to Italian identity is $a_L(D) + a_N(D) = v_L(1 - D) + v_N D e^{-c_N}$, so that $\partial[a_L(D) + a_N(D)]/\partial D = v_N e^{-c_N} - v_L$. Hence, $v_N e^{-c_N} > v_L$ implies that the increase in the return to national identity generated by diversity more than offsets the decline in the return to local identity. Diversity, therefore, raises the overall return to Italian identity. Since the return to American identity is unaffected by D , the following result follows.

Proposition D.2. (*Americanization*) *Higher within-Italian diversity decreases investment in American identity.*

The framework highlights identity as a coordination device whose value depends on community composition. High within-group diversity reduces the payoff to narrow local identities while increasing the appeal of broader national identities. Crucially, diversity reallocates identity *within* the immigrant group rather than toward the host society, generating national identity formation.

E Additional Tables

Table E.1. Comparison Between Linked and Non-Linked Migrants

	Linked	Non-Linked	Difference	
	(1)	(2)	(3)	(4)
Panel A. Comparison with Ellis Island Records				
Age at Migration	22.810	26.225	-3.416	(0.480)
Latitude of Origin Municipality	40.727	41.011	-0.284	(0.075)
Longitude of Origin Municipality	13.837	13.676	0.162	(0.046)
Area of Origin Municipality:				
North-West	0.082	0.090	-0.008	(0.004)
North-East	0.055	0.070	-0.015	(0.002)
Center	0.109	0.130	-0.021	(0.004)
South	0.484	0.467	0.018	(0.013)
Islands	0.270	0.243	0.027	(0.015)
Panel B. Comparison with Census Records				
Age at Migration	22.629	22.387	0.246	(0.351)
Is Male	0.801	0.596	0.205	(0.020)
Is Employed	0.513	0.449	0.063	(0.013)
log(Occupational Income)	3.184	3.157	0.026	(0.004)
Industry:				
Durable Manufacturing	0.119	0.115	0.004	(0.003)
Nondurable Manufacturing	0.179	0.161	0.018	(0.003)
Trade	0.184	0.169	0.014	(0.003)
Transportation	0.054	0.054	0.000	(0.002)
Agriculture	0.076	0.089	-0.013	(0.003)
Occupation:				
Professional	0.018	0.017	0.002	(0.001)
Managerial	0.073	0.065	0.008	(0.003)
Clerical	0.012	0.013	-0.001	(0.001)
Sales	0.034	0.035	-0.001	(0.001)
Skilled Manufacturing	0.281	0.248	0.033	(0.003)
Unskilled Manufacturing	0.372	0.398	-0.026	(0.006)
Census Region of Residence:				
North-East	0.593	0.585	0.008	(0.008)
Midwest	0.026	0.024	0.002	(0.002)
South	0.014	0.022	-0.008	(0.002)
West	0.061	0.057	0.004	(0.003)

Notes. This table compares migrants in the linked sample with those without a valid link. The unit of observation is an individual, observed once. Columns (1) and (2) report mean characteristics for linked and non-linked individuals. In Panel A, the non-linked group consists of all non-linked records in the Ellis Island dataset; in Panel B, it consists of all non-linked census records of Italian immigrants. Characteristics in Panel A are observed in the Ellis Island data, while those in Panel B are observed in the population census. Columns (3)–(4) report differences between linked and non-linked migrants, with standard errors in parentheses. Referenced on page: 9.

Table E.2. Validation of Local Name Index Against Contemporary Survey-Based Measures of Trust

	Trust in National Institutions								Generalized Trust		
	(1) First P.C.	(2) Army	(3) Parliament	(4) Head of State	(5) Judicial System	(6) Public Adm.	(7) Police	(8) European Union	(9) Church	(10) People	(11) Unions
Panel A: Local Name Index											
LNI	-0.371*** (0.135)	-0.120* (0.063)	-0.089* (0.053)	-0.037 (0.059)	-0.156*** (0.058)	-0.206** (0.082)	-0.183** (0.072)	-0.186** (0.073)	-0.078 (0.076)	-0.097 (0.087)	-0.062 (0.055)
Panel B: Distance-Adjusted Local Name Index											
Distance-Adjusted LNI	0.502*** (0.141)	0.148** (0.061)	0.122** (0.054)	0.083 (0.057)	0.163*** (0.059)	0.280*** (0.084)	0.190*** (0.071)	0.230*** (0.072)	0.108 (0.070)	0.128 (0.079)	0.120** (0.051)
Controls	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
# Municipalities	245	333	393	393	331	245	367	367	367	367	334
Mean Dep. Var.	0.018	0.009	0.003	0.026	0.011	0.002	0.004	0.010	0.016	0.010	0.010

Notes. This table reports the correlation between the local name index (Panel A) and the distance-adjusted local name index (Panel B) and various indicators of trust based on the ITANES survey data between 2001 and 2013. The sample is the cross-section of municipalities with at least five respondents across all waves. In column (1), the dependent variable is the first principal component of all variables displayed in columns (2–8). Each variable in columns (2–8) captures average trust in one institution on a scale between 1 (least trust) and 5 (highest trust). In columns (9–11), the dependent variables report trust in “placebo” categories: the Church, other people, and labor unions. Both the outcome variables and the independent variables are standardized to facilitate comparisons. In all regressions, we control for log-population in 1991 and for an indicator equal to one for municipalities located in the South. Referenced on pages: [13](#), [A.3](#).

***: $p < 0.01$, **: $p < 0.05$, *: $p < 0.10$

Table E.3. Movers Design and National Identity Formation

	Local Name Index (LNI)					Distance LNI
	(1)	(2)	(3)	(4)	(5)	(6)
Diversity Shock $\times$ Post-Move	-0.334** (0.139)	-0.341** (0.136)	-0.349** (0.147)		-1.386 (0.952)	0.078*** (0.024)
Diversity Shock (Cont.) $\times$ Post-Move				-16.599*** (5.392)		
Diversity Shock $\times$ Post-Move $\times$ Age					0.029 (0.025)	
Individual FE	Yes	Yes	Yes	Yes	Yes	Yes
Hexagon FE	Yes	Yes	–	–	–	–
Time Since Move FE	Yes	Yes	Yes	Yes	Yes	Yes
Cohort FE	No	Yes	–	–	–	–
Hexagon-Cohort FE	No	No	Yes	Yes	Yes	Yes
# Individuals	114,052	114,051	112,155	112,155	112,155	112,155
# Observations	410,343	410,340	402,640	402,640	402,640	402,640
Mean Dep. Var.	3.909	3.909	3.902	3.902	3.902	9.603

Notes. This table reports results from the movers design specification in (A.5). The unit of observation is an individual migrant observed in each year in which they have a child born in the United States between 1900 and 1930. The dependent variables are the local name index (columns 1–5) and the distance-adjusted local name index (column 6) defined in (1)–(2). The baseline treatment is the interaction between an indicator equal to one after an individual moves to a different hexagon (and zero otherwise) and an indicator equal to one if the difference in diversity between the Italian communities in the destination and origin districts lies in the top 20% of the distribution (and zero otherwise). Column (4) replaces this indicator with the continuous diversity difference measure. Column (5) additionally includes age and its interaction with the baseline treatment. All specifications include individual and event-time (time since move) fixed effects. Column (2) also includes cohort (year) and hexagon fixed effects, while columns (3–6) replace these with hexagon-by-cohort fixed effects. Standard errors, clustered at the district level, are reported in parentheses. Referenced on pages: 21, A.13.

***: $p < 0.01$, **: $p < 0.05$, *: $p < 0.10$

Table E.4. Minority Group Size, Italian Diversity, and Identity Formation

	Local Name Index				Foreign Name Index			
	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)
Italian Diversity	-2.181*** (0.640)	-4.416*** (1.201)	-5.973*** (1.910)	-2.200*** (0.650)	0.102*** (0.023)	0.194*** (0.051)	0.249*** (0.077)	0.103*** (0.024)
Italian Diversity $\times$ Share of Italians		0.050*** (0.013)	0.047*** (0.015)			-0.002*** (0.001)	-0.002*** (0.001)	
Italian Diversity $\times$ Share of Immigrants			0.027*** (0.009)				-0.001** (0.000)	
Italian Diversity $\times$ Share of Blacks				-0.005 (0.003)				0.000*** (0.000)
Municipality FE	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
County FE	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Year FE	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
# Individuals	209,033	209,033	209,033	209,033	210,152	210,152	210,152	210,152
# Observations	312,202	312,202	312,202	312,202	314,834	314,834	314,834	314,834
Mean Dep. Var.	3.988	3.988	3.988	3.988	0.758	0.758	0.758	0.758
K-P <i>F</i> -stat	25.597	14.391	5.681	12.365	22.148	11.342	4.492	10.749

Notes. This table reports the effect of exposure to diversity within the Italian community on name-based indices from specification (4) interacted with the size of the immigrant, Italian immigrant, and African-American communities. Columns (1–4) use the local name index defined in (1) as the dependent variable; Columns (5–8) use the foreign name index of Abramitzky et al. (2020). The unit of observation is an individual observed in the first census in which they appear and, when possible, linked across subsequent censuses between 1900 and 1930. Columns (1) and (5) report the baseline results; columns (2) and (6) include an interaction term between the baseline diversity measure and the share of Italian immigrants in the neighborhood; columns (3) and (7) further include an interaction between the baseline diversity measure and the share of immigrants in the neighborhood; in columns (4) and (8), we control for an interaction between the baseline diversity measure and the share of African Americans. All columns report 2SLS estimates. All specifications include municipality, county, and census-year fixed effects. Standard errors, clustered at the district level, are reported in parentheses. Referenced on page: 25.

***: $p < 0.01$, **: $p < 0.05$, *: $p < 0.10$

Table E.5. Exposure to Immigrant Heterogeneity and National Identity Formation, Excluding Firstborn Children

	(1)	(2)	(3)	(4)	(5)	(6)
Panel A: Local Name Index						
Italian Diversity	-0.240*** (0.054)	-0.213*** (0.052)	-0.164*** (0.052)	-0.224*** (0.053)	-0.092 (0.060)	-0.157** (0.072)
Mean Dep. Var.	10.697	10.697	10.710	10.698	10.697	10.697
Panel B: Distance-Adjusted Local Name Index						
Italian Diversity	0.528*** (0.090)	0.544*** (0.085)	0.614*** (0.080)	0.510*** (0.087)	0.464*** (0.096)	0.350*** (0.096)
Mean Dep. Var.	24.163	24.167	24.206	24.167	24.167	24.167
Panel C: Foreign Name Index						
Italian Diversity	0.065*** (0.007)	0.066*** (0.007)	0.073*** (0.007)	0.060*** (0.007)	0.055*** (0.008)	0.043*** (0.008)
Mean Dep. Var.	1.966	1.966	1.970	1.966	1.966	1.966
Municipality FE	Yes	Yes	–	Yes	Yes	Yes
County FE	Yes	–	–	–	–	–
Year FE	Yes	Yes	–	Yes	Yes	Yes
Hexagon FE	No	Yes	–	Yes	Yes	Yes
Municipality-Year FE	No	No	Yes	No	No	No
Hexagon-Year FE	No	No	Yes	No	No	No
Individual Controls	No	No	No	Yes	No	No
Demographic Controls	No	No	No	No	Yes	Yes
Economic Controls	No	No	No	No	No	Yes
# Individuals	207,673	207,499	205,832	207,484	207,499	207,499
# Observations	308,520	308,196	304,480	308,173	308,196	308,196

Notes. This table reports the effect of exposure to diversity within the Italian community on name-based indices from specification (4). Unlike in the main analysis, we exclude firstborn children when computing parents' name-based indices. Panel A uses the local name index defined in (1) as the dependent variable; Panel B uses the distance-adjusted local name index defined in (2); Panel C uses the foreign name index of Abramitzky et al. (2020). The unit of observation is an individual observed in the first census in which they appear and, when possible, linked across subsequent censuses between 1900 and 1930. All specifications include municipality and census-year fixed effects. Column (1) includes county fixed effects; columns (2) and (4–6) replace county with more granular hexagon fixed effects. Column (3) replaces municipality fixed effects with municipality-year fixed effects and hexagon fixed effects with hexagon-year fixed effects. Demographic controls (included in column 5) are measured at the district level and include population, the share of women, African Americans, immigrants, and Italian immigrants. Economic controls (included in column 6) are measured at the district level and include employment shares by industry. Standard errors, clustered at the district level, are reported in parentheses. Referenced on page: 26.

***: $p < 0.01$, **: $p < 0.05$, *: $p < 0.10$

Table E.6. Movers Design and National Identity Formation, Excluding Firstborn Children

	Local Name Index (LNI)					Distance LNI
	(1)	(2)	(3)	(4)	(5)	(6)
Diversity Shock $\times$ Post-Move	-0.439* (0.242)	-0.437* (0.243)	-0.518** (0.246)		0.362 (1.316)	0.067** (0.030)
Diversity Shock (Cont.) $\times$ Post-Move				-16.697** (7.808)		
Diversity Shock $\times$ Post-Move $\times$ Age					-0.026 (0.041)	
Individual FE	Yes	Yes	Yes	Yes	Yes	Yes
Hexagon FE	Yes	Yes	–	–	–	–
Time Since Move FE	Yes	Yes	Yes	Yes	Yes	Yes
Cohort FE	No	Yes	–	–	–	–
Hexagon-Cohort FE	No	No	Yes	Yes	Yes	Yes
# Individuals	72,360	72,360	70,792	70,792	70,792	70,792
# Observations	235,384	235,383	229,667	229,667	229,667	229,667
Mean Dep. Var.	3.913	3.913	3.905	3.905	3.905	9.603

Notes. This table reports results from the movers design specification in (A.5). Unlike in the main analysis, we exclude firstborn children from the estimation sample. The unit of observation is an individual migrant observed in each year in which they have a child born in the United States between 1900 and 1930. The dependent variables are the local name index (columns 1–5) and the distance-adjusted local name index (column 6) defined in (1)–(2). The baseline treatment is the interaction between an indicator equal to one after an individual moves to a different hexagon (and zero otherwise) and an indicator equal to one if the difference in diversity between the Italian communities in the destination and origin districts lies in the top 20% of the distribution (and zero otherwise). Column (4) replaces this indicator with the continuous diversity difference measure. Column (5) additionally includes age, which is absorbed by the fixed effects, and its interaction with the baseline treatment. All specifications include individual and event-time (time since move) fixed effects. Column (2) also includes cohort (year) and hexagon fixed effects, while columns (3–6) replace these with hexagon-by-cohort fixed effects. Standard errors, clustered at the district level, are reported in parentheses. Referenced on page: 26.

***: $p < 0.01$, **: $p < 0.05$, *: $p < 0.10$

Table E.7. Exposure to Immigrant Heterogeneity and National Identity Formation, Variants of Local Name Index and Diversity Definitions

	Italian Diversity Measured by:					
	Municipality -		Province -		Region -	
	(1) District	(2) County	(3) District	(4) County	(5) District	(6) County
Panel A: Municipality-Level Local Name Index						
Italian Diversity	-0.164*** (0.015)	-0.126** (0.051)	-0.192*** (0.022)	-0.043 (0.110)	-0.200*** (0.029)	0.116 (0.152)
Mean Dep. Var.	3.989	3.989	3.989	3.989	3.989	3.989
Panel B: Province-Level Local Name Index						
Italian Diversity	-0.105*** (0.012)	-0.067 (0.042)	-0.129*** (0.018)	-0.064 (0.096)	-0.152*** (0.025)	-0.045 (0.137)
Mean Dep. Var.	4.804	4.804	4.804	4.804	4.804	4.804
Municipality FE	Yes	Yes	Yes	Yes	Yes	Yes
Year FE	Yes	Yes	Yes	Yes	Yes	Yes
Hexagon FE	Yes	Yes	Yes	Yes	Yes	Yes
# Individuals	223,531	223,531	223,531	223,531	223,531	223,531
# Observations	334,303	334,303	334,303	334,303	334,303	334,303

Notes. This table reports the effect of exposure to diversity within the Italian community on name-based indices from specification (4). Panel A uses the local name index defined in (1) as the dependent variable; Panel B uses a variant where the name distribution is taken at the province-level instead of the municipality-level. Each column reports estimates from a variant of equation (3) that differs only in the geographic level at which we construct H . Column (1) uses the baseline municipality–district definition. Column (2) aggregates districts to the county level (municipality–county). Columns (3–4) redefine origins at the province level (province–district and province–county, respectively). Columns (5–6) analogously use regions as origins (region–district and region–county, respectively). Provinces are coarser than municipalities; Regions are coarser than provinces. Counties are coarser than districts. The unit of observation is an individual observed in the first census in which they appear and, when possible, linked across subsequent censuses between 1900 and 1930. All specifications include municipality, census-year fixed effects, and hexagon fixed effects. Standard errors, clustered at the district level, are reported in parentheses. Referenced on pages: 26, 27.

***: $p < 0.01$, **: $p < 0.05$, *: $p < 0.10$

Table E.8. Movers Design and National Identity Formation, Variants of Local Name Index and Diversity Definitions

	Italian Diversity Measured by:					
	Municipality -		Province -		Region -	
	(1) District	(2) County	(3) District	(4) County	(5) District	(6) County
Panel A: Municipality-Level Local Name Index						
Diversity Shock $\times$ Post-Move	-0.335** (0.137)	-0.234 (0.169)	-0.423*** (0.140)	-0.236 (0.176)	-0.430*** (0.133)	-0.237* (0.134)
Mean Dep. Var.	3.909	3.909	3.909	3.909	3.909	3.909
Panel B: Province-Level Local Name Index						
Diversity Shock $\times$ Post-Move	-0.348*** (0.124)	-0.123 (0.159)	-0.340*** (0.123)	-0.095 (0.167)	-0.348*** (0.122)	-0.104 (0.124)
Mean Dep. Var.	4.678	4.678	4.678	4.678	4.678	4.678
Individual FE	Yes	Yes	Yes	Yes	Yes	Yes
Time Since Move FE	Yes	Yes	Yes	Yes	Yes	Yes
# Individuals	114,052	114,052	114,052	114,052	114,052	114,052
# Observations	410,343	410,343	410,343	410,343	410,343	410,343

Notes. This table reports results from the movers design specification in (A.5). The unit of observation is an individual migrant observed in each year in which they have a child born in the United States between 1900 and 1930. Panel A uses the local name index defined in (1) as the dependent variable; Panel B uses a variant where the name distribution is taken at the province-level instead of the municipality-level. The treatment is the interaction between an indicator equal to one after an individual moves to a different hexagon (and zero otherwise) and an indicator equal to one if the difference in diversity between the Italian communities in the destination and origin districts lies in the top 20% of the distribution (and zero otherwise). Each column reports estimates from a variant of equation (3) that differs only in the geographic level at which we construct H . Column (1) uses the baseline municipality–district definition. Column (2) aggregates districts to the county level (municipality–county). Columns (3–4) redefine origins at the province level (province–district and province–county, respectively). Columns (5–6) analogously use regions as origins (region–district and region–county, respectively). Provinces are coarser than municipalities; Regions are coarser than provinces. Counties are coarser than districts. All specifications include individual, hexagon-by-cohort, and event-time (time since move) fixed effects. Standard errors, clustered at the district level, are reported in parentheses. Referenced on pages: 26, 27.

***: $p < 0.01$, **: $p < 0.05$, *: $p < 0.10$

Table E.9. Exposure to Immigrant Heterogeneity and National Identity Formation: Name Supply-Side Adjustment

	Local Name Index				Distance-Adjusted Local Name Index			
	(1) Baseline	(2) OLS	(3) Reduced Form	(4) 2SLS	(5) Baseline	(6) OLS	(7) Reduced Form	(8) 2SLS
Italian Diversity	-0.205*** (0.018)	-0.069*** (0.008)		-2.622** (1.161)	0.023*** (0.003)	0.010*** (0.001)		0.279* (0.169)
Ship Diversity			-0.035*** (0.009)				0.004** (0.002)	
Expected LNI		0.871*** (0.016)	0.877*** (0.017)	0.699*** (0.098)				
Expected D-LNI						1.153*** (0.019)	1.153*** (0.021)	1.055*** (0.077)
Municipality FE	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Hexagon FE	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Year FE	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
# Individuals	209,390	209,390	195,886	195,886	209,390	209,390	195,886	195,886
# Observations	304,898	304,898	284,988	284,988	304,898	304,898	284,988	284,988
Mean Dep. Var.	3.962	3.962	3.960	3.960	9.596	9.596	9.597	9.597
K-P <i>F</i> -stat				10.998				11.700

Notes. This table reports the effect of exposure to diversity within the Italian community on name-based indices from specification (4). Columns (1–4) employ the local name index defined in (1) as the dependent variable; Columns (5–8) use the distance-adjusted local name index defined in (2). Columns (1) and (5) report the baseline results; columns (2) and (6) include the expected LNI and DLNI defined in Appendix Section A.10 as a control; columns (3) and (7) report reduced-form instrumental variable estimates; columns (4) and (8) report two-stage least squares estimates. The main treatment of interest is Italian diversity, as defined in (3), and the instrument is defined in (5). The unit of observation is an individual observed in the first census in which they appear and, when possible, linked across subsequent censuses between 1900 and 1930. All specifications include municipality, census-year fixed effects, and hexagon fixed effects. Standard errors, clustered at the district level, are reported in parentheses. Referenced on pages: 27, A.14.

***: $p < 0.01$, **: $p < 0.05$, *: $p < 0.10$

Table E.10. Exposure to Immigrant Heterogeneity and National Identity Formation, Fractionalization Index

	(1)	(2)	(3)	(4)	(5)	(6)
Panel A: Local Name Index						
Italian Diversity (Fractionalization)	-0.549*** (0.071)	-0.523*** (0.078)	-0.474*** (0.085)	-0.495*** (0.073)	-0.093* (0.055)	-0.108* (0.061)
Mean Dep. Var.	3.990	3.989	3.989	3.989	3.989	3.989
Panel B: Distance-Adjusted Local Name Index						
Italian Diversity (Fractionalization)	0.073*** (0.011)	0.072*** (0.012)	0.064*** (0.013)	0.065*** (0.011)	0.024** (0.010)	0.021* (0.011)
Mean Dep. Var.	9.593	9.593	9.593	9.593	9.593	9.593
Panel C: Foreign Name Index						
Italian Diversity (Fractionalization)	0.080*** (0.004)	0.078*** (0.005)	0.078*** (0.005)	0.068*** (0.004)	0.043*** (0.004)	0.038*** (0.004)
Mean Dep. Var.	0.758	0.758	0.758	0.758	0.758	0.758
Municipality FE	Yes	Yes	–	Yes	Yes	Yes
County FE	Yes	–	–	–	–	–
Year FE	Yes	Yes	–	Yes	Yes	Yes
Hexagon FE	No	Yes	–	Yes	Yes	Yes
Municipality-Year FE	No	No	Yes	No	No	No
Hexagon-Year FE	No	No	Yes	No	No	No
Individual Controls	No	No	No	Yes	No	No
Demographic Controls	No	No	No	No	Yes	Yes
Economic Controls	No	No	No	No	No	Yes
# Individuals	223,721	223,531	221,913	223,521	223,531	223,531
# Observations	334,668	334,303	330,615	334,283	334,303	334,303

Notes. This table reports the effect of exposure to diversity within the Italian community on name-based indices from specification (4). Unlike in the main analysis, we use the fractionalization index ($= 1 - \text{Hirschman-Herfindahl Index}$) to measure the diversity of the Italian immigrant community. Panel A uses the local name index defined in (1) as the dependent variable; Panel B uses the distance-adjusted local name index defined in (2); Panel C uses the foreign name index of Abramitzky et al. (2020). The unit of observation is an individual observed in the first census in which they appear and, when possible, linked across subsequent censuses between 1900 and 1930. All specifications include municipality and census-year fixed effects. Column (1) includes county fixed effects; columns (2) and (4–6) replace county with more granular hexagon fixed effects. Column (3) replaces municipality fixed effects with municipality-year fixed effects and hexagon fixed effects with hexagon-year fixed effects. Demographic controls (included in column 5) are measured at the district level and include population, the share of women, African Americans, immigrants, and Italian immigrants. Economic controls (included in column 6) are measured at the district level and include employment shares by industry. Standard errors, clustered at the district level, are reported in parentheses. Referenced on page: 27.

***: $p < 0.01$, **: $p < 0.05$, *: $p < 0.10$

Table E.11. Movers Design and National Identity Formation, Fractionalization Index

	Local Name Index (LNI)					Distance LNI
	(1)	(2)	(3)	(4)	(5)	(6)
Diversity Shock $\times$ Post-Move	-0.443*** (0.147)	-0.441*** (0.146)	-0.579*** (0.155)		-0.939 (0.895)	0.087*** (0.022)
Diversity Shock (Cont.) $\times$ Post-Move				-0.491** (0.219)		
Age					-0.000 (0.013)	
Diversity Shock $\times$ Post-Move $\times$ Age					0.010 (0.024)	
Individual FE	Yes	Yes	Yes	Yes	Yes	Yes
Hexagon FE	Yes	Yes	–	–	–	–
Time Since Move FE	Yes	Yes	Yes	Yes	Yes	Yes
Cohort FE	No	Yes	–	–	–	–
Hexagon-Cohort FE	No	No	Yes	Yes	Yes	Yes
# Individuals	114,085	114,084	112,155	112,155	112,155	112,155
# Observations	410,478	410,475	402,640	402,640	402,640	402,640
Mean Dep. Var.	3.909	3.909	3.902	3.902	3.902	9.603

Notes. This table reports results from the movers design specification in (A.5). Unlike in the main analysis, we use the fractionalization index ($= 1 - \text{Hirschman-Herfindahl Index}$) to measure the diversity of the Italian immigrant community. The unit of observation is an individual migrant observed in each year in which they have a child born in the United States between 1900 and 1930. The dependent variables are the local name index (columns 1–5) and the distance-adjusted local name index (column 6) defined in (1)–(2). The baseline treatment is the interaction between an indicator equal to one after an individual moves to a different hexagon (and zero otherwise) and an indicator equal to one if the difference in diversity between the Italian communities in the destination and origin districts lies in the top 20% of the distribution (and zero otherwise). Column (4) replaces this indicator with the continuous diversity difference measure. Column (5) additionally includes age, which is absorbed by the fixed effects, and its interaction with the baseline treatment. All specifications include individual and event-time (time since move) fixed effects. Column (2) also includes cohort (year) and hexagon fixed effects, while columns (3–6) replace these with hexagon-by-cohort fixed effects. Standard errors, clustered at the district level, are reported in parentheses. Referenced on page: 27.

***: $p < 0.01$, **: $p < 0.05$, *: $p < 0.10$

Table E.12. Name Particularism in Italy by Exposure to Immigrant Diversity, Alternative Mass Migration Threshold

	Threshold of Emigration Rate for Treatment:					
	1 ‰		2 ‰		5 ‰	
	(1) LNI	(2) D-LNI	(3) LNI	(4) D-LNI	(5) LNI	(6) D-LNI
Diversity Exposure × Post Mass Migration	-0.464*** (0.091)	0.045*** (0.010)	-0.634*** (0.125)	0.062*** (0.014)	-0.824*** (0.177)	0.077*** (0.020)
Post Mass Migration	0.010 (0.025)	-0.001 (0.003)	-0.008 (0.029)	0.000 (0.003)	-0.060 (0.038)	0.006 (0.004)
Municipality FE	Yes	Yes	Yes	Yes	Yes	Yes
Year FE	Yes	Yes	Yes	Yes	Yes	Yes
# Municipalities	7,823	7,823	7,823	7,823	7,823	7,823
# Observations	200,774	200,774	200,774	200,774	200,774	200,774
Mean Dep. Var.	0.879	9.909	0.879	9.909	0.879	9.909

Notes. This table reports the effect of exposure to diversity within the Italian migrant community on name-based indices in Italian municipalities from regression (7). The unit of observation is a municipality observed at a yearly frequency between 1880 and 1930. The dependent variables are the local name index defined in (1) in columns (1), (3), and (5), and the distance-adjusted local name index defined in (2) in columns (2), (4), and (6). The baseline treatment is defined as the share of a municipality's U.S. emigrants who settled in districts in the top quintile of the diversity distribution (3) interacted with a post indicator equals one in years at or after the municipality-specific onset of mass migration (defined as the first year with at least one (columns 1–2), two (columns 3–4), or five (columns 5–6) emigrant per 1,000 inhabitants). The table reports the coefficient of interaction terms between the diversity exposure measure and the post indicator. All regressions include municipality and year fixed effects. Standard errors are clustered at the municipality level and reported in parentheses. Referenced on page: 28.

***: $p < 0.01$, **: $p < 0.05$, *: $p < 0.10$

Table E.13. Monuments by Designation in Italy

	National Monuments					Other Monuments				
	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)	(9)	(10)
Diversity Exposure $\times I(1870 \leq t < 1880)$	-0.002 (0.007)	0.004 (0.004)	0.000 (0.005)	-0.003 (0.007)	-0.001 (0.008)	0.055 (0.034)	0.074* (0.041)	0.066 (0.041)	0.046 (0.035)	0.036 (0.033)
Diversity Exposure $\times I(1880 \leq t < 1890)$	-0.005 (0.010)	0.005 (0.007)	0.002 (0.008)	-0.010 (0.010)	-0.011 (0.011)	0.040 (0.042)	0.092** (0.046)	0.087* (0.046)	0.047 (0.042)	0.034 (0.043)
Diversity Exposure $\times I(1890 \leq t < 1900)$	-0.004 (0.007)	0.002 (0.004)	-0.000 (0.005)	-0.004 (0.007)	-0.004 (0.007)	0.083* (0.043)	0.108** (0.051)	0.090* (0.051)	0.083* (0.043)	0.068 (0.045)
Diversity Exposure $\times I(1900 \leq t < 1910)$	-0.012 (0.009)	-0.016 (0.010)	-0.016 (0.011)	-0.014* (0.009)	-0.014 (0.009)	0.019 (0.045)	0.090* (0.053)	0.064 (0.053)	0.013 (0.046)	-0.041 (0.042)
Diversity Exposure $\times I(1910 \leq t < 1920)$	0.082** (0.038)	0.019 (0.045)	0.011 (0.047)	0.075** (0.038)	0.094** (0.045)	0.053 (0.040)	0.112** (0.047)	0.100** (0.047)	0.048 (0.041)	0.037 (0.043)
Diversity Exposure $\times I(1920 \leq t < 1930)$	0.139** (0.066)	0.111 (0.072)	0.076 (0.070)	0.136** (0.066)	0.148** (0.075)	-0.038 (0.041)	0.040 (0.048)	0.031 (0.048)	-0.040 (0.041)	-0.067 (0.042)
Diversity Exposure $\times I(1930 \leq t < 1940)$	-0.023 (0.028)	-0.034 (0.028)	-0.039 (0.028)	-0.035 (0.028)	-0.039 (0.033)	-0.032 (0.027)	0.015 (0.028)	0.006 (0.028)	-0.043 (0.028)	-0.048 (0.031)
Diversity Exposure $\times I(1940 \leq t < 1950)$	0.019 (0.021)	0.024 (0.028)	0.001 (0.028)	0.015 (0.022)	0.022 (0.024)	-0.035 (0.024)	-0.012 (0.027)	-0.016 (0.026)	-0.033 (0.024)	-0.042 (0.029)
Diversity Exposure $\times I(1950 \leq t < 1960)$	0.122*** (0.032)	0.109*** (0.036)	0.104*** (0.035)	0.107*** (0.033)	0.131*** (0.037)	-0.043 (0.031)	-0.031 (0.033)	-0.033 (0.033)	-0.037 (0.032)	-0.046 (0.036)
Diversity Exposure $\times I(1960 \leq t < 1970)$	-0.015 (0.023)	-0.004 (0.025)	-0.013 (0.025)	-0.011 (0.024)	-0.004 (0.026)	-0.002 (0.023)	0.004 (0.027)	0.010 (0.027)	0.013 (0.023)	0.013 (0.027)
Diversity Exposure $\times I(1970 \leq t < 1980)$	-0.027 (0.036)	0.000 (0.039)	-0.014 (0.038)	-0.024 (0.036)	-0.021 (0.033)	-0.028 (0.021)	-0.034 (0.021)	-0.028 (0.021)	-0.014 (0.022)	-0.019 (0.023)
Diversity Exposure $\times I(1980 \leq t < 1990)$	-0.010 (0.022)	-0.020 (0.024)	-0.030 (0.024)	-0.009 (0.022)	-0.006 (0.025)	0.023 (0.024)	-0.008 (0.025)	-0.004 (0.024)	0.036 (0.025)	0.023 (0.028)
Municipality FE	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Decade FE	Yes	–	–	Yes	Yes	Yes	–	–	Yes	Yes
Region-Decade FE	No	Yes	–	No	No	No	Yes	–	No	No
Province-Decade FE	No	No	Yes	No	No	No	No	Yes	No	No
US Emigrants $\times$ Decade FE	No	No	No	Yes	Yes	No	No	No	Yes	Yes
Controls $\times$ Decade FE	No	No	No	No	Yes	No	No	No	No	Yes
# Municipalities	2,549	2,549	2,549	2,549	2,399	2,549	2,549	2,549	2,549	2,399
# Observations	33,137	33,137	33,137	33,137	31,187	33,137	33,137	33,137	33,137	31,187
Mean Dep. Var.	0.043	0.043	0.043	0.043	0.043	0.041	0.041	0.041	0.041	0.042

Notes. This table reports dynamic estimates obtained from (9) of the effect of exposure to diversity within the Italian migrant community on monument construction in Southern Italy. The unit of observation is a municipality-decade for municipalities observed between 1861 and 2000. In columns (1–5), the dependent variable is an indicator equal to one if the municipality has at least one national-identity monument in a given decade. In columns (6–10), the dependent variable is an indicator equal to one if the municipality has at least one non-national-identity monument in a given decade. The baseline treatment is an interaction term between decade indicators and the share of U.S. emigrants from each municipality who settled in districts in the top quintile of the diversity measure defined in (3). In columns (1), (4), (5), (6), (9), and (10), the specification includes municipality fixed effects and decade fixed effects; in columns (2) and (7), we include region-decade fixed effects; in columns (3) and (8), we add province-decade fixed effects; in columns (4) and (9), we control for the number of U.S. emigrants interacted with decade dummies; in columns (5) and (10), we further control for the population in 1881 and in 1991, the number of WW1 deaths, and an indicator for municipalities on the coast, interacted with decade dummies. Standard errors are clustered at the municipality level and are reported in parentheses. Referenced on page: 35.

***: $p < 0.01$, **: $p < 0.05$, *: $p < 0.10$

Table E.14. List of Americanisms

Americanism (1)	Variants (2)	Source (3)	Municipalities (4)
arinò	aranò; airanò	I don't know	2
assicurazione	—	insurance	1
azzò	ezzò	that's all	8
bbeccausu	bbaccausu	backhouse	4
bbisinissi	bbisinissi; bbusinissi	business	12
bbossu	bbuossu	boss	12
cottu	—	coat	5
farma	—	farm	2
fatturia	—	factory	6
frènzà	frìnza; friènza	fence	11
gassulinu	—	gasoline	1
ggiòbba	gghiòbba; ggiubba	job	25
ghella	chella	girl	9
grussaria	—	grocery	1
mascina	maccina; macina	machine	14

Notes. This table lists words that appear in the *Vocabolario Siciliano* and have English origins. The list is produced by GPT 5.5. Column (1) reports the word indicated by GPT 5.5; column (2) reports variants that appear in the *Vocabolario*; column (3) indicates the source; column (4) reports the number of municipalities where each word appears. Referenced on pages: [37](#), [A.10](#).

***: $p < 0.01$, **: $p < 0.05$, *: $p < 0.10$

Table E.15. Dialect Contamination and Exposure to Immigrant Diversity, Additional Measures

	(log) Number of Words Spoken in Other Municipalities		(log) Number of Words Spoken in Municipalities Within:			(log) Number of Words Spoken in ... Other Municipalities		
	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)
	Unweighted	Distance Weighted	50 Km	50–100 Km	100+ Km	2–4	5–9	10+
High Diversity Exposure	0.614*** (0.163)	0.635*** (0.168)	0.684*** (0.160)	0.688*** (0.163)	0.576*** (0.163)	0.666*** (0.175)	0.632*** (0.163)	0.512*** (0.151)
US Emigration (Cont.)	-0.216 (0.133)	-0.221 (0.137)	-0.170 (0.121)	-0.185 (0.127)	-0.206 (0.134)	-0.241* (0.142)	-0.218 (0.141)	-0.168 (0.116)
Demographic Controls	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Geographic Controls	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Province FE	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
# Municipalities	378	378	378	378	378	378	378	378
Mean Dep. Var.	4.622	4.355	4.398	4.288	4.286	3.342	3.293	3.850

Notes. This table reports the cross-sectional association between dialect contamination and emigration to diverse emigrant communities in the United States. The unit of observation is a municipality. The sample comprises the universe of Sicilian municipalities. The dependent variables are: in column (1), the baseline measure (the number of words mentioned in a municipality that appear in other municipalities), in column (2), its distance-weighted analog where words share by more distant municipalities receive a larger weight; in columns (3) to (5), the number of words shared by municipalities within 50 Km, within 50 to 100 Km, and more distant than 100 Km. All dependent variables are expressed as zero-inflated log counts. The main explanatory variable is an indicator equal to one in municipalities with an above-median number of US emigrants who settled in districts in the top quintile of the diversity distribution (3). Each regression includes the number of U.S. emigrants, population in 1901, geographic controls—elevation, an indicator for municipalities on the coast, latitude, and longitude—and province fixed effects. Standard errors are clustered at the municipality level and reported in parentheses. Referenced on page: 38.

***: $p < 0.01$, **: $p < 0.05$, *: $p < 0.10$

Table E.16. Dialect Contamination and Exposure to Immigrant Diversity, Falsification

	Number of Recorded Words	(log) Number of Recorded Words	(log) Number of Local-Only Words
	(1)	(2)	(3)
Diversity Exposure (Cont.)	-1416.168 (1566.079)	-1.697 (2.970)	-1.877 (3.292)
US Emigration (Cont.)	248.656 (350.700)	0.295 (0.660)	0.336 (0.739)
Demographic Controls	Yes	Yes	Yes
Geographic Controls	Yes	Yes	Yes
Province FE	Yes	Yes	Yes
# Municipalities	378	378	378
Mean Dep. Var.	319.249	4.781	2.989

Notes. This table reports falsification exercises for the cross-sectional association between dialect contamination and emigration to diverse emigrant communities in the United States. The unit of observation is a municipality. The sample comprises the universe of Sicilian municipalities. The dependent variables are the total number of words recorded in the *Vocabolario Siciliano* (column 1), its zero-inflated logged value (column 2), and the zero-inflated log number of words that appear exclusively in the municipality (column 3). The main explanatory variables are the number of US emigrants who settled in districts in the top quintile of the diversity distribution (3) and the number of US emigrants. Each regression includes population in 1901, geographic controls—elevation, an indicator for municipalities on the coast, latitude, and longitude—and province fixed effects. Standard errors are clustered at the municipality level and reported in parentheses. Referenced on page: 39.

***: $p < 0.01$, **: $p < 0.05$, *: $p < 0.10$

Table E.17. Political Participation of First- and Second-Generation Emigrants**(a) Second-Generation Emigrants**

	Foreign-Born Population			Italian-Born Population			Over-Representation	
	(1) Number of Officers	(2) Population	(3) Share (%)	(4) Number of Officers	(5) Population	(6) Share (%)	(7) Officer rate ratio	(8) Bootstrap 90% C.I.
Foreign-born	179	210,937	0.085	14,429	56,345,974	0.026	3.311	[2.918, 3.705]
US-born	29	18,546	0.156	14,429	56,345,974	0.026	6.096	[4.203, 7.983]

(b) First-Generation Emigrants

	Number of Officers		Attrition Correction		Over-Representation	
	(1) Italian	(2) US Emigrants	(3) Correction factor	(4) Corrected Share	(5) Officer rate ratio	(6) Bootstrap 90% C.I.
Observed matched sample	14,562	51	1.00	0.344	0.127	[0.270, 0.428]
Assumed Attrition:						
95%	14,562	51	34.42	11.856	4.386	[9.306, 14.730]
90%	14,562	51	17.21	5.928	2.193	[4.653, 7.365]
85%	14,562	51	11.47	3.952	1.462	[3.102, 4.910]

Notes. Table E.17a compares second-generation emigrants, measured as U.S.-born individuals observed among local officers in Italy, with both the foreign-born and the Italian-born population in Italy. For each group, the table reports the number of officers, the corresponding population count, and the implied officer share. The last columns report over-representation, defined as the ratio between the officer's share of the group of interest and the officer's share of Italian-born individuals, with 90% confidence intervals obtained by bootstrap. Bootstrapped confidence intervals are constructed by repeatedly resampling the observed data with replacement, recomputing the statistic of interest in each resample, and taking the central 90 percent of the resulting distribution of estimates. Table E.17b compares the presence of first-generation emigrants, identified as local officers matched to U.S. emigrant records, with the presence of non-emigrant Italian-born officers in the same sample. Columns (1) and (2) report the numbers of Italian officers and matched U.S. emigrants, respectively. Columns (3) and (4) apply alternative assumptions about match attrition to recover the implied share of U.S. emigrants among local officers. Columns (5) and (6) report the corresponding over-representation of emigrants in political office, measured relative to the officer rate of Italians, together with 90% confidence intervals obtained by bootstrap. Referenced on page: 39.

Table E.18. U.S. Emigrants' Participation in WW1 and Diversity Exposure

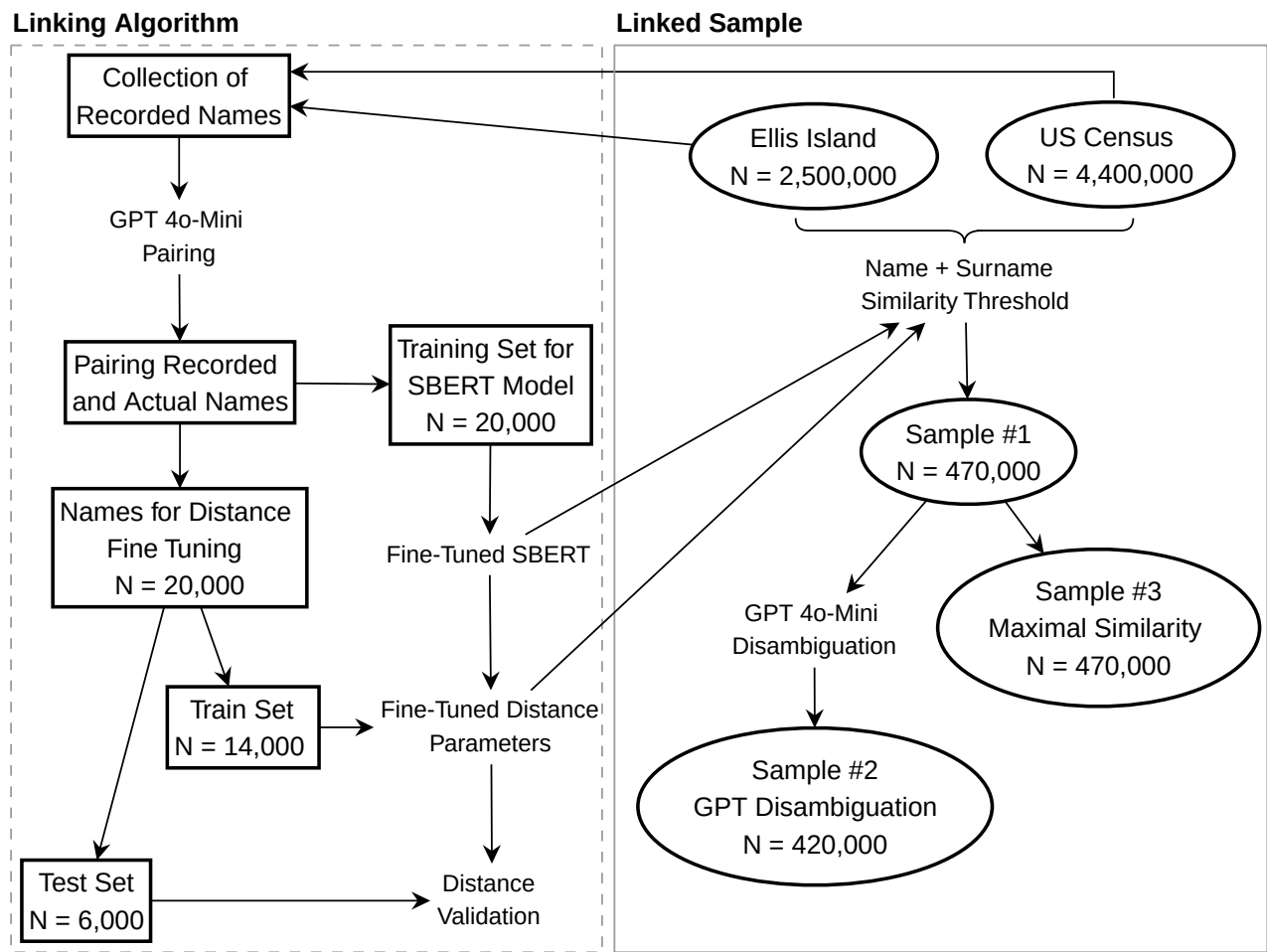
	Death Rate Among US Emigrants (%)				Overall Death Rate (%)
	(1)	(2)	(3)	(4)	(5)
Diversity Exposure	4.527*** (0.576)	4.046*** (0.579)	4.019*** (0.584)	3.979*** (0.591)	0.127 (0.098)
US Emigrants (1,000s)			0.186 (0.179)	-0.287 (0.205)	-0.100 (0.064)
Region FE	Yes	–	–	–	–
Province FE	No	Yes	Yes	Yes	Yes
Controls	No	No	No	Yes	Yes
# Municipalities	8,100	8,100	8,100	8,023	8,023
Mean Dep. Var.	1.926	1.926	1.926	1.940	1.979

Notes. This table reports the correlation between the death rate during WW1 of U.S. emigrants (columns 1–4), the overall WW1 death rate (column 5), and exposure to diversity, as in (10). The unit of observation is a municipality. Each municipality is observed once; the outcome variable is measured after WW1; the control variables are measured before 1914. The dependent variables are the share of U.S. emigrants who died during WW1 (columns 1–4), and the WW1 death rate, computed as the ratio between the number of deaths and population in 1881. The baseline treatment is defined as the share of a municipality's U.S. emigrants who settled in districts in the top quintile of the diversity distribution (3). In columns (4–5), we further control for the number of U.S. emigrants, in thousands. In column (1), we include region fixed effects; in columns (2–5), we include province fixed effects. Columns (4–5) further include municipality-level controls: population in 1881, area, and altitude. Standard errors are clustered at the municipality level and reported in parentheses. Referenced on page: 40.

***: $p < 0.01$, **: $p < 0.05$, *: $p < 0.10$

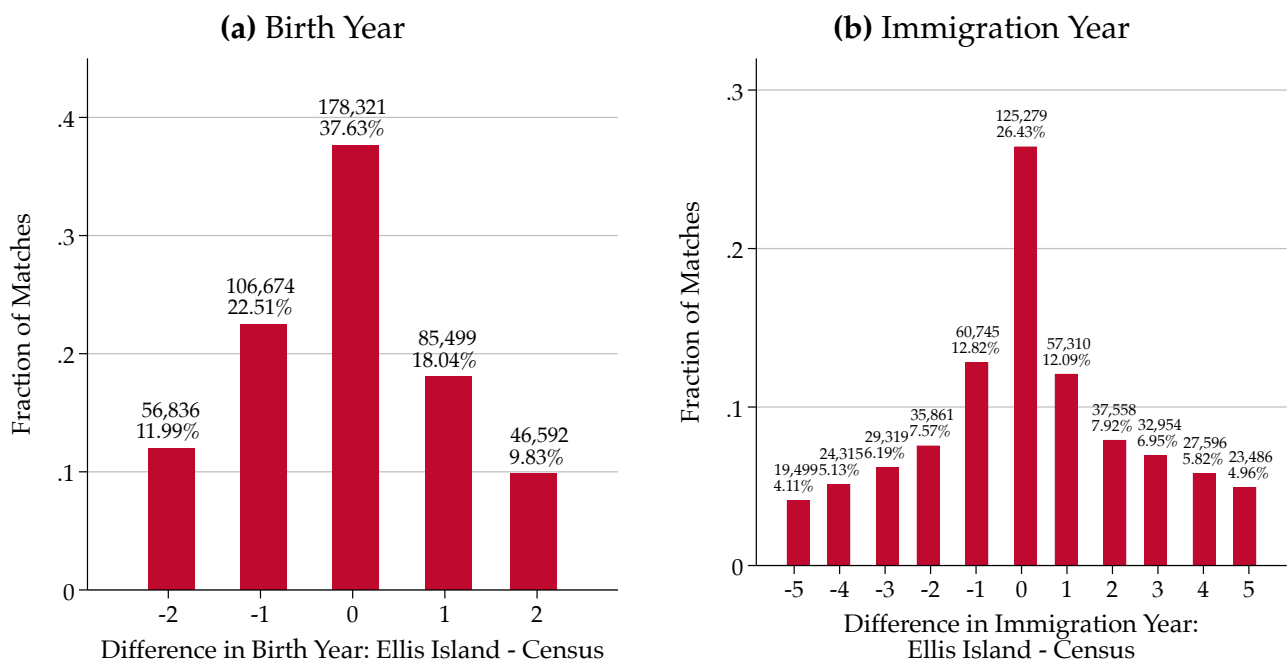
F Additional Figures

Figure F.1. Visual Representation of the Linking Algorithm



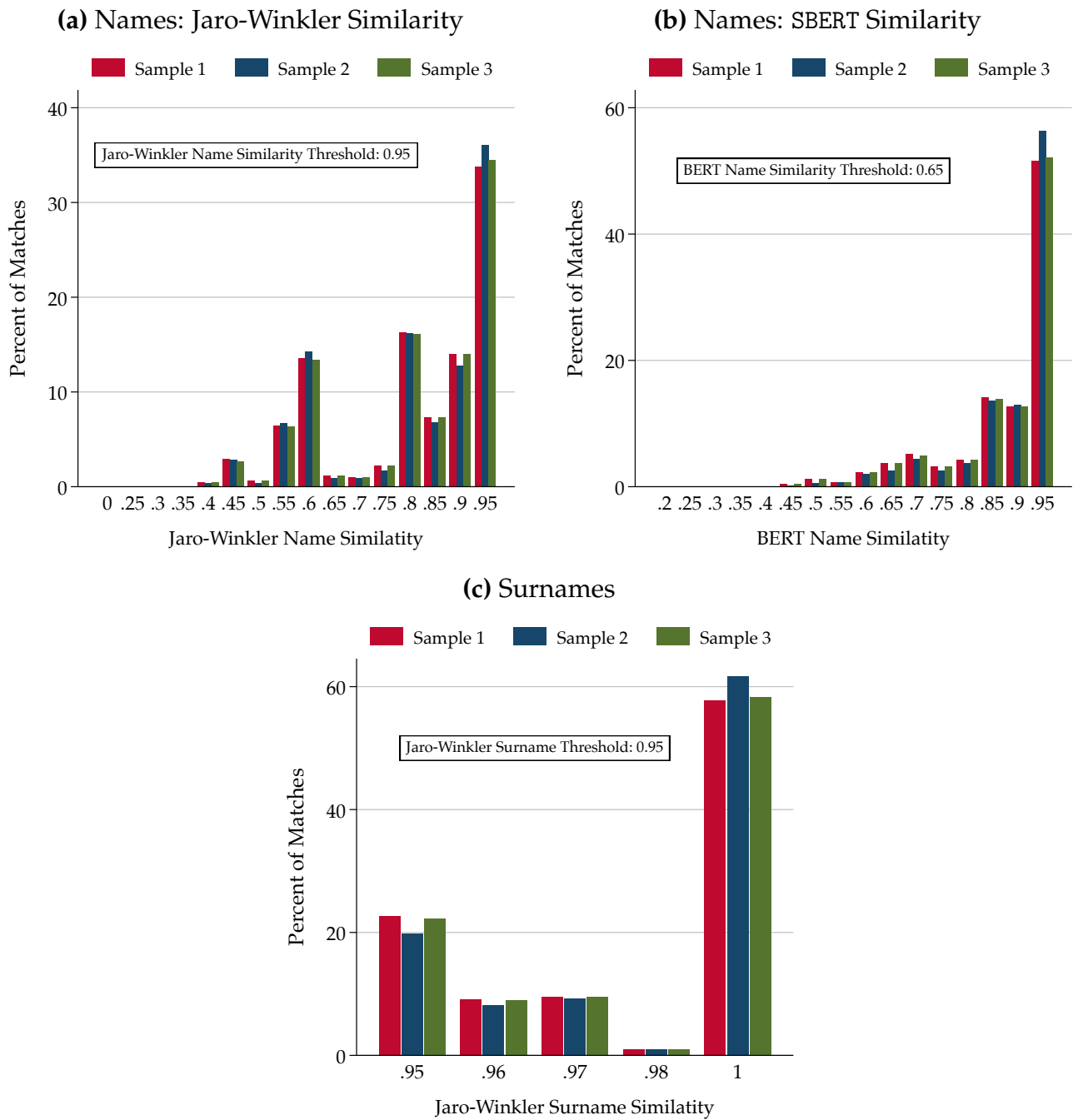
Notes. This Figure provides a graphical overview of the linking algorithm. The round boxes depict the starting data (Ellis Island and U.S. Census) and the final linked sample output. The square boxes denote the dataset we constructed from the original datasets to implement the training and validation of the SBERT and Jaro-Winkler name similarities. The text not embedded in squares or circles denotes logical operations applied to the source dataset to reach the target output. Sample sizes are approximate. Referenced on pages: 8, A.17.

Figure F.2. Linked Sample: Blocking Variables Discrepancy



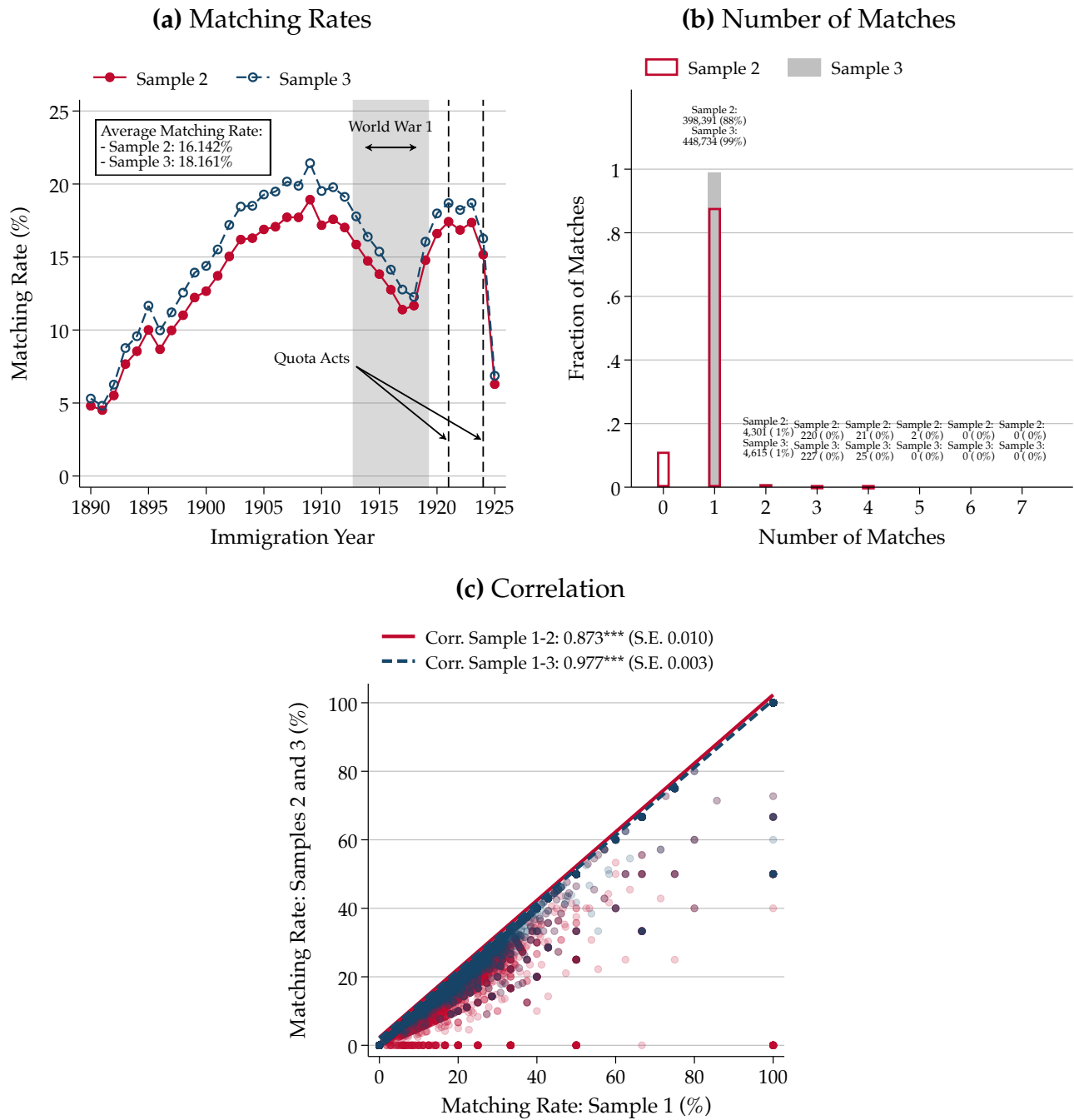
Notes. This figure displays differences between birth year (Panel F.2a) and immigration year (Panel F.2b) as recorded in the census and the Ellis Island datasets. To account for misreporting in the linkage procedure, matches are allowed to differ by up to two years in birth year and up to five years in immigration year. Each bar reports the number of matches in each difference bin and the corresponding share of the total sample. Referenced on page: A.21.

Figure F.3. Linked Sample: Match Quality



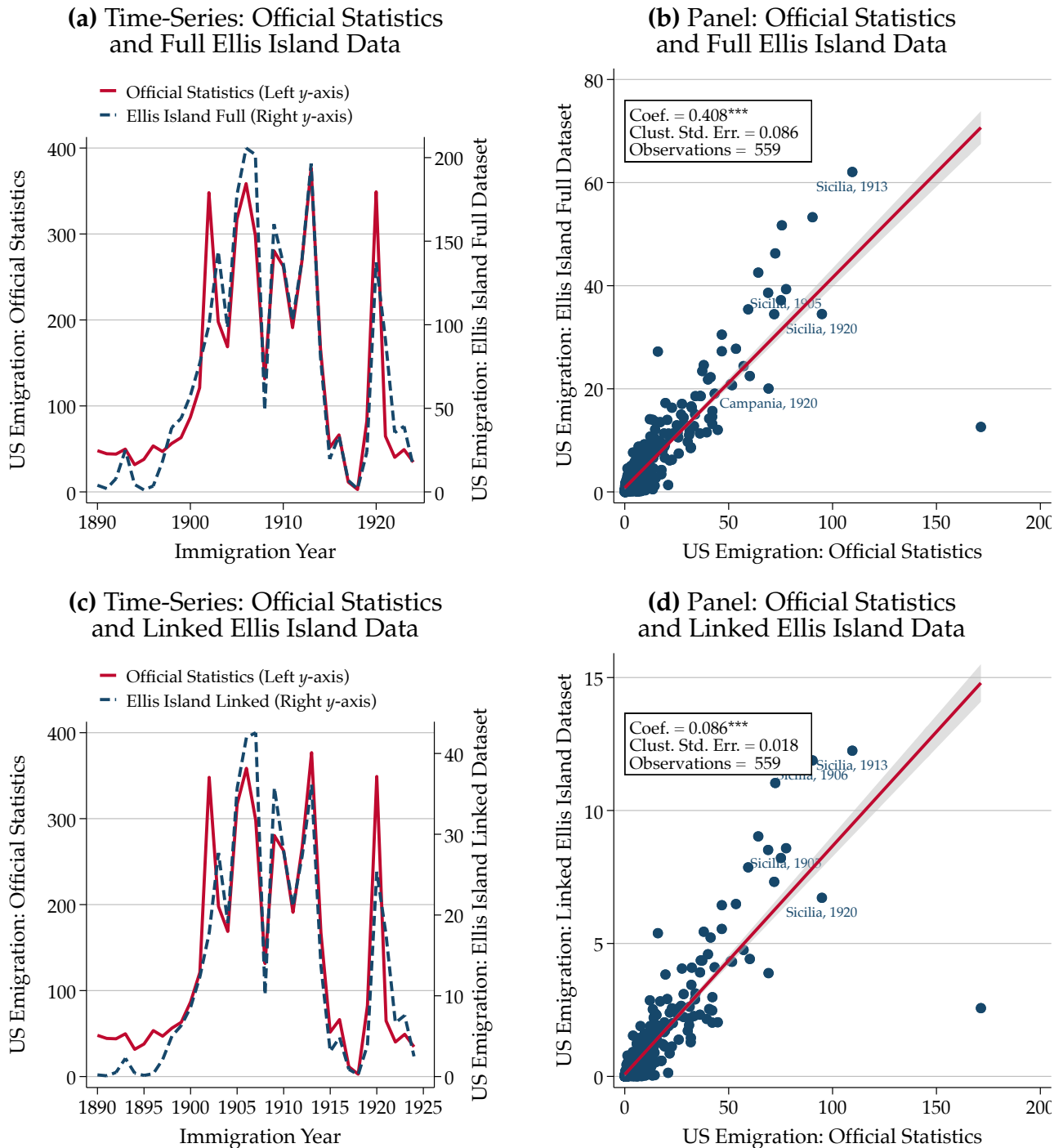
Notes. This figure reports the quality of matches produced by the linking algorithm. Red bars correspond to the baseline linked sample (Sample 1), while blue and green bars correspond to refinements 2 and 3. Panel F.3a shows the distribution of Jaro–Winkler similarities between first names recorded in the census and in the Ellis Island data; matches below the Jaro–Winkler threshold (0.95) satisfy the SBERT similarity criterion. Panel F.3b reports the distribution of SBERT similarities; matches below the SBERT threshold (0.65) satisfy the Jaro–Winkler rule. Panel F.3c reports the distribution of Jaro–Winkler similarities for last names; all matches exceed the Jaro–Winkler threshold (0.95). The number of matches is expressed in thousands. Referenced on page: A.22.

Figure F.4. Linked Sample: Comparisons Between Samples



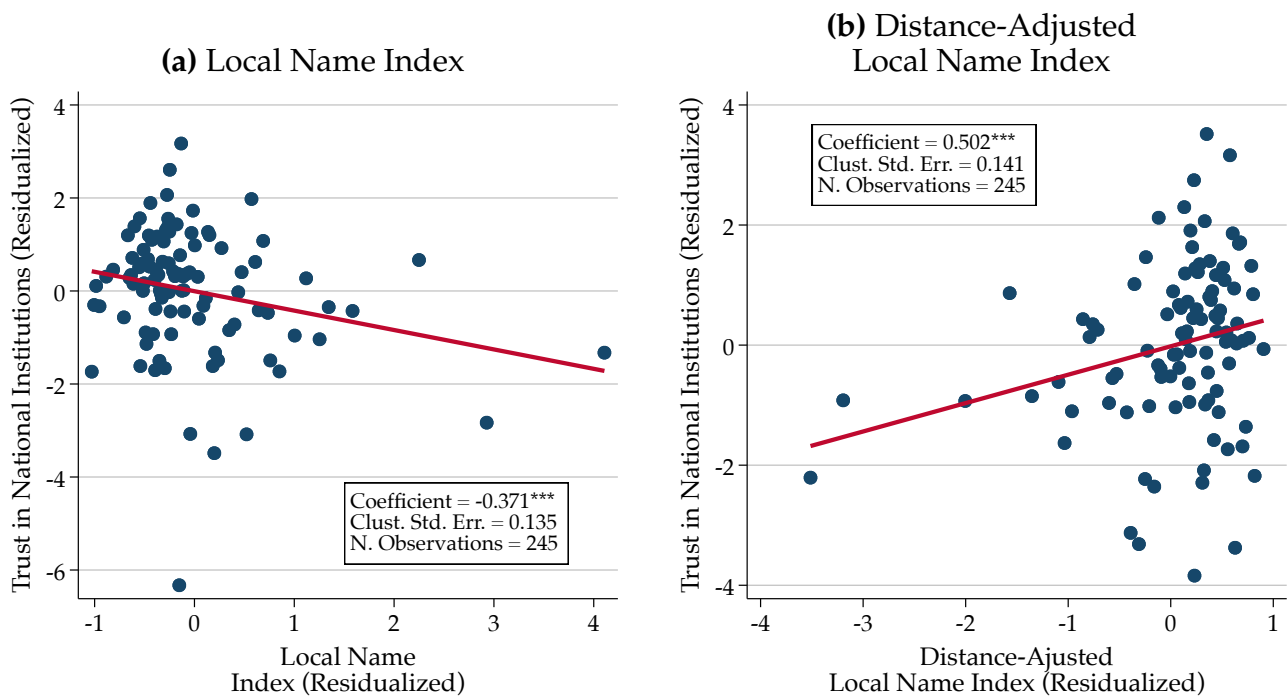
Notes. This figure compares the baseline linked sample with two refinements described in Appendix Section B.3. Red dots correspond to the refinement based on GPT 4-mini ("Sample 2"), and blue dots to the maximal-similarity refinement ("Sample 3"). Panel F.4a reports matching rates over time and on average. Panel F.4b shows the distribution of the number of census matches per Ellis Island record and the corresponding shares of the total sample. Panel F.4c plots, at the municipality level, the correlation between the (log) number of migrants in the baseline sample ("Sample 1," x-axis) and in Samples 2 and 3 (y-axis). Reported values give the correlation and its standard error. The solid red and dashed blue lines report the fitted values from a linear relationship between the two series in the two samples. Referenced on pages: 11, A.22.

Figure F.5. Linked Sample: Validation with Official Statistics



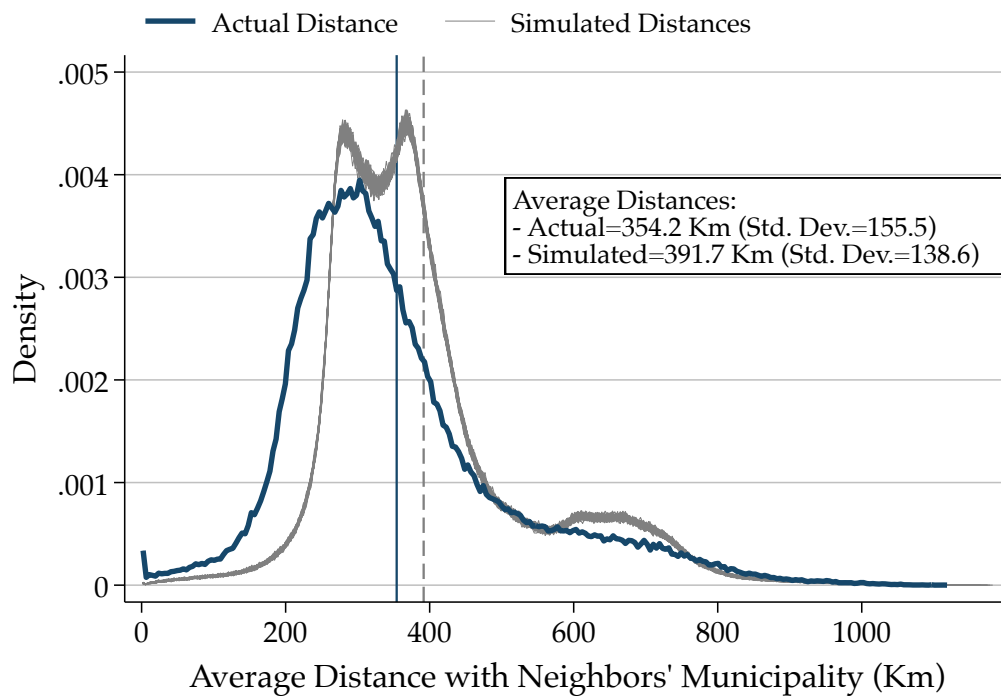
Notes. This figure reports the correlation between the Ellis Island dataset and data compiled from the *Annuario Statistico della Emigrazione Italiana dal 1876 al 1925* published by the General Commissioner for Emigration and containing regional data on U.S. emigration between 1876 and 1925. Panels F.5a and F.5c display the total number of U.S. emigrants recorded in the *Annuario* and the immigrants recorded in the full and linked Ellis Island dataset, respectively. Panels F.5b and F.5d display the regression slopes between the linked rates in Sample 1 and in Samples 2 and 3, respectively. In these figures, a dot is a region-year pair, and each panel separately reports the correlation between data in the *Annuario* and in the Ellis Island full and linked datasets, along with their standard errors clustered by region. The red lines overlay the fitted values from the linear regression, and the gray bands display 95% confidence intervals. Referenced on page: A.23.

Figure F.6. Validation of Local Name Index Against Contemporary Survey-Based Measures of Trust



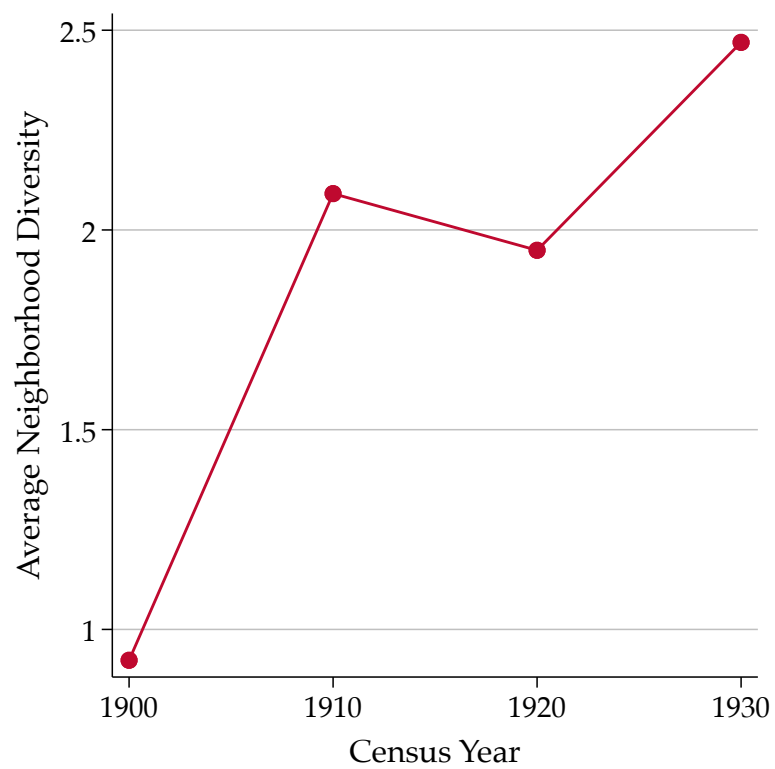
Notes. This figure reports the correlation between the local name index (Panel F.6a) and the distance-adjusted local name index (Panel F.6b) and the first principal component constructed over a set of indicators of trust in national institutions. The sample is the cross-section of municipalities with at least five respondents across all waves. Both the outcome variables and the independent variables are standardized to facilitate comparisons. The figures are binned scatter plots, and overlay a linear fit between the two variables. In both regressions, we control for log-population in 1991 and for an indicator equal to one for municipalities located in the South. We report the regression coefficient along with its standard error, clustered at the municipality level. Referenced on pages: 13, A.3.

Figure F.7. Distance from Neighbors' Municipality of Origin



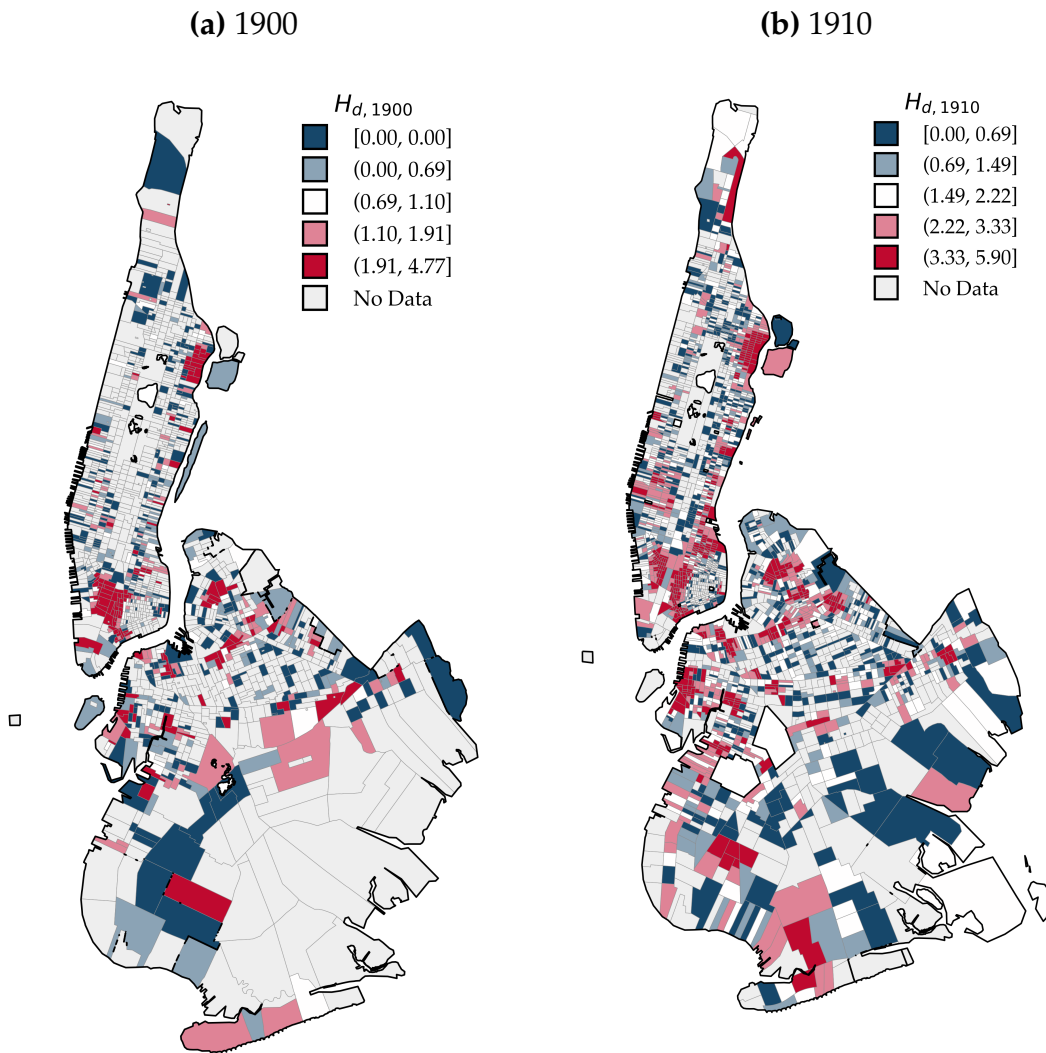
Notes. This figure displays the average distribution of distances between individuals' municipalities of origin within each enumeration district (blue). Gray lines report results from 1,000 simulations in which individuals are randomly allocated across districts; in each simulation, all members of the same household are excluded before reassignment. Excess mass of the blue distribution on the left of the random-matching counterfactual indicates that individuals disproportionately co-locate with immigrants originating from municipalities geographically close to their own. Referenced on page: [17](#).

Figure F.8. Average Neighborhood Diversity Over Time



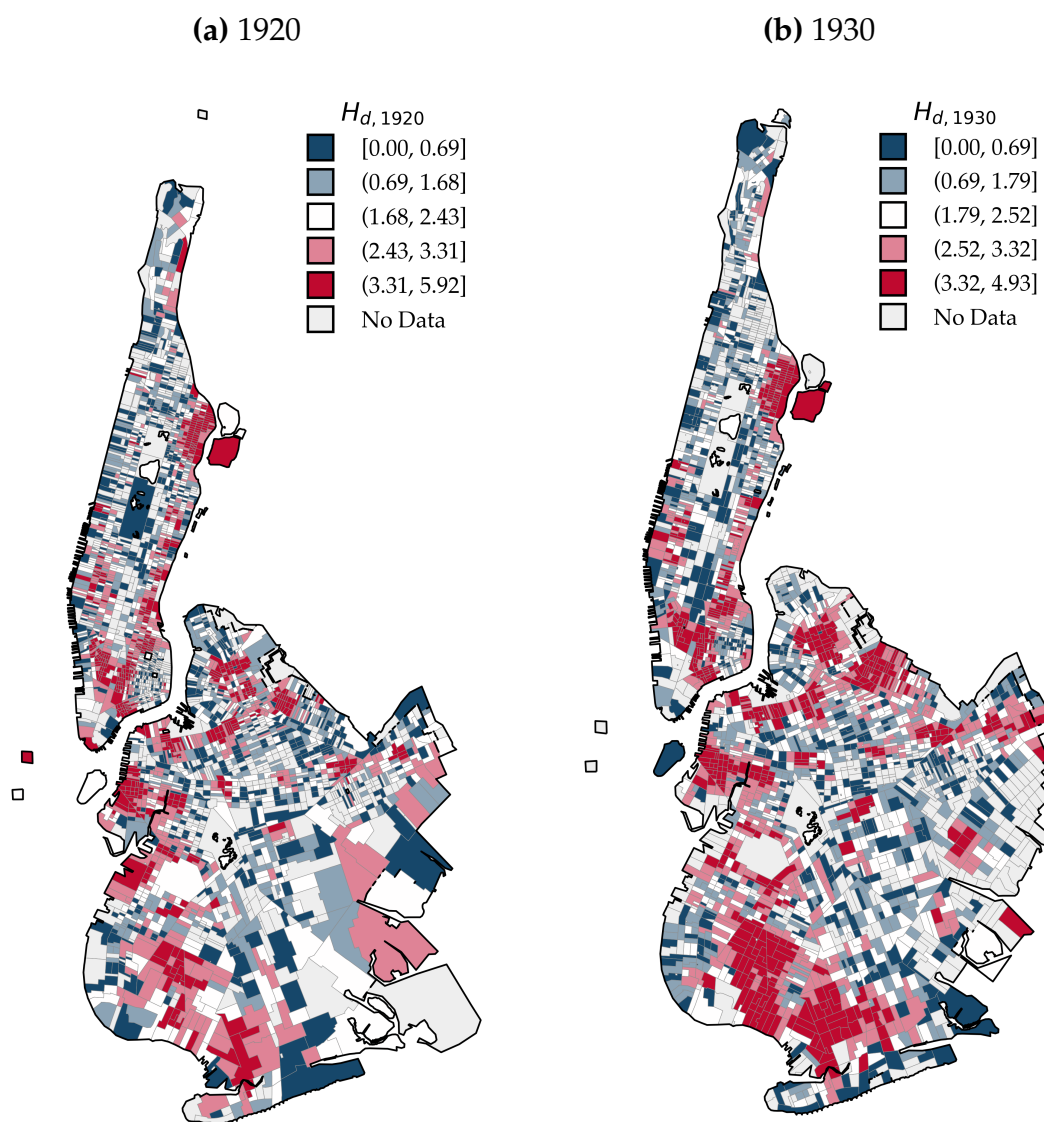
Notes. This figure displays the average diversity of the Italian community, as defined in (3) at the census-decade frequency between 1900 and 1930, i.e., the decades in our analysis sample. Referenced on page: 17.

Figure E.9. Diversity of Italian Communities in New York City, 1900–1930



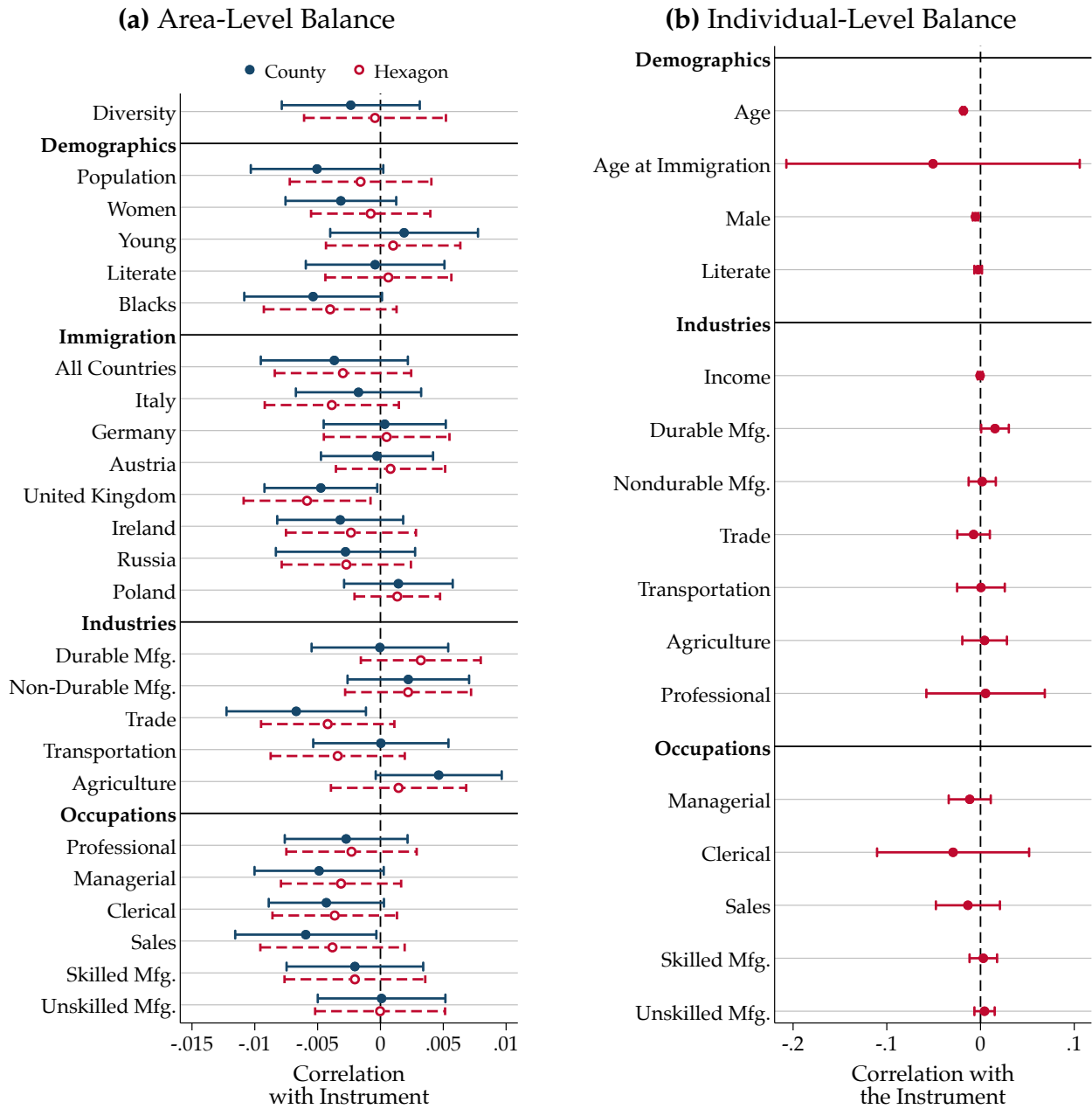
Notes. This figure displays the enumeration-district-level diversity of the Italian community in New York City (Manhattan and Brooklyn boroughs), as defined in (3), between 1900 and 1930. District boundary files are from Shertzer et al. (2016). Gray districts are districts with no Italians present. Blue to red colors signify increasing diversity of the community. Referenced on page: 17.

Figure F.9. Diversity of Italian Communities in New York City, 1900–1930 (Continued)



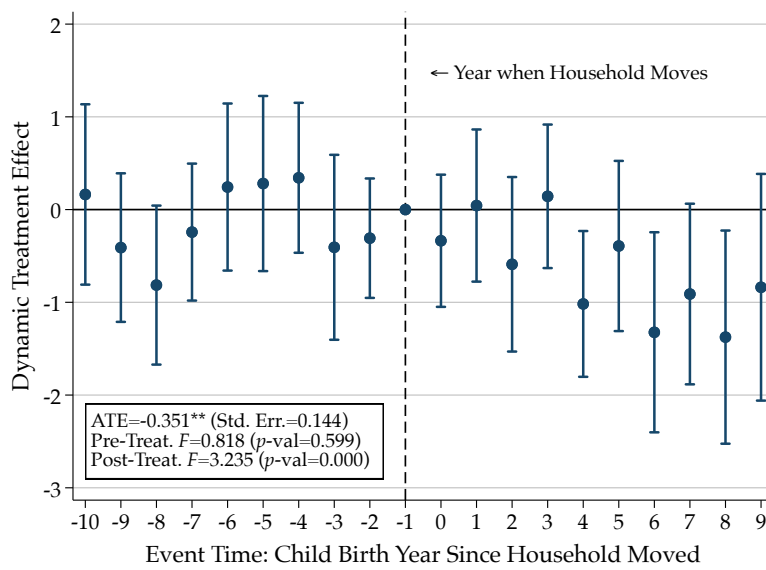
Notes. This figure displays the enumeration-district-level diversity of the Italian community in New York City (Manhattan and Brooklyn boroughs), as defined in (3), between 1900 and 1930. District boundary files are from Shertzer et al. (2016). Gray districts are districts with no Italians present. Blue to red colors signify increasing diversity of the community. Referenced on page: 17.

Figure F.10. Tests of Instrument Balance



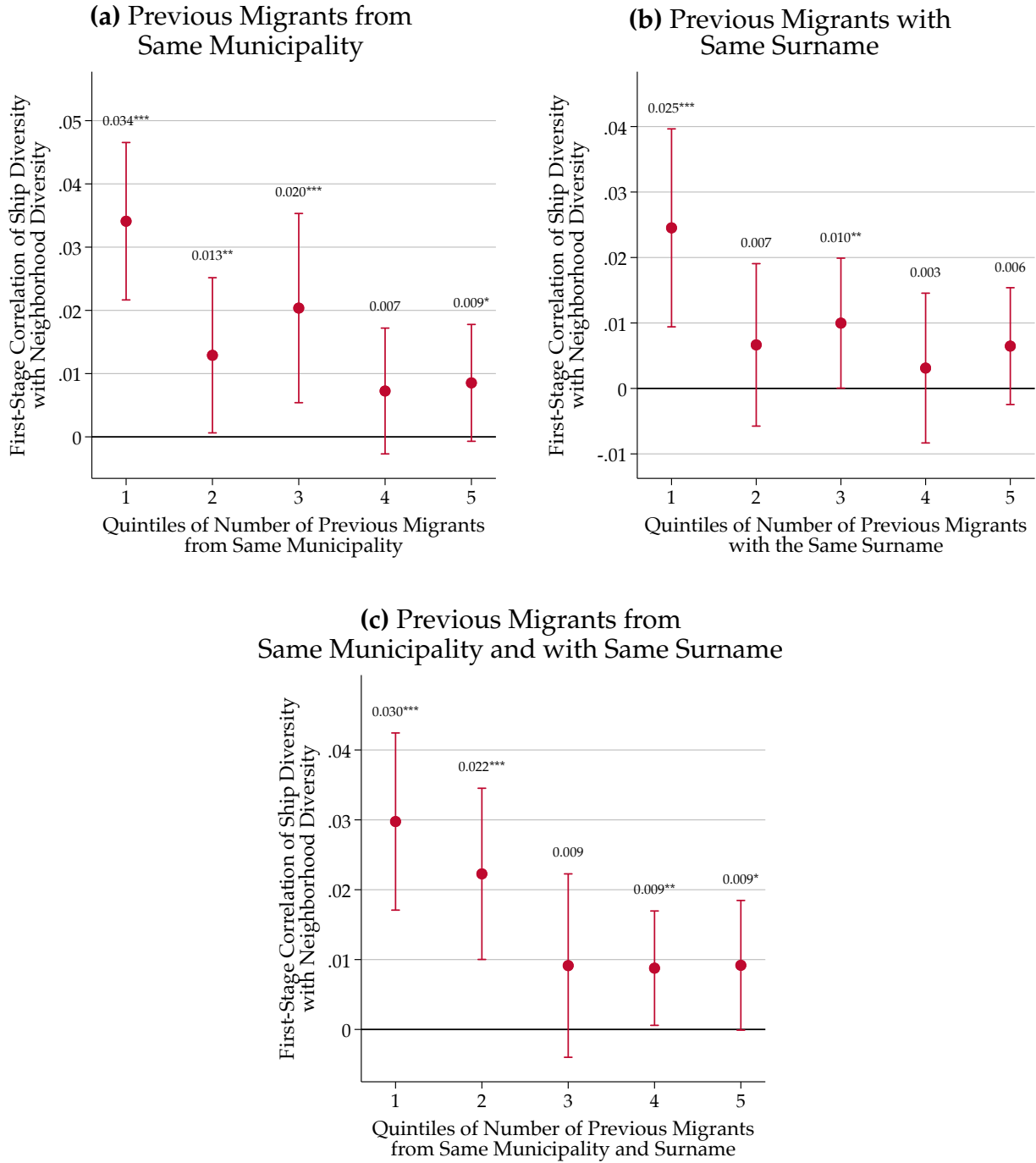
Notes. This figure reports correlations between the area-diversity instrument defined in (5) and area-level (Panel F.10a) and individual-level (Panel F.10b) characteristics. The unit of observation in both panels is an individual observed between 1900 (or the year of immigration, if later) and 1930. In Panel F.10a, each point reports the coefficient from a separate regression of a predetermined county-level (blue) or hexagon-level (red) characteristic, lagged one census period, against the ship-based instrument. In Panel F.10b, each point reports the coefficient from a separate regression of a predetermined individual characteristic, forwarded one census period, on the ship-based instrument. All specifications include year fixed effects, municipality-of-origin fixed effects, and individual controls (sex, birth year, and immigration year). Panel F.10a additionally includes county or hexagon fixed effects, while Panel F.10b includes hexagon fixed effects. Coefficients are standardized by the standard deviation (Panel F.10a) or the mean (Panel F.10b) of the dependent variable. Standard errors are clustered at the county level for county characteristics and at the hexagon level for hexagon characteristics; bars report 95% confidence intervals. Referenced on page: 20.

Figure F.11. Movers Design Event-Study Estimates



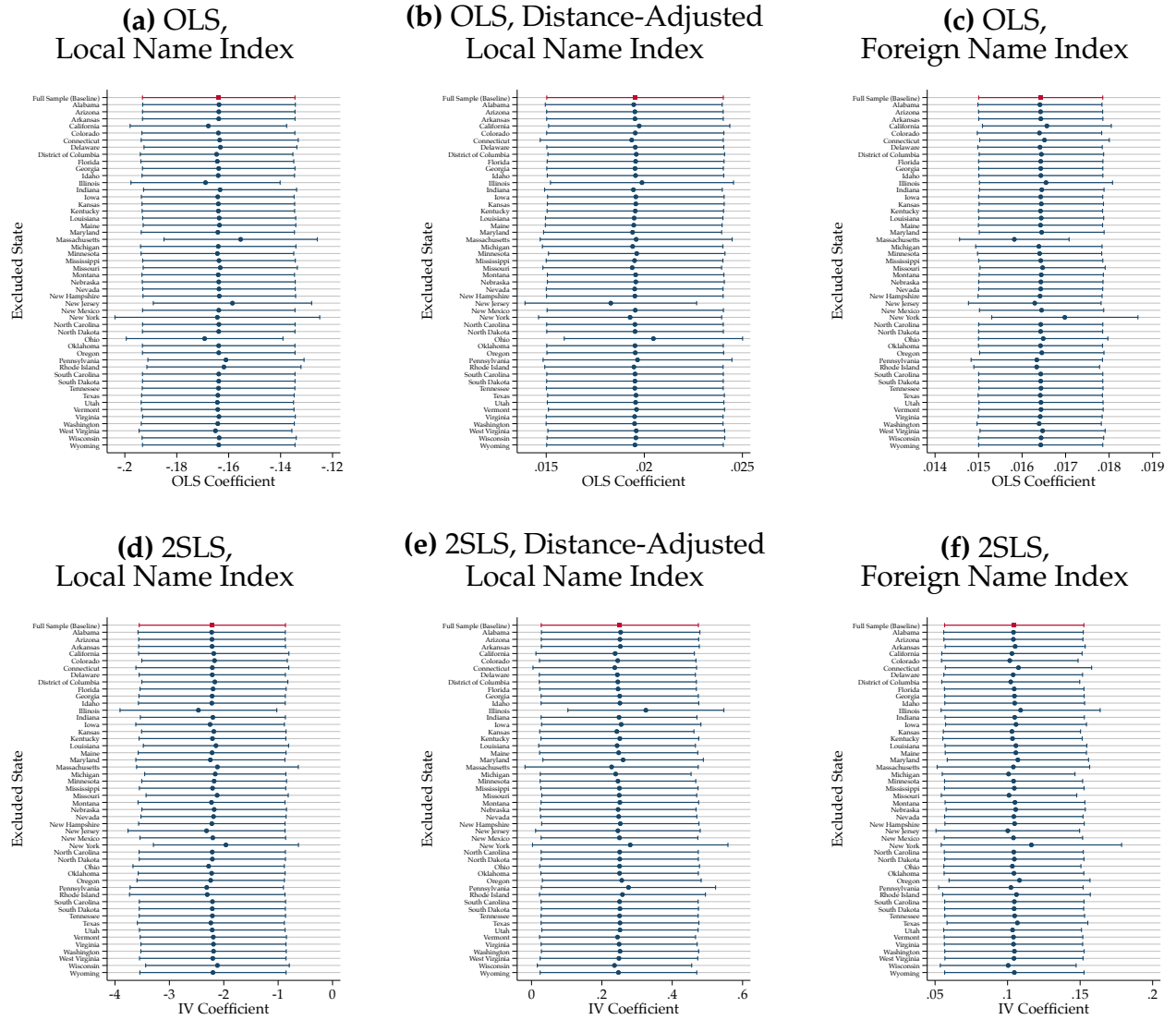
Notes. This figure reports results from the movers design specification in (A.5). The unit of observation is an individual migrant observed in each year in which they have a child born in the United States between 1900 and 1930. The dependent variable is the local name index defined in (1). The treatment is an interaction between an indicator equal to one for individuals experiencing a large diversity shock, defined as a move to a district where the change in Italian-community diversity between the destination and origin lies in the top quintile of the distribution, and a set of event-time indicators relative to the year of the move. The specification includes individual and event-time fixed effects, as well as hexagon-by-birth-year fixed effects. Standard errors are clustered at the hexagon level. Bands report 95% confidence intervals. We report the average treatment effect as well as joint F -tests for the joint significance of pre- and post-treatment coefficients. Referenced on pages: 21, A.13.

Figure F.12. Selection into Compliance



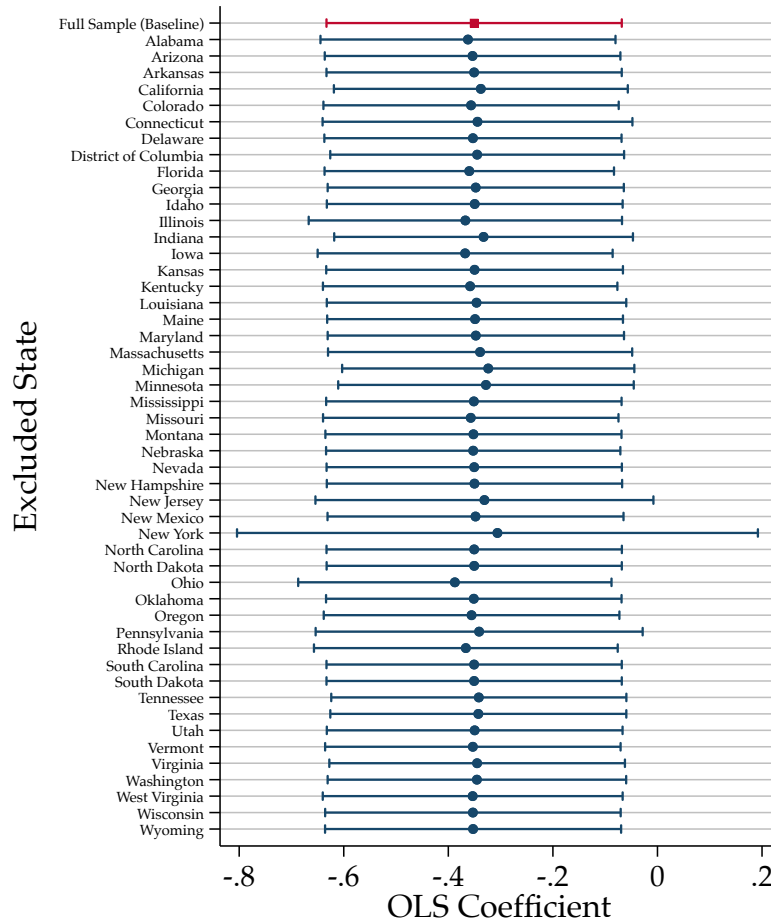
Notes. This figure reports the relevance of the ship-based instrument defined in (5) across the distribution of emigrants' social networks upon migration. The unit of observation is an individual observed in the first census in which they appear and, when possible, linked across subsequent censuses between 1900 and 1930. To construct each plot, we regress the baseline diversity measure (3) against the instrument, quintiles of various proxies for emigrants' social networks, and their interactions. Each dot reports the association for each quintile. We consider three proxies for emigrants' social networks: the number of migrants from the same municipality (Panel F.12a), the number of migrants with the same surname (Panel F.12b), and the number of migrants with the same surname who originated from the same municipality (Panel F.12c). All these measures are fixed at baseline, that is, for each immigrant, we construct them using exclusively those who migrated before them. Each regression includes hexagon, origin municipality, and census-year fixed effects. Standard errors are clustered at the hexagon level; confidence bands report 95% confidence intervals. Referenced on page: 24.

Figure F.13. State Leave-Out: OLS and 2SLS Panel Regressions



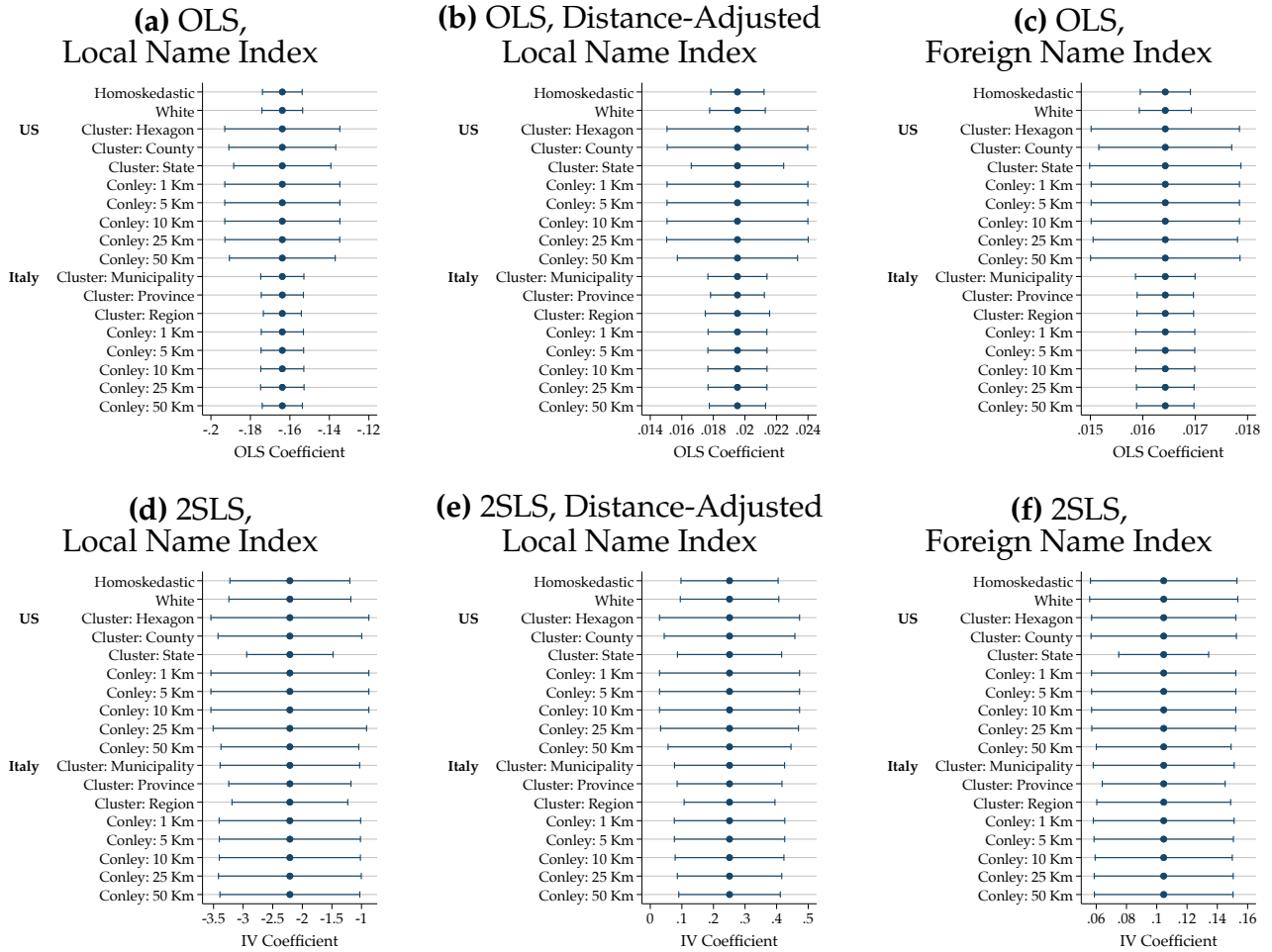
Notes. This figure reports the effect of exposure to diversity within the Italian community on name-based indices. Panels F.13a and F.13d use the local name index defined in (1) as the dependent variable; Panels F.13b and F.13e use the distance-adjusted local name index defined in (2); Panels F.13c and F.13f use the foreign name index of Abramitzky et al. (2020). The unit of observation is an individual observed in the first census in which they appear and, when possible, linked across subsequent censuses between 1900 and 1930. Panels F.13a, F.13b, and F.13c report ordinary least squares estimates; Panels F.13d, F.13e, and F.13f report two-stage least squares estimates. The first marker in each graph (in red) reports the baseline estimate; the subsequent markers report the estimates obtained when dropping one state at a time from the estimation sample. All specifications include municipality, census year, and hexagon fixed effects. Bands report 95% standard errors, clustered at the district level. Referenced on page: 27.

Figure F.14. State Leave-Out: Movers Regressions



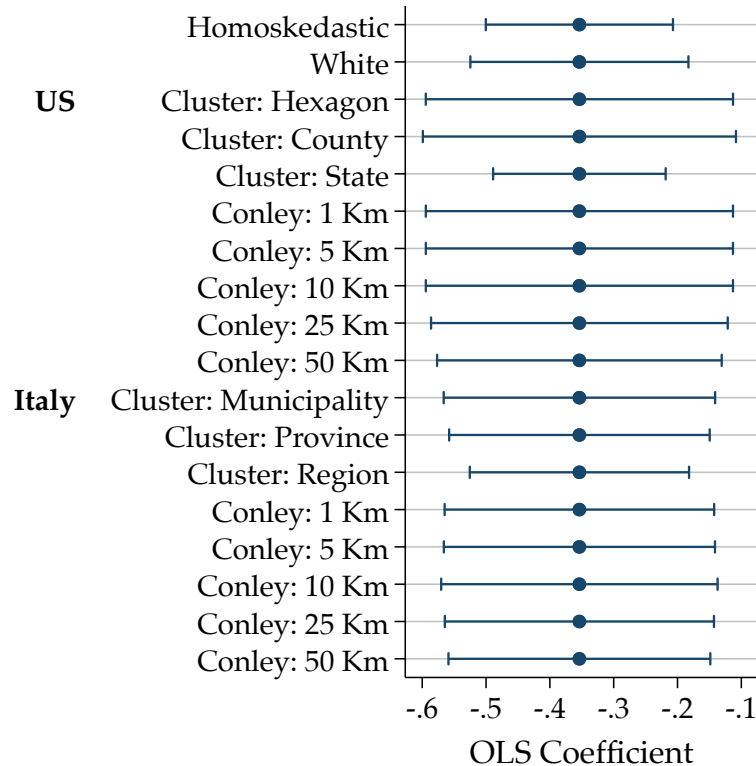
Notes. This figure reports results from the movers design specification in (A.5). The unit of observation is an individual migrant observed in each year in which they have a child born in the United States between 1900 and 1930. The dependent variable is the local name index defined in (1). The treatment is the interaction between an indicator equal to one after an individual moves to a different hexagon (and zero otherwise) and an indicator equal to one if the difference in diversity between the Italian communities in the destination and origin districts lies in the top 20% of the distribution (and zero otherwise). The first marker (in red) reports the baseline estimate; the subsequent markers report the estimates obtained when dropping one state at a time from the estimation sample. The specification includes individual, event-time (time since move), and hexagon-by-cohort fixed effects. Bands report 95% standard errors, clustered at the district level. Referenced on page: 27.

Figure F.15. Alternative Standard Errors: OLS and 2SLS Panel Regressions



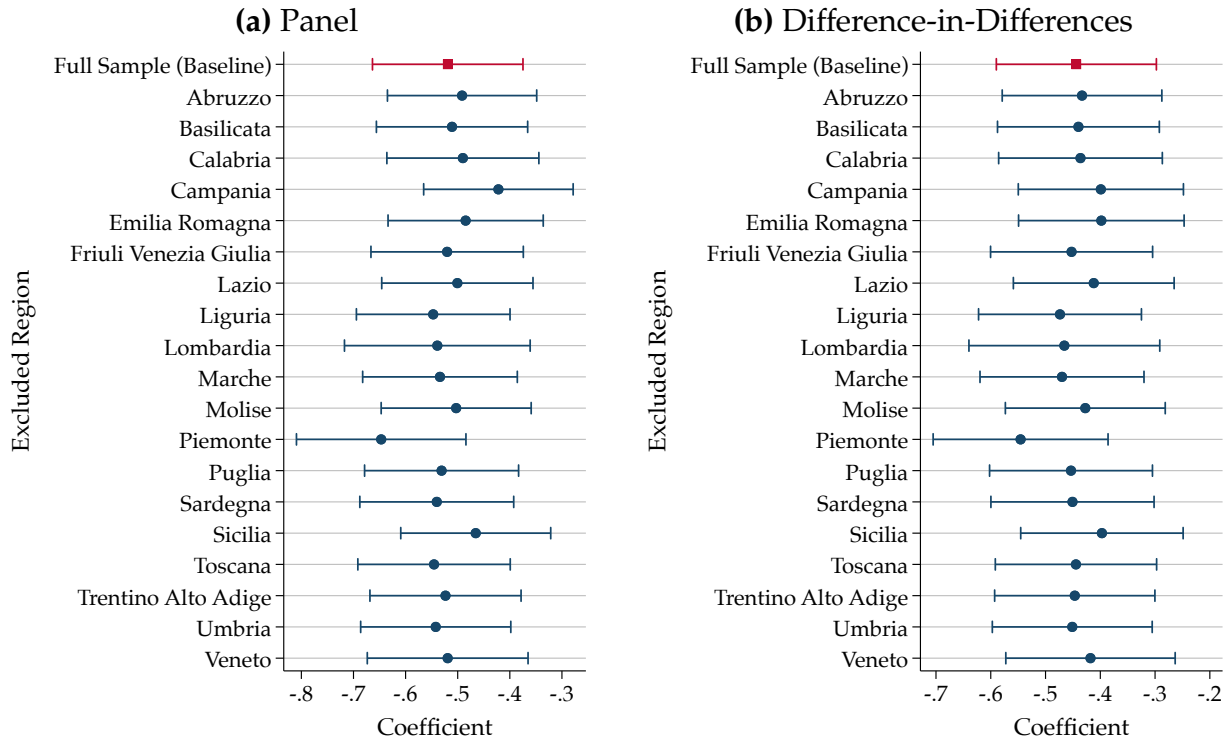
Notes. This figure reports the effect of exposure to diversity within the Italian community on name-based indices. Panels F.15a and F.15d use the local name index defined in (1) as the dependent variable; Panels F.15b and F.15e use the distance-adjusted local name index defined in (2); Panels F.15c and F.15f use the foreign name index of Abramitzky et al. (2020). The unit of observation is an individual observed in the first census in which they appear and, when possible, linked across subsequent censuses between 1900 and 1930. Panels F.15a, F.15b, and F.15c report ordinary least squares estimates; Panels F.15d, F.15e, and F.15f report two-stage least squares estimates. In each regression, we compute the standard errors using different estimators: unadjusted, White, clustered at different U.S. geographical layers, Conley-corrected using various thresholds centered on the hexagon's centroid, clustered at different Italy geographical layers, and Conley-corrected using various thresholds centered on the origin municipality's centroid. All specifications include municipality, census year, and hexagon fixed effects. Bands report 95% standard errors. Referenced on page: 27.

Figure F.16. Alternative Standard Errors: Movers Regressions



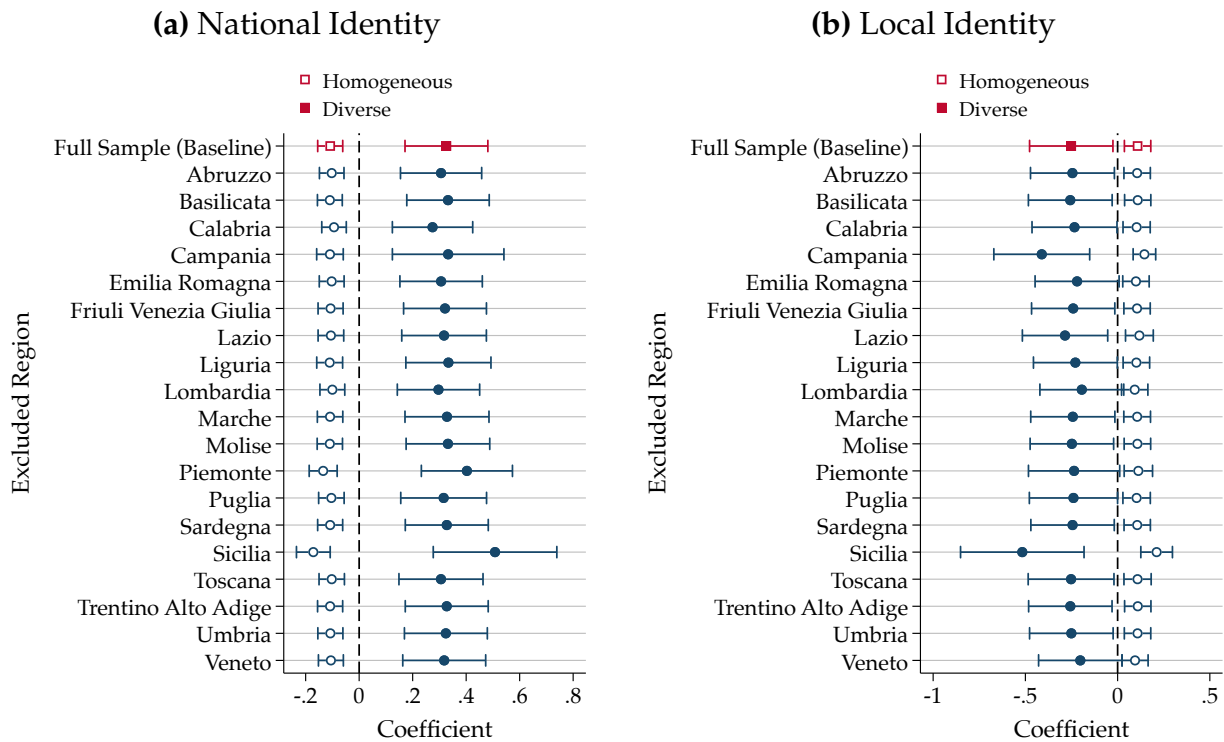
Notes. This figure reports results from the movers design specification in (A.5). The unit of observation is an individual migrant observed in each year in which they have a child born in the United States between 1900 and 1930. The dependent variable is the local name index defined in (1). The treatment is the interaction between an indicator equal to one after an individual moves to a different hexagon (and zero otherwise) and an indicator equal to one if the difference in diversity between the Italian communities in the destination and origin districts lies in the top 20% of the distribution (and zero otherwise). In each regression, we compute the standard errors using different estimators: unadjusted, White, clustered at different U.S. geographical layers, Conley-corrected using various thresholds centered on the hexagon's centroid, clustered at different Italy geographical layers, and Conley-corrected using various thresholds centered on the origin municipality's centroid. The specification includes individual, event-time (time since move), and hexagon-by-cohort fixed effects. Bands report 95% standard errors. Referenced on page: 27.

Figure F.17. Region Leave-Out: Name Particularism in Italy



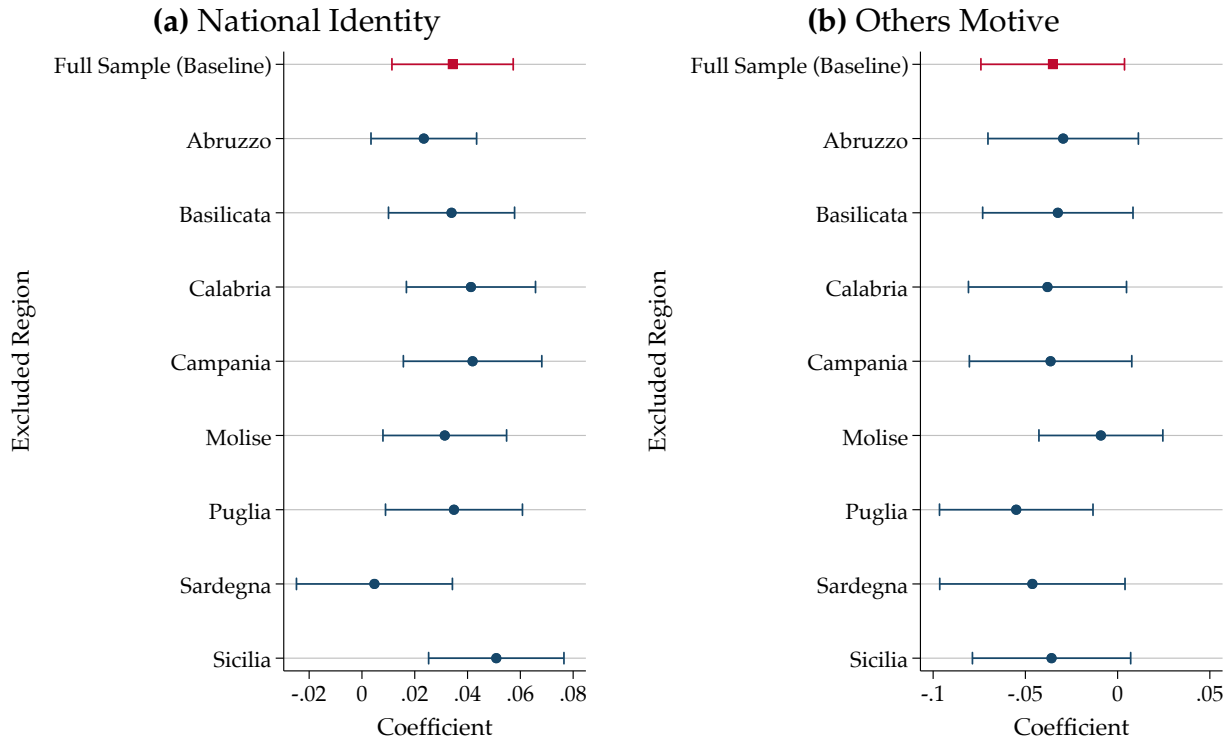
Notes. This figure reports the effect of exposure to diversity within the Italian migrant community on name-based indices in Italian municipalities from equations (6)–(7). The unit of observation is a municipality observed at a decennial frequency between 1880 and 1930. The dependent variable is the local name index defined in (1). In Panel F.17a, each dot reports the coefficient of a treatment term defined as the share of a municipality’s U.S. emigrants who settled in districts in the top quintile of the diversity distribution (3) interacted with a post-1900 indicator. In Panel F.17b, each dot reports an interaction term between a post indicator equal to one in years at or after the municipality-specific onset of mass migration (defined as the first year with at least one emigrant per 1,000 inhabitants) and the share of the municipality’s U.S. emigrants who settled in districts in the top quintile of the diversity distribution (3). The first marker (in red) reports the baseline estimate; the subsequent markers report the estimates obtained when dropping one region at a time from the estimation sample. All regressions include municipality and year fixed effects. Bands report 95% confidence intervals; Standard errors are clustered at the municipality level. Referenced on page: 35.

Figure F.18. Region Leave-Out: Street Naming in Italy



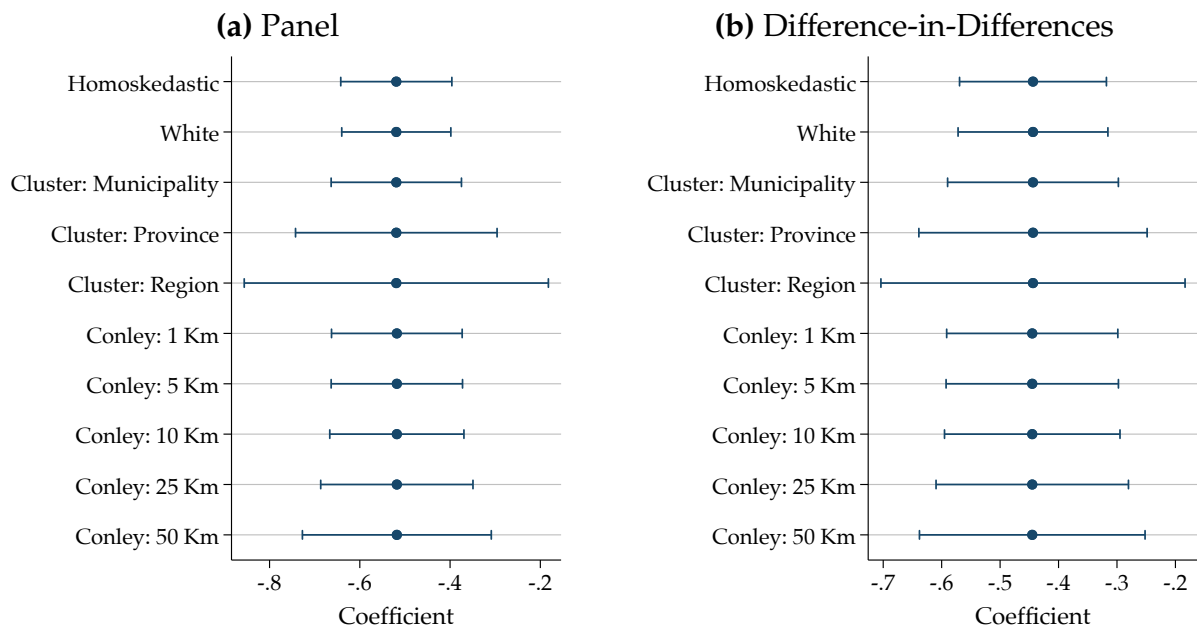
Notes. This figure reports cross-sectional estimates of specification (8) between emigration to diverse Italian communities in the United States and street-naming patterns in Italy. The unit of observation is a municipality. The dependent variables are the shares of streets whose eponyms reflect national identity (Panel F.18a) and local identity (Panel F.18b). The main explanatory variables are the number of emigrants to diverse neighborhoods (displayed by the hollow markers) and the number of emigrants to homogeneous neighborhoods (displayed by the colored markers). The first markers (in red) report the baseline estimates; the subsequent markers report the estimates obtained when dropping one region at a time from the estimation sample. The outcomes are normalized by the total number of streets in the municipality. All specifications include region fixed effects. Bands report 95% confidence intervals; Standard errors are clustered at the municipality level. Referenced on page: 35.

Figure F.19. Region Leave-Out: Monuments in Italy



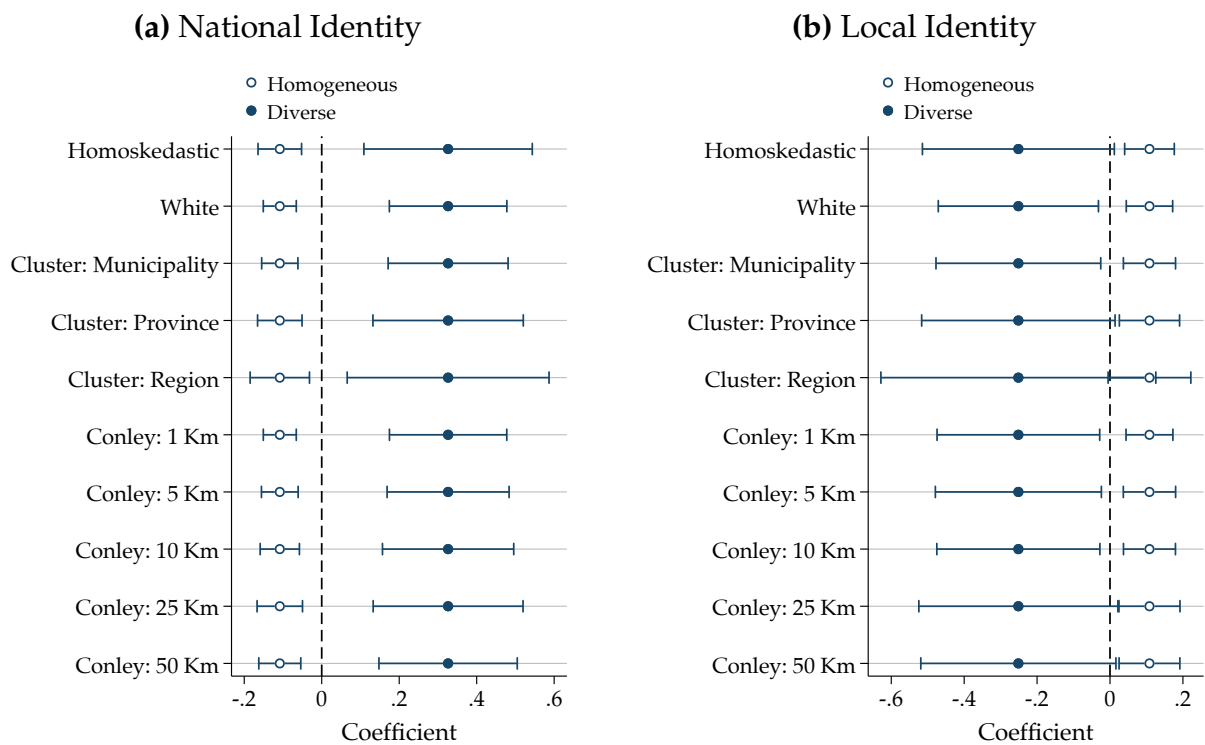
Notes. This figure reports estimates obtained from (9) of the effect of exposure to diversity within the Italian migrant community on monument construction in Southern Italy. The unit of observation is a municipality-decade for municipalities observed between 1861 and 2000. In Panel F.19a, the dependent variable is an indicator equal to one if the municipality has at least one national-identity monument in a given decade. The baseline treatment is an interaction term between a post-1900 indicator and an indicator for districts in the top-quintile of the diversity measure defined in (3). In Panel F.19b, the dependent variable is an indicator equal to one if the municipality has at least one non-national-identity monument in a given decade. The baseline treatment is an interaction term between a post-1900 indicator and the share of U.S. emigrants from each municipality who settled in districts in the top quintile of the diversity measure defined in (3). The first markers (in red) report the baseline estimates; the subsequent markers report the estimates obtained when dropping one region at a time from the estimation sample. Each specification includes municipality fixed effects and decade fixed effects. Bands report 95% confidence intervals; Standard errors are clustered at the municipality level. Referenced on page: 35.

Figure F.20. Alternative Standard Errors: Name Particularism in Italy



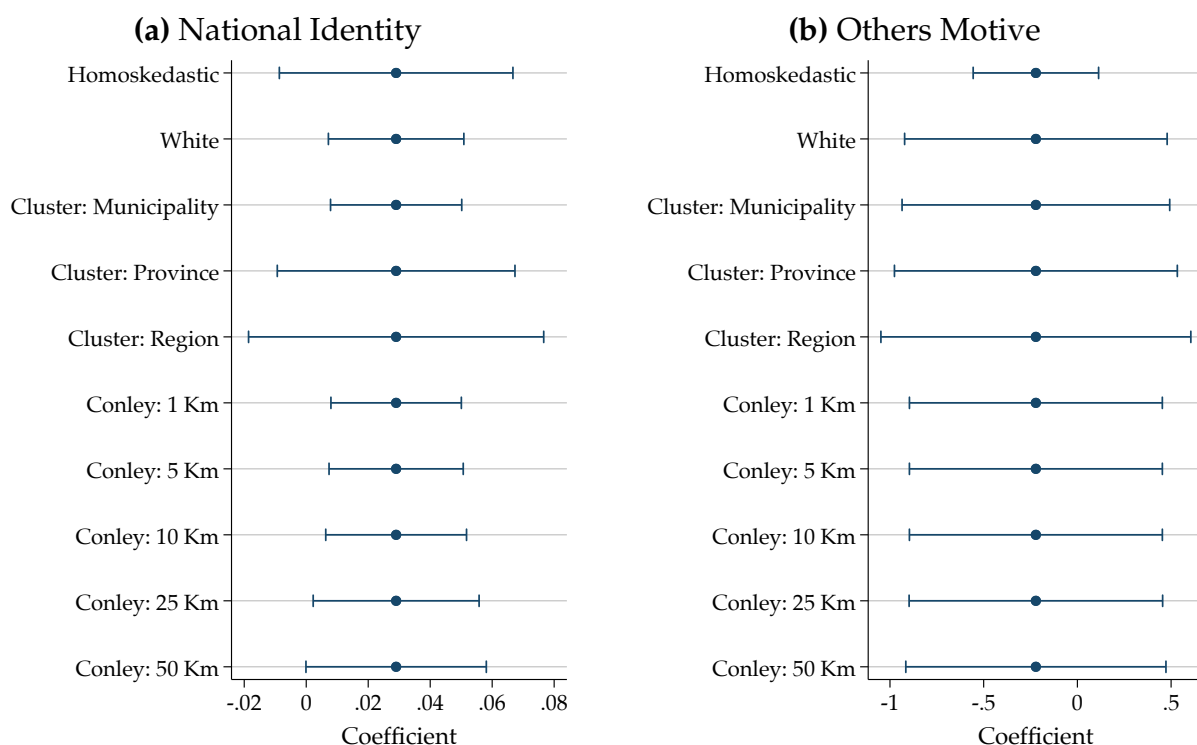
Notes. This figure reports the effect of exposure to diversity within the Italian migrant community on name-based indices in Italian municipalities from equations (6)–(7). The unit of observation is a municipality observed at a decennial frequency between 1880 and 1930. The dependent variable is the local name index defined in (1). In Panel F.20a, each dot reports the coefficient of a treatment term defined as the share of a municipality’s U.S. emigrants who settled in districts in the top quintile of the diversity distribution (3) interacted with a post-1900 indicator. In Panel F.20b, each dot reports an interaction term between a post indicator equal to one in years at or after the municipality-specific onset of mass migration (defined as the first year with at least one emigrant per 1,000 inhabitants) and the share of the municipality’s U.S. emigrants who settled in districts in the top quintile of the diversity distribution (3). In each regression, we compute the standard errors using different estimators: unadjusted, White, clustered at different geographical layers, and Conley-corrected using various thresholds centered on the municipality’s centroid. All regressions include municipality and year fixed effects. Bands report 95% confidence intervals. Referenced on page: 35.

Figure F.21. Alternative Standard Errors: Street Naming in Italy



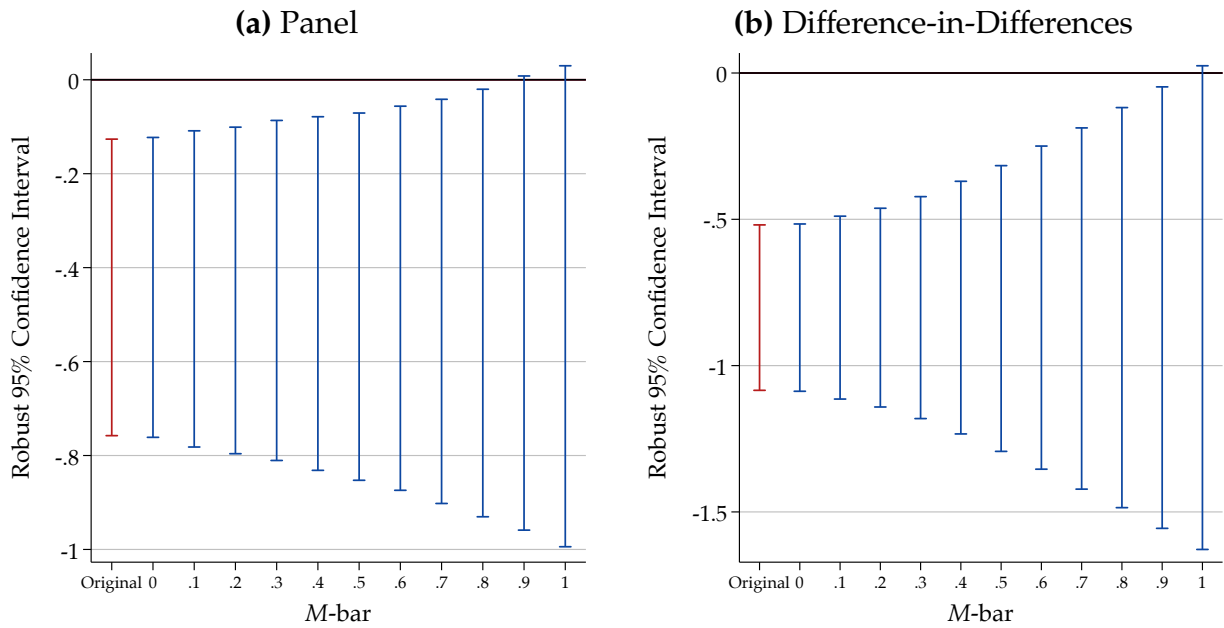
Notes. This figure reports cross-sectional estimates of specification (8) between emigration to diverse Italian communities in the United States and street-naming patterns in Italy. The unit of observation is a municipality. The dependent variables are the shares of streets whose eponyms reflect national identity (Panel F.21a) and local identity (Panel F.21b). The main explanatory variables are the number of emigrants to diverse neighborhoods (displayed by the hollow markers) and the number of emigrants to homogeneous neighborhoods (displayed by the colored markers). In each regression, we compute the standard errors using different estimators: unadjusted, White, clustered at different geographical layers, and Conley-corrected using various thresholds centered on the municipality's centroid. The outcomes are normalized by the total number of streets in the municipality. All specifications include region fixed effects. Bands report 95% confidence intervals. Referenced on page: 35.

Figure F.22. Alternative Standard Errors: Monuments in Italy



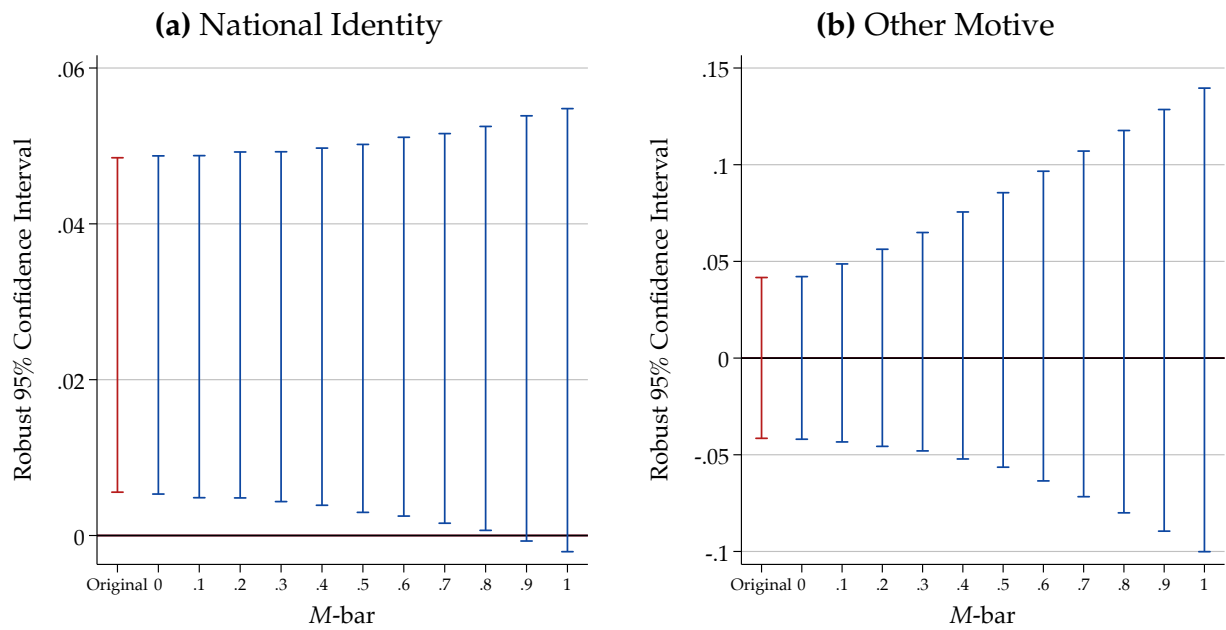
Notes. This figure reports estimates obtained from (9) of the effect of exposure to diversity within the Italian migrant community on monument construction in Southern Italy. The unit of observation is a municipality-decade for municipalities observed between 1861 and 2000. In Panel F.22a, the dependent variable is an indicator equal to one if the municipality has at least one national-identity monument in a given decade. In Panel F.22b, the dependent variable is an indicator equal to one if the municipality has at least one non-national-identity monument in a given decade. The baseline treatment is an interaction term between a post-1900 indicator and the share of U.S. emigrants from each municipality who settled in districts in the top quintile of the diversity measure defined in (3). In each regression, we compute the standard errors using different estimators: unadjusted, White, clustered at different geographical layers, and Conley-corrected using various thresholds centered on the municipality's centroid. Each specification includes municipality fixed effects and decade fixed effects. Bands report 95% confidence intervals. Referenced on page: 35.

Figure F.23. Robustness to Violations of Parallel Trends: Naming in Italy



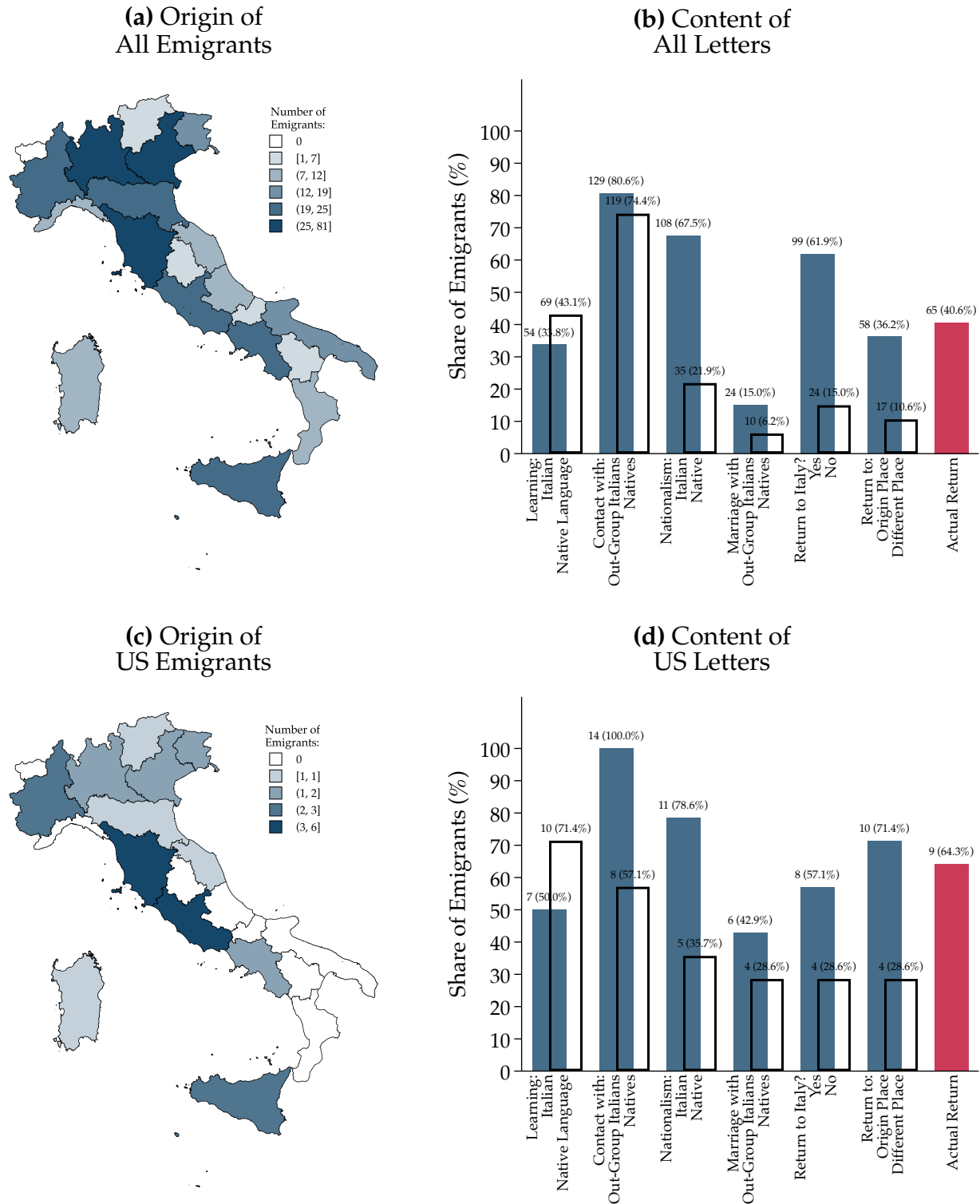
Notes. This figure reports alternative confidence intervals for the effect of exposure to diversity within the Italian migrant community on name-based indices in Italian municipalities from equations (6)–(7). The bands report 95% robust confidence intervals constructed following the method developed by Rambachan and Roth (2023). The unit of observation is a municipality observed at a decennial frequency between 1880 and 1930. The dependent variable is the local name index defined in (1). In Panel F.23a, each dot reports the coefficient of a treatment term defined as the share of a municipality’s U.S. emigrants who settled in districts in the top quintile of the diversity distribution (3) interacted with a post-1900 indicator. In Panel F.23b, each dot reports an interaction term between a post indicator equal to one in years at or after the municipality-specific onset of mass migration (defined as the first year with at least one emigrant per 1,000 inhabitants) and the share of the municipality’s U.S. emigrants who settled in districts in the top quintile of the diversity distribution (3). All regressions include municipality and year fixed effects. Referenced on page: 36.

Figure F.24. Robustness to Violations of Parallel Trends: Monuments in Italy



Notes. This figure reports alternative confidence intervals obtained from (9) on the effect of exposure to diversity within the Italian migrant community on monument construction in Southern Italy. The unit of observation is a municipality-decade for municipalities observed between 1861 and 2000. The bands report 95% robust confidence intervals constructed following the method developed by Rambachan and Roth (2023). In Panel F.24a, the dependent variable is an indicator equal to one if the municipality has at least one national-identity monument in a given decade. In Panel F.24b, the dependent variable is an indicator equal to one if the municipality has at least one non-national-identity monument in a given decade. The baseline treatment is an interaction term between a post-1900 indicator and the share of U.S. emigrants from each municipality who settled in districts in the top quintile of the diversity measure defined in (3). Referenced on page: 36.

Figure F.25. Analysis of Emigrants' Letters



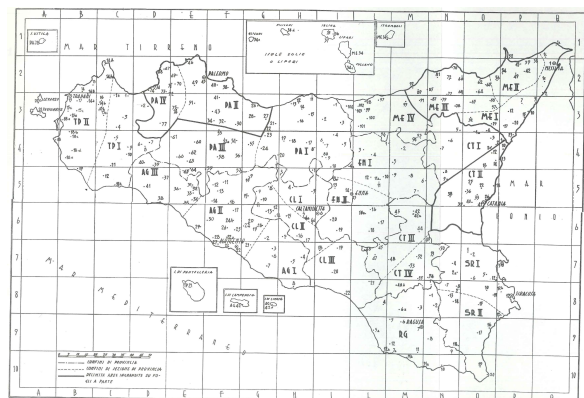
Notes. This Figure presents the results of a descriptive analysis of letters from Italian emigrants to their families in Italy between 1861 and 1939. Panel F.25a (resp. F.25c) reports the number of emigrants by region of origin across all destination countries (resp. for emigrants to the United States). Panel F.25b (resp. F.25d) reports the share of all (resp. U.S.) emigrants whose letters mention each topic. The red column reports the share of emigrants who effectively returned, as inferred from their biographies. On top of each bar, we report the number and share of emigrants within each category. Referenced on pages: 36, 37, A.8, A.9.

Figure F.26. The *Vocabolario Siciliano*

(a) Sample Page

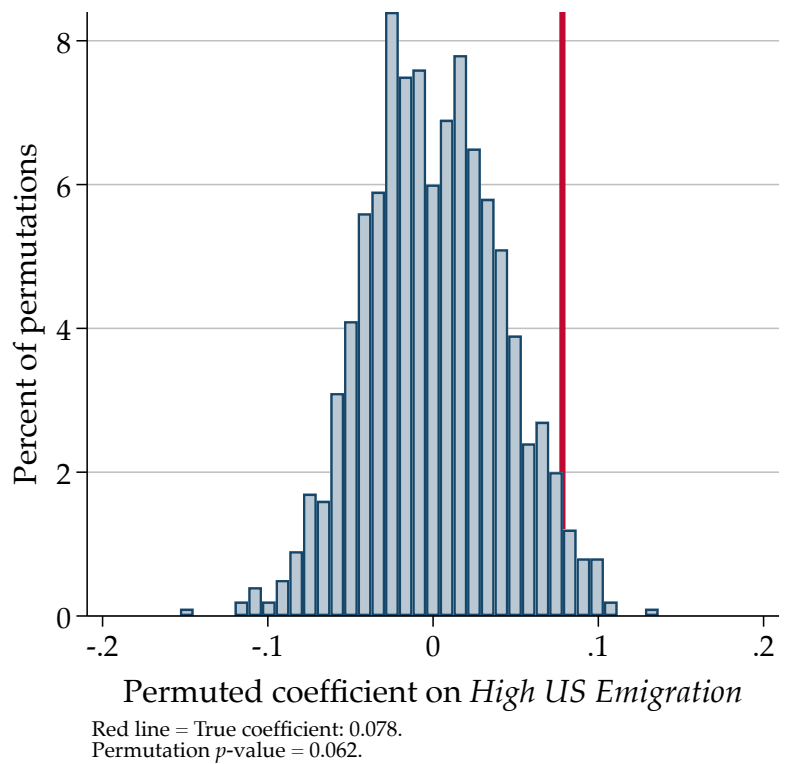
bbordu	— 435 —	bbötina
orlatura, guarnizione. 3. (Tr.¹, NU., Tri.) sopraggitto. 4. (CT 12, ALI: TP I) t. calz. tramezza tra suola e tomaio. 5. (CT II) <i>fil'i bb.</i> striscia di cuio disposta circolarmente sulla suola, in corrispondenza del calcagno, per rendere piana la base del tacco; anche <i>bbuòrdilu</i>. 6. (Man.) <i>tèniri a bb.</i> fig. tenere a bada. 7. pl. (Av.) <i>cc'hè bbordi</i> di persona molto in gamba; di cosa eccellente, coi fiocchi. V. anche <i>bbuordu</i>. <i>bbordu</i>³ (ME 42) v. <i>abbordu</i>². <i>bbordu</i>⁴ (Tr., Av.) m. <i>a-pprimu bb.</i> di primo acchito. <i>bborè</i> (PA 59) in <i>piru bb.</i> varietà di pera. Anche <i>bburè</i>. <i>bbörgia</i> (CT I e 12) f. pappagorgia. Anche <i>bböggia</i>. Cfr. <i>bbürgiu</i> 5. <i>bbòria</i>¹ f. (Ar.) maltempo provocato da vento di tramontana. 2. (Tri., ME 20 e 22, CT 1-4) venticello; (ME 61) brezza. 3. (ALM: CT 26a) colpo di vento. 4. (ALI: ME 104a, AG 8) tempesta di mare, fortunale. 5. (ME 94) vento di sud-est. 6. (EN 5) vento che spira da mezzogiorno. 7. (AA.) <i>bb. di nivi</i> nuvola che annunzia un'abbondante nevicata. 8. nebbia. V. anche <i>bbòira</i>, <i>bbuori</i> e <i>uòria</i>. <i>bbòria</i>² f. rar. (AA., Mal., VS., ecc., CT 7 e 43, EN 5) boria, alterigia; anche <i>bbuòria</i>. 2. (Cas.) <i>fari passari la bb. a unu</i> rintuzzare la superbia di q. <i>bboriata</i> (Tri.) f. folata, raffica improvvisa di vento. 2. infreddatura; <i>pighiàrisi na bb.</i> raffreddarsi. 3. sfuriata. Cfr. <i>bbòria</i>¹. <i>bbòrica</i> agg. femm. in <i>acqua bb.</i> soluzione di acqua e acido bórico. <i>bborisca</i> (DB.) v. <i>bborissa</i>. <i>bborissa</i> (Pa., Tr.) f. t. bot. lunaria odorata: <i>Lunaria annua</i>. 2. denararia: <i>Lysimachia nummularia</i>. Anche <i>bborisca</i> e <i>bborizza</i>. <i>bborizza</i> (DB.) v. <i>bborissa</i>. <i>bbornussu</i> (Tr.) v. <i>bbarnusu</i>. <i>bborra</i> (Tr.¹) f. fascio di canapa. <i>bborru</i>¹ m. (Tr., Ma., Av., Man., NU.) bozza, minuta, brutta copia; anche <i>bburruni</i>. 2. (CT 7) contratto di compravendita.		<i>bborru</i>² (Tri.) m. t. bot. tamarice: <i>Tamarix gallica</i>. <i>bborru</i>³ (Man.) m. scrocco. 2. frode. Cfr. <i>porru</i>². <i>bborsa</i> f. rec. borsa; borsetta. <i>bboru</i> (CT 7) v. <i>bbolu</i>². <i>bbosa</i> (ALI: CT 47, EN 19 e 20) f. sterco bovino. Anche <i>bbusa</i>². <i>bboscainu</i> (Tri.) agg. zotico, rozzo. V. anche <i>bbuscainu</i>. <i>bbòscia</i> (EN 5) f. vescichetta prodotta da scottatura. <i>bbosciuletti</i> (ALM: ME 34) v. <i>bbusciulitti</i>. <i>bboscù</i> (Mal., DB., ecc., Tri., ME I e II, CT 1-7, CT 47, ALI: TP 15) m. bosco. Anche <i>bbuoscu</i> e <i>voscù</i>. <i>bbòsica</i> (ME 76) f. altalena. Anche <i>bbòcica</i> e <i>bbàsica</i>. <i>bbòsicchia</i> (EN 5) f. terreno lasciato incolto per molto tempo. Cfr. <i>bbòsicchia</i>¹. <i>bbossa</i> (RG 4) f. rar. donna piuttosto grassa. Cfr. <i>bbossu</i>. <i>bbossamù</i> (EN 11) escl. mi dia! Anche <i>ossamù</i> e <i>vossamù</i>. <i>bbossi</i> (RG 4) rar. v. <i>bbossu</i> 12. <i>bbossu</i> e <i>bbuosu</i> m. padrone, principale. 2. sorvegliante; campiere. 3. (ME 93) chi si preoccupa dell'andamento di un'iniziativa e, in particolare, di quello della famiglia. 4. (EN 1) persona influente in paese. 5. (PA 43) capomafia. 6. (AG 15) persona che non vuol lavorare e ama piuttosto soprintendere o sindacare. 7. (ME 93) tipo autoritario o che si dà l'aria di persona importante e autorevole. 8. (ME 96) persona tronfia. 9. (PA 1) persona autorevole e facoltosa. 10. (PA 50) pezzo grosso arricchitosi col mercato nero nel periodo dell'occupazione angloamericana in Sicilia. 11. (PA 24, AG 14) persona grassa e panciuta. 12. agg. (RG 3 e 5, EN 2) corpacciuto; tarchiato. V. anche <i>bbossa</i> e <i>capubbossu</i>. <i>bbota</i> (RG 10) f. risolto di un indumento. V. anche <i>sbota</i>. <i>bbötina</i> f. (Av.) salume fatto con la schiena del tonno. 2. (SR 16) la parte più colorita della carne del tonno. V. anche <i>bbönira</i>.

(b) Geography of the Vocabulary



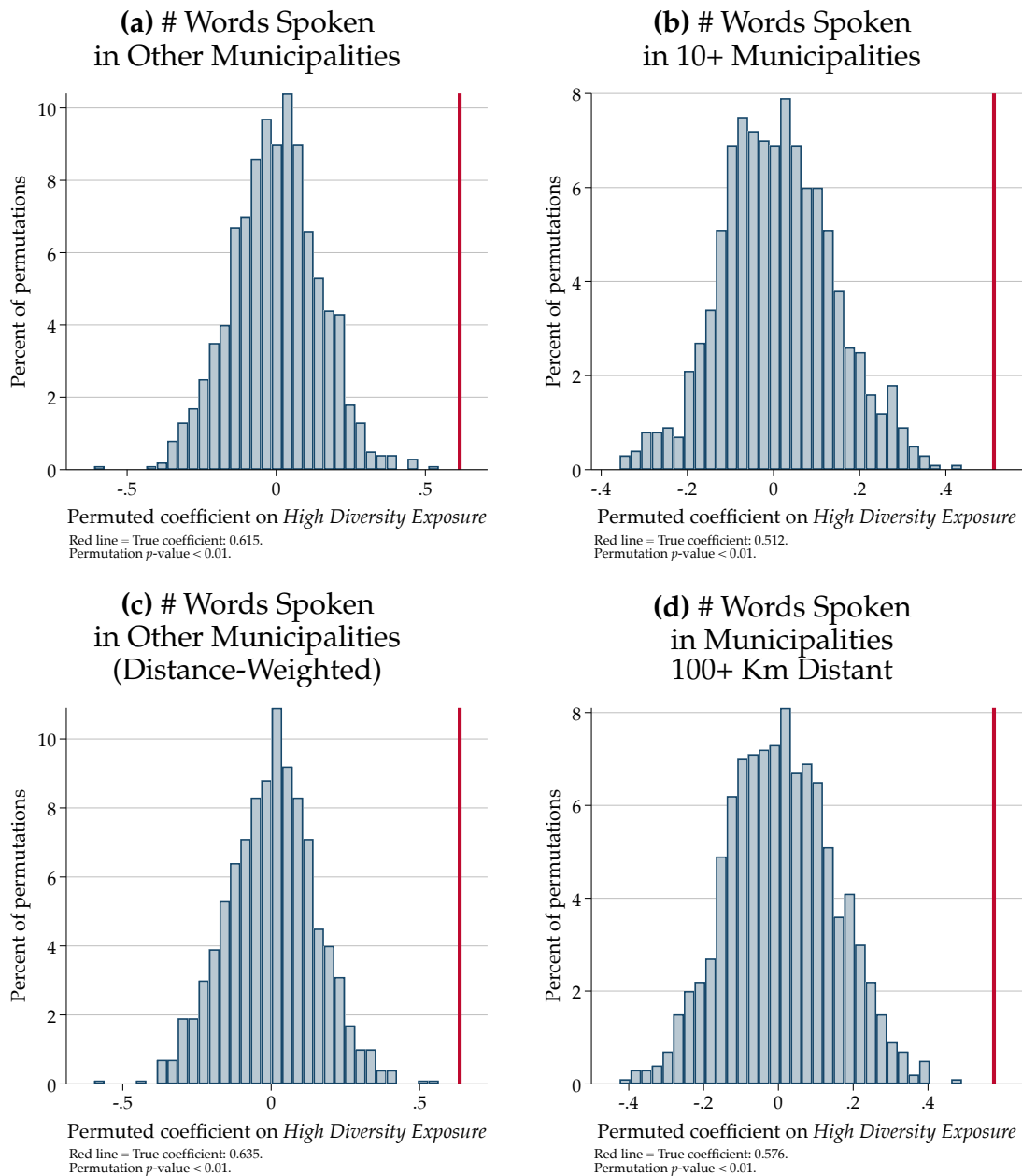
Notes. This figure displays one sample page (F.26a) of Volume I of the *Vocabolario Siciliano* and the map (F.26b) of the enumerated municipalities (Piccitto, 1977). The structure of the other two volumes is analogous to the ones displayed here. The municipality codes appear in parentheses within each entry (e.g., EN 1 is the code for the municipality of Enna). The codes are mapped to the municipalities displayed in Figure F.26b. Referenced on pages: 37, A.9.

Figure F.27. Americanisms in Dialects and Exposure to Immigrant Diversity, Permutation Analysis



Notes. This figure reports a permutation analysis of the cross-sectional association between the number of Americanisms in local dialects and emigration to the United States. The unit of observation is a municipality. The sample comprises the universe of Sicilian municipalities. The dependent variable is the zero-inflated log number of Americanisms that, in the *Vocabolario Siciliano*, appear in the municipality's dialect. The baseline explanatory variable is an indicator for municipalities with above-median US emigration. The histograms show the distribution of coefficients obtained from 1,000 random permutations of US emigration across municipalities, which is then dichotomized as in the baseline analysis. The vertical red line marks the coefficient estimated in the actual data. The permutation p -value is the share of permuted coefficients that are at least as extreme as the observed coefficient, based on a two-sided test. Each regression includes the number of emigrants to districts in the top quintile of the diversity distribution (3), population in 1901, and geographic controls—elevation, an indicator for municipalities on the coast, latitude, and longitude—and province fixed effects. Referenced on page: 39.

Figure F.28. Dialect Contamination and Exposure to Immigrant Diversity, Permutation Analysis



Notes. This figure reports a permutation analysis of the cross-sectional association between four measures of dialect contamination and emigration to diverse emigrant communities in the United States. The unit of observation is a municipality. The sample comprises the universe of Sicilian municipalities. The dependent variables are the zero-inflated log number of words that, in the *Vocabolario Siciliano*, appear in the municipality's dialect and in other municipalities (panel F.28a), and in 10 or more other municipalities (panel F.28b). In panel F.28c, the dependent variable is the distance-weighted analog of the baseline measure where words receive a larger weight if they are shared by more distant municipalities; in panel F.28d, the dependent variable is the log number of words shared with municipalities distant more than 100 Km. The main explanatory variable is an indicator equal to one for municipalities where the number of US emigrants who settled in districts in the top quintile of the diversity distribution (3) is above the median and zero otherwise. The histograms show the distribution of coefficients obtained from 1,000 random permutations of the continuous emigration measure across municipalities, which is then dichotomized as in the baseline analysis. The vertical red line marks the coefficient estimated in the actual data. The permutation p -value is the share of permuted coefficients that are at least as extreme as the observed coefficient, based on a two-sided test. Each regression includes the number of US emigrants, population in 1901, and geographic controls—elevation, an indicator for municipalities on the coast, latitude, and longitude—and province fixed effects. Referenced on page: 39.

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